



SAP LeanIX

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SAP LeanIX | CLOUD

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User and Access Management

Efficiently manage user access and authorization to ensure the right people have the appropriate permissions.

You can start by inviting users and assigning roles directly in SAP LeanIX. As your platform grows, enable single sign-on (SSO) to streamline user management through your identity provider (IdP).

Key capabilities include:

- Inviting users and assigning viewer, member, or admin roles
- Enabling SSO for centralized user authentication
- Configuring SCIM provisioning to automate user synchronization
- Setting up virtual workspaces to control access to specific fact sheets

Whether you're just getting started or have a mature SAP LeanIX implementation, these features allow you to securely manage your user base.

Managing Users

Learn how to manage user access and roles with and without single sign-on (SSO).

As you begin your journey with SAP LeanIX, managing user access and roles directly within the application is a straightforward way to get started. Later, you can set up SSO for a more efficient and streamlined user management experience. You can view workspace users in the [Users](#) section of the administration area. For more information, see [Users Overview](#).

Authentication and Authorization

Authentication

Authentication is the process of verifying the identity of a user or system attempting to access SAP LeanIX. Users provide credentials to confirm their identity and gain access.

You have two options for managing authentication:

- **SAP LeanIX (without SSO):** You manage access directly in SAP LeanIX. Users sign in to your workspace using their email and password on the sign-in page.
- **SSO:** You manage access through your identity provider (IdP) system. Users sign in through your IdP, which eliminates the need for separate credentials.

Authorization

Authorization determines what actions an authenticated user is allowed to perform in SAP LeanIX. It involves assigning specific roles and permissions that define the data and features users can access.

In SAP LeanIX, you manage permissions by assigning user roles. These roles define each user's access level within the workspace. For more details, see [User Roles and Permissions](#).

You have two options for managing authorization:

- **SAP LeanIX:** You manage user roles directly in SAP LeanIX.
 - If SSO is not enabled in your workspace, you always manage user roles in SAP LeanIX.
 - If SSO is used only for authentication, you manage user roles in SAP LeanIX.
- **SSO:** If you use SSO for both authentication and authorization, you manage user roles in your IdP system.

i Note

When setting up SSO, decide whether to use it for authentication only or for both authentication and authorization. For more information, see [Single Sign-On \(SSO\)](#).

Managing Users Without SSO

Without SSO, you can manage users for your workspace directly within SAP LeanIX. Users sign in to the application with their email and password.

Managing Users with SSO

When SSO is enabled for your organization, you can manage user access through your identity provider (IdP). Users sign in to SAP LeanIX through your IdP. To learn how to set up SSO, see [Single Sign-On \(SSO\)](#).

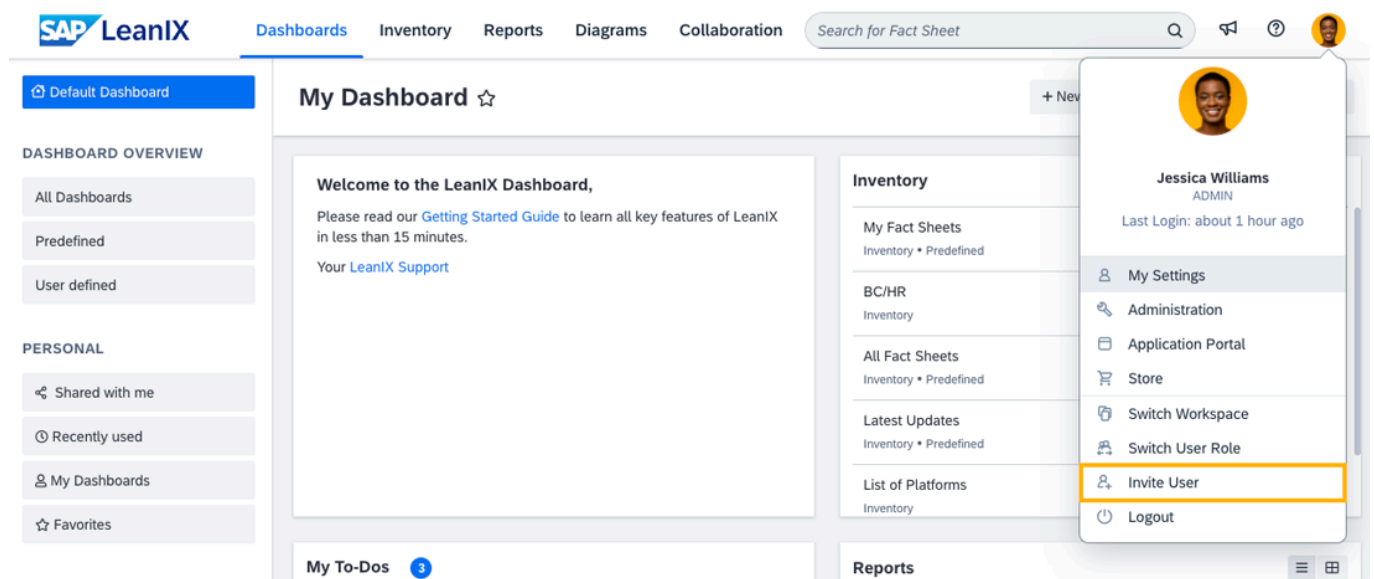
Inviting Users

To invite a user, you need to provide them with access permissions in your IdP. This enables the user to access all workspaces within the organization's SSO domain.

If you enabled the **Invite only** flow for a workspace during the SSO setup, in addition to granting access permissions in your IdP, you need to invite users through the **Invite User** option in the user profile menu. If a user is granted access permissions in your IdP but is not explicitly invited, they cannot access the workspace.

To invite a user to a workspace, follow these steps:

1. In the user profile menu in the upper-right corner, select **Invite User**.



Selecting "Invite User" in the User Profile Menu

2. In the overlay that appears, enter the user's email address, select a role, then click **Invite**. You can invite multiple users at once.

Invite to Workspace



E-mail addresses *

john.doe@example.com ✕

james.smith@example.com ✕

▼

User Rights *

Member

▼

Message

I would like to invite you to SAP LeanIX, a tool for managing our IT landscape in a simple yet effective way. Please use the link below to join our workspace.

Thank you,
Jessica

A header and the link is automatically added to the e-mail.

How to invite?

Please provide the e-mail addresses of the users you would like to invite to this workspace. Hit enter after typing each e-mail address.

The user will receive an e-mail from LeanIX (support@leanix.net) in order to complete their account details (full name and password).

Cancel

Invite

Inviting Users to a Workspace

Invited users receive email invitations to join the workspace.

Managing Roles

When inviting users to a workspace, you need to assign them a role. If needed, you can later change a user's role in the [Users](#) section of the administration area. To do this, click on the user's email address, adjust their role, and then click [Save](#).

← Edit User

User Details

User Subscriptions

E-mail

jessica.williams@organization.com

Username

jessica.williams@organization.com

Last Name

Jessica

First Name

Williams

Status

ACTIVE

Archive

Role

ADMIN

MEMBER

ADMIN

VIEWER

Last Login

Changing a User's Role

Depending on your SSO setup, you can manage user roles either within SAP LeanIX or within your identity provider. For more information, see [Managing User Roles with SSO](#).

Archiving a User

To delete a user and remove their access, you archive the user.

i Note

Archived users are permanently deleted after 90 days. To reactivate a user, you must do so within the 90-day archival period.

When SAP LeanIX permanently deletes a user, only the user's account is removed. Associated data is retained, such as fact sheet subscriptions and activity history.

During the 90-day archival period, the user name remains visible but is marked as an archived user. After permanent deletion, the user name is no longer visible and associated data shows **deleted user** instead of a name.

To learn more about user account deletion, see [Archiving Users and Automatic Account Deletion](#)

1. Go to **Administration > Users**.
2. Search for the user or go to one of the tabs, for example, **Active** or **Invited**.
3. Open the user you want to archive.
4. Select **Archive**.

Search by item name

MY SETTINGS

Notifications

Password

Profile

BASIC SETTINGS

Adoption KPIs

Branding

General

Meta Model Configuration

Subscription Roles

Tagging

Users

← Edit User

User DetailsUser SubscriptionsPII Change Audit Report

E-Mail

Username

Last NameWilliams

First NameJessica

StatusACTIVE

Archive

RoleVIEWER

Last Login2026-06-04 18:40

Cancel

Save

Reactivating an Archived User

You can reactivate an archived user at any time within the 90-day period before automatic deletion occurs.

⚠ Caution

After the 90-day period expires, deletion is permanent. You cannot reactivate a user who has been automatically deleted. Invite the user again to create a new user account.

1. Go to [Administration > Users](#).
2. Open the [Archived](#) tab.
3. Open the user you want to reactivate.
4. Select [Activate](#).

After reactivation, the user regains access to the workspace.

Users Overview

View workspace users and manage their access permissions and roles.

i Note

Depending on whether single sign-on (SSO) is enabled for your organization or not, the procedures for managing user access and roles vary. For more information on each scenario, see [Managing Users](#).

To learn how to enable SSO, see [Single Sign-On \(SSO\)](#).

In the [Users](#) section of the administration area, you can do the following:

- View and manage workspace users and contacts
- View user details and change their roles
- Export user data by clicking [Export](#)

Search by item name

BASIC SETTINGS

Branding

General

Meta Model Configuration

Subscription Roles

Tagging

Users

User Roles and Permissions

Dashboards

Users

All 20Active 16Invited 3Requested 0Not invited 1Archived 5

Search for name, role or status...

Export

E-mail	First Name	Last Name	Role	Status	Customer Roles	Access Control Entities	Last Login
jessica.williams@example.com	Jessica	Williams	Admin	Active			2024-09-23 18:02:13
carol.brown@example.com	Carol	Brown	Admin	Active			
david.koop@example.com	David	Koop	Viewer	Active			
james.smith@example.com	James	Smith	Contact	Not invited			
frank.martin@example.com	Frank	Martin	Admin	Active			

Invite

Users Section in the Administration Area

User Statuses

The following table lists the tabs that appear on the **Users** page, each corresponding to different user statuses.

Tab (User Status)	Description
All	All workspaces users, except for archived ones.
Active	Users whose permission for the workspace is active.
Invited	Invited users who haven't yet accepted the invitation request. You can reinvoke a user by clicking Re-invite .
Requested	Invited users who haven't yet been approved by an admin. This admin invitation mode can only be enabled by SAP LeanIX Support.
Not invited	Users subscribed to fact sheets as contacts. To learn more about contacts, see Contacts .
Archived	The user's access to the workspace is removed. Archived users are permanently deleted 90 days after archival. Within the 90-day archival period, admins can reactivate an archived user from the user management.

i

Note

On the **Users** page, the count of users on each tab also includes technical users, but these users are not listed. For additional information, see [Technical Users](#).

Contacts

In SAP LeanIX, users who are subscribed to fact sheets but haven't yet been invited to the workspace and don't have an SAP LeanIX account are called contacts. Contacts are created when non-SAP-LeanIX users are added as fact sheet subscribers. For more information, see [Adding Non-SAP-LeanIX Users as Subscribers](#).

The table below highlights the differences between contacts and workspace users:

Criteria	Contact	Workspace User
Status To learn more about statuses, see User Statuses .	Not invited	Active

Criteria	Contact	Workspace User
		Invited (users who haven't yet accepted the invitation request)
Ways to add users	- Excel import (subscriber) - Inline table editing (subscriber) - API request	- Email invitation - SSO - SCIM - API request
Can be added as a subscriber	Yes	Yes
A permission role is assigned	No	Yes
Access to the workspace	No	Yes
Receives notifications and surveys from SAP LeanIX	No	Yes

You can view all contacts on the **Not invited** tab in the **Users** section of the administration area. When needed, you can invite contacts to a workspace individually or in bulk by clicking **Invite** on the right. For SSO-enabled workspaces, to prevent sending invitation emails, clear the **Send Invitation Email** checkbox in the invitation overlay.

You can also invite contacts from the **Subscriptions** tab on a specific fact sheet or by using the **Invite User** option in the user profile menu.

Search by item name

General

Meta Model Configuration

Subscription Roles

Tagging

Users

User Roles and Permissions

Dashboards

Users

All 22Active 16Invited 4Requested 0Not invited 2Archived 5

Search for name, role or status...

☐ E-mail	First Name	Last Name	Role	Status	Customer Roles	Access Control Entities	Last Login	
☐ james.smith@example.com	James	Smith	Contact	Not invited				Invite
☐ john.doe@example.com	John	Doe	Contact	Not invited				Invite

2 Items < >

Contacts on the Not Invited Tab in the Users Section of the Administration Area

User Details

To view information for a specific user, on the **Users** page, select a user by clicking their email address. On the user details page, you can:

- View user details:** This includes the user's email address, name, status, role, and last login time.
- Archive and delete a user:** Workspace admins can archive users and users can also delete their own accounts through self-service. After a 90-day archival period, SAP LeanIX automatically and permanently deletes archived users. You cannot manually delete users, the process always start with archiving.
- Change the user's role:** If SSO isn't enabled or you're managing roles within SAP LeanIX with SSO, you can change a user's role on the user details page. Select a different role from the **Roles** list and click **Save**. The new role takes effect upon the user's next sign-in.
- View user's fact sheet subscriptions:** On the **User Subscriptions** tab, you can view all fact sheets a user is subscribed to, along with their subscription type and role for each fact sheet.

← Edit User

User Details

User Subscriptions

E-mail

jessica.williams@example.com

Last Name

Jessica

First Name

Williams

Status

ACTIVE

Archive

Role

ADMIN

Last Login

2024-09-23 18:02

Cancel

Save

User Details Page

Archiving Users and Automatic Account Deletion

Understand how archiving users and automatic user deletion works in SAP LeanIX, what happens to user data, and how to reactivate accounts.

Overview

SAP LeanIX automatically deletes archived users to comply with the EU General Data Protection Regulation (GDPR) and similar privacy legislation. These regulations require that personal data isn't retained longer than necessary and that data subjects have the right to erasure.

Deletion Process

In SAP LeanIX, you can't directly delete a user. Instead, you archive the user. Archiving a user removes their access. You can still see the user in the user management and reactivate the account.

Exactly 90 days after their individual archival date, users are permanently deleted. This action is irreversible and the user account can't be restored.

Usage Notes

- Deleting a user applies only to a single workspace, not across workspaces.
- The user deletion process can't be disabled because it's required for GDPR and similar privacy legislation.

What Is Deleted

When you permanently delete a user, SAP LeanIX deletes only the user's account and profile. All associated data is retained, including fact sheet subscriptions, activity history, and audit logs.

The user name remains visible during the 90-day archival period but is marked as an archived user. After permanent deletion, the user name is no longer visible, and associated data shows **deleted user** instead of a name.

How Archiving Users and Automatic Deletion Works

Archiving and Deletion Actions			
Action	Who	Permanent Deletion	Details
Delete account: User deletes their own account from profile settings	User	After 90 days	Delete Your User Account
Archive user: Admin removes workspace access	Admin	After 90 days	Archiving a User
Reactivate user: Admin restores access	Admin	Removed from deletion, account set to active	Reactivating an Archived User

Related Information

- [Managing Users](#)
- [Users Overview](#)
- [SCIM Provisioning](#)
- [Delete Your User Account](#)

User Roles and Permissions

Adjust role-based permissions and define custom roles.

In the **User Roles and Permissions** section of the administration area, you can:

- Create custom user roles — if you're managing roles within your single sign-on (SSO) identity provider (IdP)
- Edit translations for user roles
- Modify role-based permissions unrelated to fact sheets

User Roles

In SAP LeanIX, the following standard user roles are available by default: Viewer, Member, and Admin. If SSO is enabled for your organization and you're managing roles within your IdP, you can create custom roles in addition to standard ones.

Standard User Roles

The following table lists standard user roles available in SAP LeanIX.

User Role	Default Fact Sheet Permissions
Viewer	Can view, subscribe to, and comment on all fact sheets.
Member	Can view, create, and modify all fact sheets.
Admin	Has all permissions of a member user as well as administrator permissions for a workspace.

As an admin, you can adjust permissions for non-admin roles. For more information, see [Role-Based Permissions](#).

i Note

You cannot delete standard user roles.

Custom User Roles

When configuring SSO for SAP LeanIX, you choose how you want to manage user roles: internally within SAP LeanIX or externally within your IdP. For more information, see [Managing User Roles with SSO](#).

The possibility to create custom user roles depends on how you manage roles:

- **Managing user roles internally within SAP LeanIX:** Only standard roles are available: Viewer, Member, and Admin. You can not create custom user roles.
- **Managing user roles externally within your IdP:** You can create custom user roles in addition to standard ones. This lets you assign permissions that are more granular than those provided by standard roles.

i Note

- A standard role is still required for access, but the custom role takes precedence.
- If a user is assigned multiple custom roles, their permissions are aggregated.

Creating Custom User Roles

Follow these steps:

1. In your IdP, add claims `role` and `customer_roles`, then create the corresponding roles.

- `role`: Create the following roles using uppercase letters: VIEWER, MEMBER, and ADMIN.
- `customer_roles`: Create custom user roles using uppercase letters, for example, AUDITOR.

To learn more about attribute mapping, see [SSO Attribute Overview](#).

2. In SAP LeanIX, create the corresponding user roles in the **User Roles** section of the administration area.

- Click **New User Role**. You land on the role configuration page.
- In the **Technical Name** field, enter the role name using uppercase letters as specified in your IdP, for example, AUDITOR. The name serves as the unique role ID.
- Optional: Clone permissions for the custom role from an existing role. This is a one-time snapshot, not a dynamic relation that is actively maintained. If you skip this option, the new custom role gets a set of default permissions, ensuring that the workspace functions as expected.

- d. Optional: Add translations for the technical name to provide a more user-friendly format for the role label. Translations appear in various areas of the application. Select languages, then enter a translation and description for each language.
- e. Click [Add](#).
- f. Create more custom roles to match the role matrix in your IdP.

← Add User Role

Technical Name *

AUDITOR

Clone From

MEMBER x v

Translations

☒ English

Translated Name

Auditor

Description

Users with this role have additional permissions for auditing specific fields

☒ German

Translated Name

Prüfer

Description

Benutzer mit dieser Rolle haben zusätzliche Berechtigungen zur Überprüfung bestimmter Audit-Felder

☐ Spanish

☐ French

☐ Portuguese

Cancel

Add

Creating a Custom User Role

3. In SAP LeanIX, configure permissions for each custom role. To learn more about permissions, see [Role-Based Permissions](#).

After you’ve configured custom roles, you can assign these roles to users in your IdP. Users get access to SAP LeanIX with the permissions that you configured.

With this configuration, you can’t assign custom roles to users in your SAP LeanIX workspace. You can only do it in your IdP.

Role-Based Permissions

In SAP LeanIX, user permissions are based on their assigned roles. As an admin, you can configure permissions for non-admin user roles as needed. Permissions fall into two main categories:

- **Fact sheet permissions:** Configure these in the [Meta Model Configuration](#) section of the administration area. For more details, see [Fact Sheet Permissions](#).
- **Permissions unrelated to fact sheets:** Configure these in the [User Roles and Permissions](#) section of the administration area. For an overview of permissions, refer to the following table. Permissions for the admin role are read-only to ensure all workspace admins can view and modify items.

The table below lists permissions unrelated to fact sheets.

Feature	Permissions	Related Information
Architecture executive dashboard	• Read: View the architecture executive dashboard.	Architecture Executive Dashboard

Feature	Permissions	Related Information
	• Update: Update the architecture executive dashboard.	
KPIs	• Update: Update existing KPIs. • Create: Create new custom KPIs. • Delete: Delete custom KPIs.	KPIs
Portals	Open, delete, create, and update portals	Portals
Collections	Manage collections and content within collections	Collections
Surveys	Create and manage surveys	Surveys
Diagram templates	• Read, create, update, delete, unlock, and change the owner for diagram templates • Generate diagrams from templates	Diagram Templates
Fact sheet shape templates	Manage fact sheet shape templates in diagrams: view, delete, create, update, change the owner, change default settings, and unlock custom templates	Fact Sheet Shape Templates
Presentations	Manage presentations	Presentations
Discovery inbox	Access to the SAP landscape discovery inbox and SaaS discovery inbox	SAP Landscape Discover Inbox SaaS Discovery Inbox
Architecture decisions (available to customers using SAP LeanIX Architecture and Road Map Planning)	• View, create, update, and delete architecture decisions • Manage architecture decision templates	Architecture Decisions
Software bill of materials (SBOM)	• Read: View SBOM explorer and SBOM fact sheet widget • Upload: Upload SBOM files	Software Bill of Materials (SBOM)
Tech stack discovery	• View tech stack discovery • Create and update tech stacks	Tech Stack Discovery from SBOMs
Transformations	Read, create, update, delete, and execute transformations	Transformations

Feature	Permissions	Related Information
Developer tools	• Access the GraphQL editor. • Enable the Copy as JSON Query option in the filter bar in the inventory to copy the current query.	GraphQL API

→ Tip

As a best practice, grant full permissions sparingly and avoid assigning all available permissions to every role. By ensuring that only users with the appropriate roles have the ability to modify data, you can maintain its quality and prevent unintended changes.

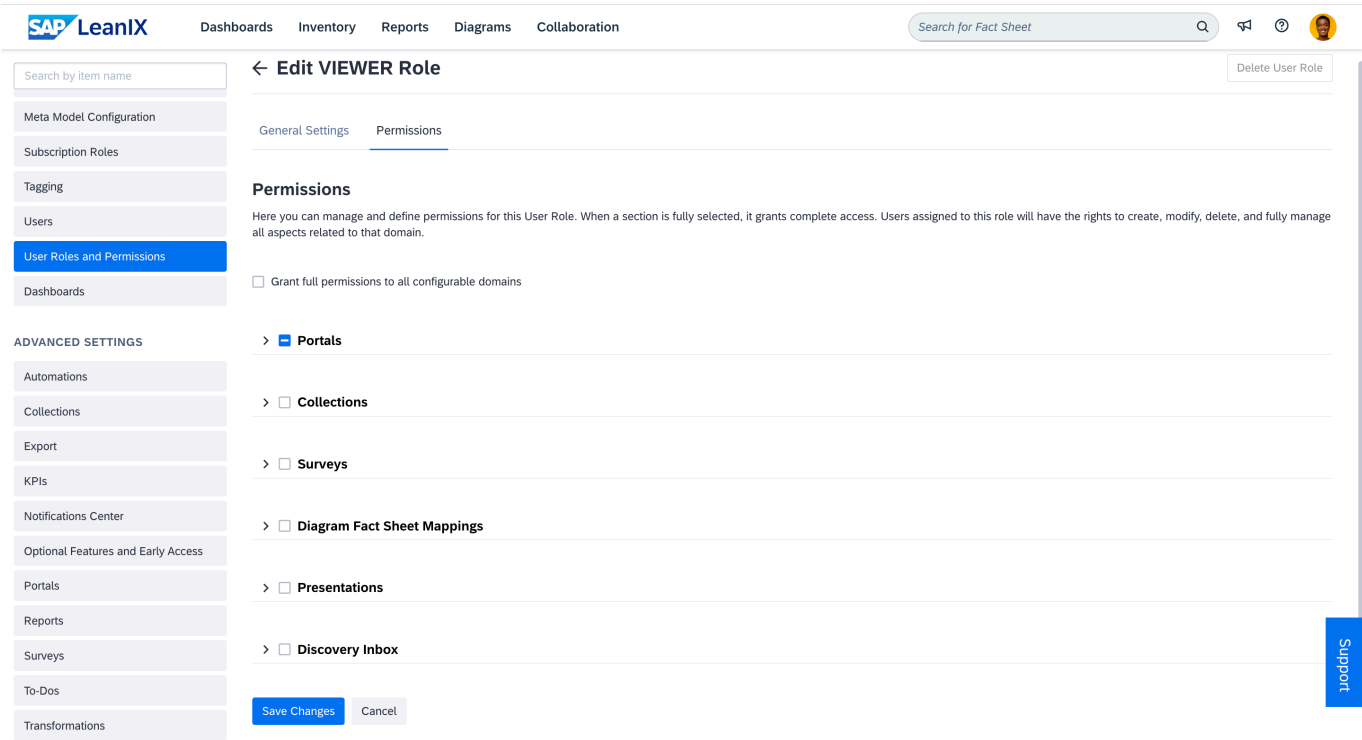
Configuring Role-Based Permissions

→ Tip

Configure fact sheet permissions for non-admin roles in the [Meta Model Configuration](#) section of the administration area. For more information, see [Fact Sheet Permissions](#).

To configure role-based permissions unrelated to fact sheets, follow these steps:

- 1. In the administration area, navigate to the [User Roles and Permissions](#) section.
- 2. Select a role for which you want to configure permissions.
- 3. On the role editing page, navigate to the [Permissions](#) tab.
- 4. Adjust permissions as needed using relevant checkboxes.
- 5. Save the changes.



Adjusting Permissions Unrelated to Fact Sheets for the Viewer Role

Authorization Model

Overview

The SAP LeanIX out-of-the-box Authorization model (part of the overall meta model) includes three default roles, VIEWER, MEMBER, and ADMIN. **In general, these roles govern access to read (view), create (add), update (edit), and delete information at the Fact Sheet TYPE level. This "information" includes but is not limited to attributes, relations, and tags.** This means that the Authorization Model is set and defined for the entire Fact Sheet TYPES, for example, IT Component. There are also some capabilities surrounding Import, Export, Saved Searches, and more.

Please see the [meta model](#) for basic information on what a Fact Sheet TYPE is and what is the SAP LeanIX meta model.

You can request that these settings be different per role, VIEWER, MEMBER, and or ADMIN. In addition, you can also request that new roles be added to the authorization model; they would exist in addition to the standard roles.

The ability to configure the Authorization model is contingent upon the SAP LeanIX package that you have. As such, please discuss any authorization model changes that you are considering with your CSM, and they can inform you if your package covers authorization model changes. This document is not meant to be exhaustive in nature, as such, having a conversation with your CSM regarding your use case, especially if it's not covered in this document, could be very useful.

i Note

Virtual Workspaces

If your organization is looking to control access at the level of **individual Fact Sheets**, your Use Case may be accommodated by [Virtual Workspaces](#). The standard authorization model cannot control access to individual Fact Sheets.

! Restriction

New Authorization Model roles can only be created and utilized when your organization has your Identity Provider configured as external IdP. To learn more about setting up single sign-on (SSO), see [Single Sign-On \(SSO\)](#).

Any customer can request that the out-of-the-Box roles be updated (no new roles being created), this does not require a particular IP configuration (assuming that your package allows Authorization model changes).

Terminology

Please see the chart below for helpful terminology:

Terminology	Definition
read	To be able to "see", to make visible. For example, this can be defined for a particular attribute and or relation.
create	To be able to associate a response to an attribute, relation, etc that is "null"/empty/not filled out.
delete	To be able to remove/delete information that has been entered AND SAVED against a particular Fact Sheet within a particular attribute, relation, etc. Such as a particular relation, e.g. the Data Objects relation on the Application Fact Sheet type.
update	To be able to edit information that is already existing. For example, the Description of a Fact Sheet reads, "A new version is coming out in 2022". If a user is assigned to an authorization role that has

Terminology	Definition
	access to update this attribute, they would be able to CHANGE the answer to anything else they wish, such as "A new version is coming out in 2024"

Fact Sheet Attributes

The most commonly requested authorization model configuration changes pertain to controlling read (view), create (add), update (edit), and delete access on attributes. As previously explained, for the most part, the Authorization Model works on the level of Fact Sheet types. If you need clarification on what an attribute or relation is, please see [Meta Model](#).

For each attribute and or "relation" on a Fact Sheet type, it is possible to define whether a role has a view, edit, and or delete access to it. These properties can be combined into several combinations, that is, as long as it is logical. For example, you would not set the Description attribute to be write(edit) access, but NOT read(access) for the MEMBER role or any role. After all, a user cannot edit what they cannot see.

In the example below, the Description attribute has been set to be read-only (view only) for any user assigned to the Member role. You can see that the attribute is visible but greyed out, which indicates to the user that they do not have access to edit this attribute.

LeanIX

DashboardInventoryReportsDiagrams

Jump to a Fact Sheet

+ Invite

What's new

AC Management Cloud7%

Set your role

+

i

ApplicationPublic CloudInvestEdit tags

Fact SheetSubscriptionsCommentsResourcesMetricsSurveysLast Update (25 days ago)

Information33%

Name & Description

Name

AC Management Cloud

The name is used to identify this Fact Sheet in the Inventory, Reporting and Search.

Alias

Use this field for an alternative name, which is also used in full-text search.

Description

Please provide a meaningful description to enable other users to understand the main purpose.

Release

External ID

Please enter an external ID...

Store your External ID here which uniquely identifies this Fact Sheet.

Recommended Business Capabilities

We have no recommendation for you at this time. Add more information to the description or make the descriptions of your Business Capabilities more descriptive.

LeanIX v3 ID

100000109

Cancel

Save & Next

Save

»

INTEGRATIONS

Pivio

TO-DO

Remind me

RECENTLY VIEWED

AC Management Cloud

Acoco Application Hosting

The same can be done for any attribute and or relation, whether it's an out-of-the-box attribute such as a description or a custom attribute (an attribute that was created by the customer using self-configuration).

The same logic applies when setting an attribute to write(edit), delete, and or update properties. You can see in the screenshot above that any attribute that is available to write (edit) has a textbox that is white instead of grey.

Below is an example where create (add) was restricted for Members on the Application Fact Sheet type, Application to IT Component relation. You can see that there is a lock symbol within that subsection indicating to the user that they cannot "add"/create a new relation.

Data Management0%

Data Objects

Which Data Objects are created, modified or required by this Application?

Provided Interfaces

Which Interfaces are provided by this Application?

Consumed Interfaces

Which Interfaces are consumed by this Application?

Sourcing0%

Technical Fit

In what shape is this Application from a technical point of view?

IT Components

Which services, software & hardware (e.g. Hosting, ASP, SaaS) is this Application using? [Show in Inventory](#)

Common Use Case: Hide Costs

A common use case is making cost and or other financial information hidden from specific authorization roles. The "Total Annual Cost" is an out-of-the-box attribute that can be found in several relations, including the Application to IT Component relation. This can be done by removing "read" (view) access for that particular attribute and or relation.

Below you can see that the Total Annual Cost **is not visible** to the user who signed in under the MEMBER authorization role:

IT Components

IT Component

Search for Fact Sheet

Active from

yyyy-mm-dd

Active until

yyyy-mm-dd

Technical Fit

Please select

Please estimate if there is any need to replace this IT Component with respect to the business requirements today and in the near future (next 2 years).

Description

Service level

Cancel

Save & New

Save & Next

Save

RECENTLY VIEWED

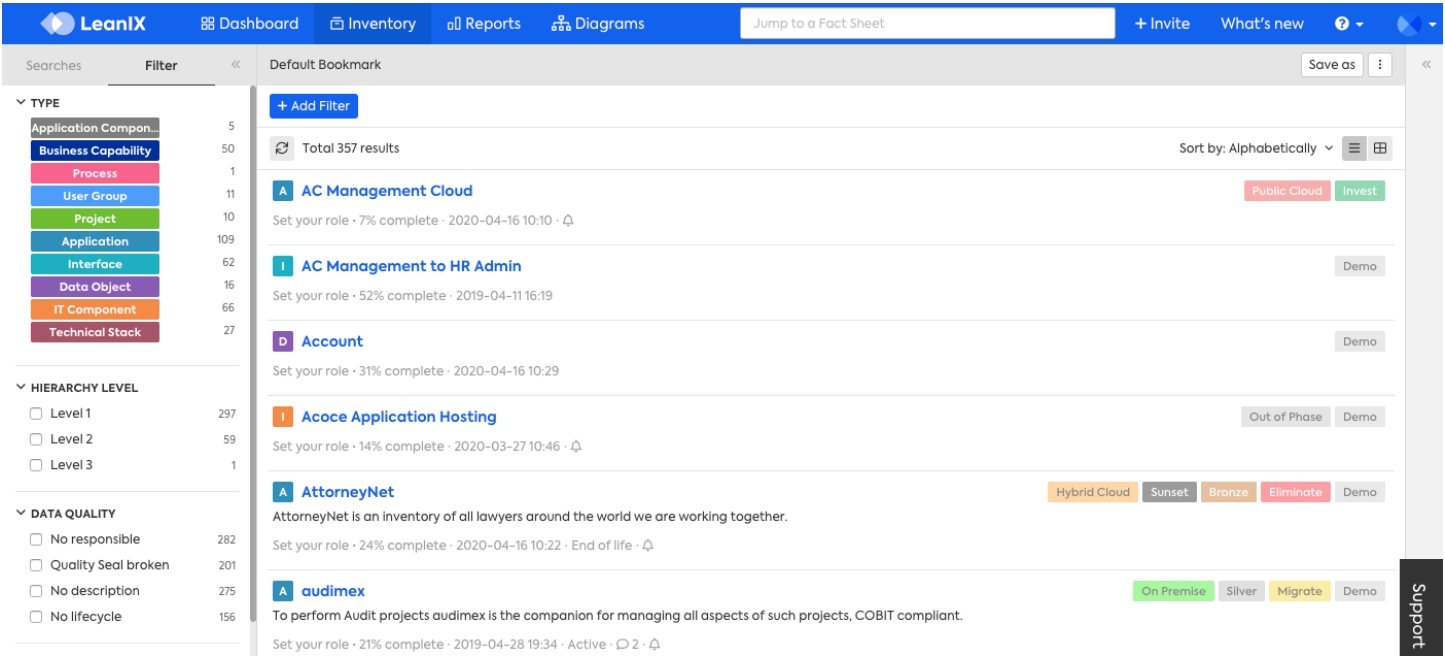
AC Management Cloud

Acocce Application Hosting

Fact Sheet Type Level

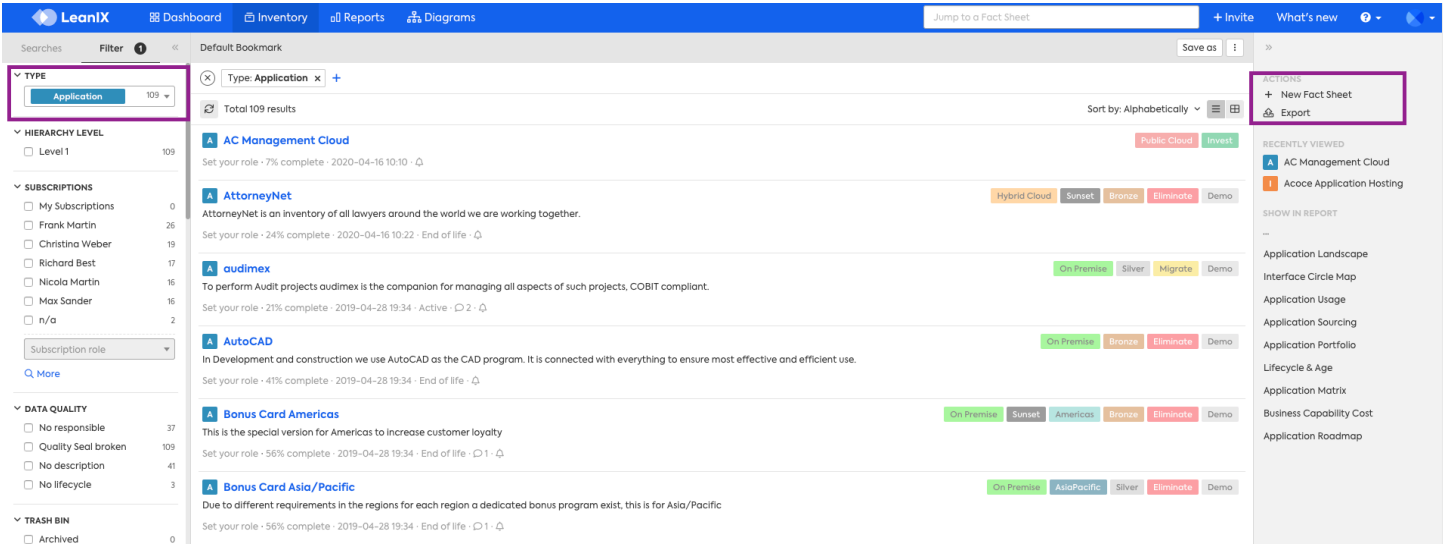
Entire Fact Sheet types can be set to be read, created, updated, and or deleted per role. In the example below, take a look at the upper left-hand side. Notice that the Provider Fact Sheet type is no longer visible to any users assigned to the Member role. This was done by removing read (view) access.

Similarly, entire Fact Sheet types can be set to be read-only (view) by specific roles. This means that those roles would only be able to view the information for any Fact Sheets under that particular Fact Sheet TYPE, but they would not be able to delete, update (edit), nor create any information. This is helpful for Fact Sheet types that represent fixed models or taxonomies like Business Capabilities and or Tech Category.

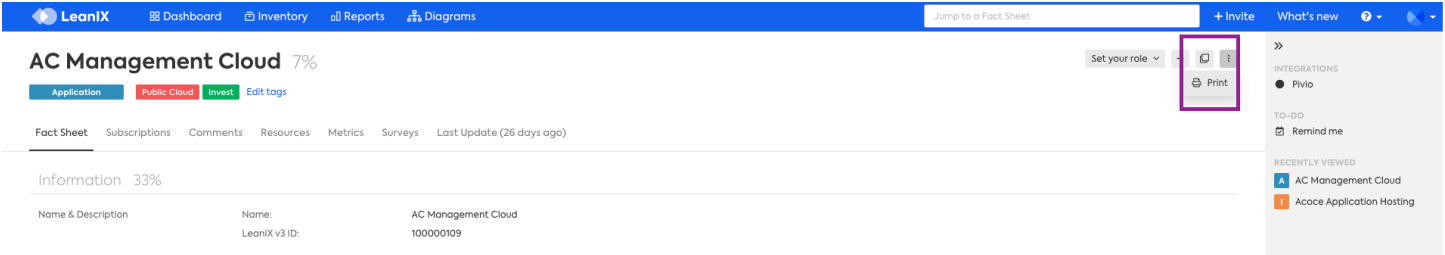


In addition, the Export, Import, and Inline Edit functionality can also be restricted. In the below example, Import and Table Edit have been restricted for MEMBERS for the **Application Fact Sheet Type**. As you can see, the user does NOT have access to the Import functionality, which would normally be in the highlighted area on the right-hand side.

Please see below, and notice that the Import button is not available for the Application Fact Sheet type.



Below is an example where the MEMBER role has been restricted from "Archiving" any Application Fact Sheets:

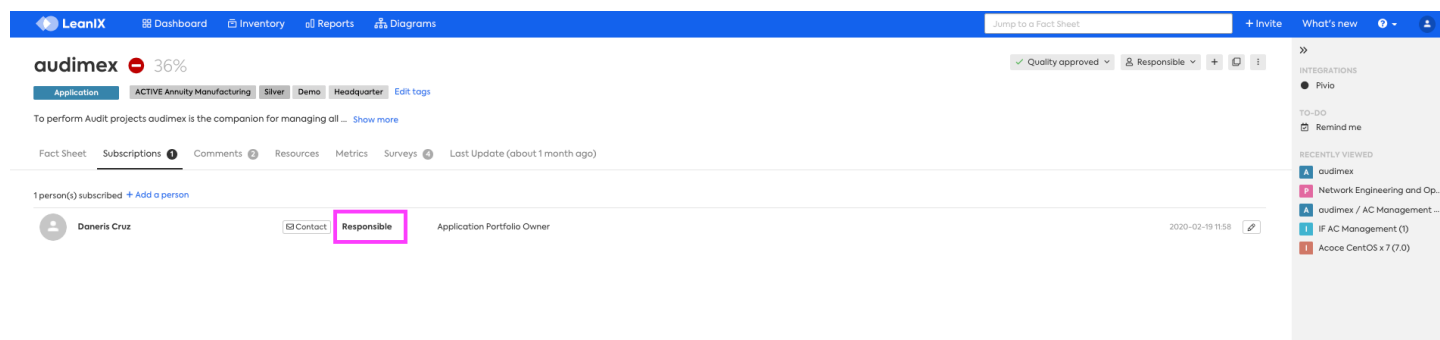


Subscription Type Check

Any attribute or relation can have a "subscription type check" against it. If you are not familiar with Subscription Types, please see the [Manage Subscription Roles](#).

A subscription type check allows you to define read, create, delete, and or update access based on that user's subscription **type** for that particular Fact Sheet TYPE. The actual check occurs at the Fact Sheet level, this is because subscriptions are assigned to the necessary users, on each Fact Sheet, via the Subscriptions tab of that Fact Sheet.

In the example below the audimex Fact Sheet has one subscription type assigned, which is the subscription type of Responsible. **Please take note that it is not possible to complete a check by "Application Portfolio Owner".** Application Portfolio Owner is the **Subscription Role**, and the type is "Responsible".



Saved Searches

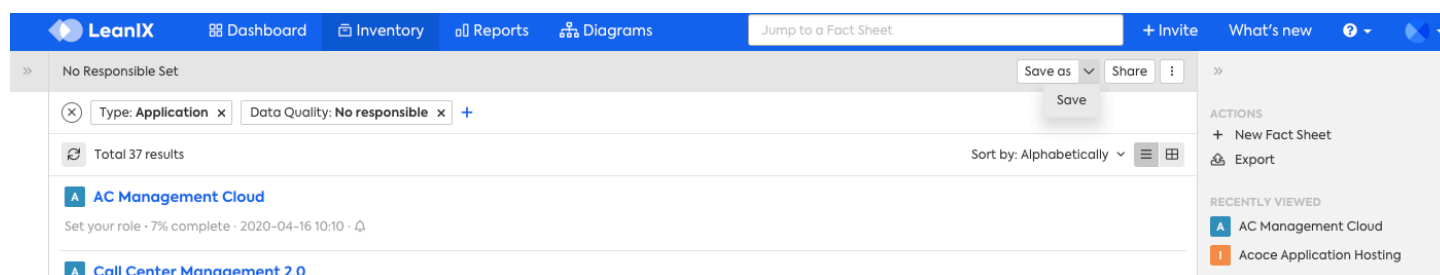
Saved Searches access can also be controlled, but these changes apply to All Fact Sheet types. It is not possible to define access differently for Saved Searches by Fact Sheet Type. The same options previously described are available read (view), create (add), update (edit), and delete.

In the example below, as a Member I have read, create, update, and delete access. This Saved Search is named "No responsible set". You can see in the upper right-hand corner, along with the grey box, that I have access to Save As (UPDATE) and Save (CREATE)

i Note

Saved Searches Permission Settings

The Authorization model will respect the specific Saved Search settings, defined at the time that the Saved Search is created and Saved. For example if "User B", creates a search and sets it so that only they have access to edit it. I as "User A", cannot update that Saved Search even if the role I have has access to update Saved Searches.



Granular Access

In addition, "change-owner", and "manage predefined" can be set. Change Owner is referring to the "Change Owner" button available within "Manage Searches". In the default Authorization model, the only role that has access to these two properties is ADMIN.

Manage Saved Searches

0 Saved Searches selected

Change OwnerShare

Search

Personalized onlyPredefined only

☐	Name	Owner	Last Update
☐	All Fact Sheets	LeanIX Default	2019-11-19
☐	My Fact Sheets	LeanIX Default	2019-11-19
☐	Latest Updates	LeanIX Default	2019-11-21

General Categories:

Type	Details
Published	
Private	
Shared	
System	This is referring to predefined Saved Searches, that all users have set by default.

Saved Searches can be defined at another level of granularity:

- Published[own]
- Private[own]
- Shared[own]

In the above examples, the functionality is restricted to the users "own" those Saved Searches. Meaning the searches that they themselves created.

Summary of Functionality

The Authorization Model allows for the configuration of many different combinations of access both within Fact Sheet types, and with other Fact Sheet related functionality such as Saved Searches.

Please see the tables in the following sections for a capabilities summary of the Authorization model.

Please refer back to the table available in the "Terminology" section, which provides details on the specific access properties that can be defined, they are read, create, update, and delete.

Routines

The table below focuses on executable programs, such as the Export routine, that you as a user can trigger, as opposed to fields/attributes within a Fact Sheet. As such, the normal CRUD operations (create, read, update, and delete) don't quite apply here. In general, if a user can see a routine, then it follows that they can run/execute that routine.

Functionality/Area of Interest	Is it possible to configure	Details/Example
Import	Yes	This is can be set by Fact Sheet TYPE. Meaning that this can be set to be viewable and usable on all Fact Sheet Types except IT Component (as an example).

Functionality/Area of Interest	Is it possible to configure	Details/Example
Export	Yes	This can be set by Fact Sheet TYPE. Meaning that this can be set to be viewable and usable on all Fact Sheet Types except IT Component (as an example).
Inline Edit	Yes	For detailed information on inline editing, see Inline Editing . Please note that Inline Edit DOES respect the Authorization model. For example, the MEMBER role is only able to read but Not update or create, the attribute Description for Application Fact Sheet types. In addition, the MEMBER role does have access to Inline Edit. This means that they will be able to click on Inline Edit and utilize it, but they will only be able to "read" the Description for Application Fact Sheet but not update, delete, or create. This can be set by Fact Sheet TYPE. For example, for a specific role(s) this could be set to be viewable and usable on all Fact Sheet Types except Provider.
Archive	Yes	Please see the Searching and Filtering in the Inventory , for information on what Archiving is. This is set per Fact Sheet TYPE.
Saved Searches (previously named bookmarks)	Yes	Example, "set the role NONMANAGER to be able to CREATE Saved Searches, but NOT update (edit)". This would apply to any and all Saved Searches for any user assigned to the NONMANAGER role. However, it does respect, the specific settings defined on the Saved Search. For example I can create a Saved Search and set it so that only I have access to edit it, this is defined on the Saved Search itself when I create and save it. The Authorization model will respect this. In addition, "change-owner", and "manage predefined". Are also available as properties that can be set. In the default Authorization model, the only role that has access to these two properties is ADMIN. Saved Searches can be defined at a more granular level: General Categories: Published, Shared, System(this is referring to predefined searches) Further granularity: Published[own], Shared[own]
Invite	Yes, but not via the Authorization model	Please reach out to support at SAP LeanIX Support or reach out to your CSM for more details.
Cloning	No	For more information, see Cloning a Fact Sheet
Printing	No	

Fact Sheet Specific

Functionality/Area of Interest	Is it possible to configure	Details/Example	Create	Read	Update	Delete
Fact Sheet Types	Yes	This is referring to entire Fact Sheet Types e.g Application, IT Component etc. Example set the "custom" role MANAGER to only READ (view) Application Fact Sheet types. This means that this role would not be able to create, update, or delete any Application Fact Sheets.	X	X	X	X
Fact Sheets (as in individual Fact Sheets such as Audimex)	No	No, access to individual Fact Sheets cannot be controlled via the Authorization model. One example of such a request is : can the Fact Sheet Audimex be completely hidden for MEMBERS. The answer is no. However, this functionality is available within Virtual Workspaces .				
Tags (<i><referring to associating tags to individual Fact Sheets></i>)	Yes	This is not referring to creating Tags in the Administration area, but rather associating tags to Fact Sheets.* In addition, this is referring to all tags. It is not possible to define a particular set of		X	X	

Functionality/Area of Interest	Is it possible to configure	Details/Example	Create	Read	Update	Delete
		tags that the role(s) should have access to.				
Tag Groups	No	Example, set user NONMANAGER to only be able to see, the tag group "Cloud". This is not possible. You might find it beneficial to remove some Tag Groups and instead set them as attributes. The benefit to this is that attributes can be set to create, read, update and or delete. This is a possibility that might be optimal depending on your Use Case. As such please contact your CSM in order to discuss this.				
Attributes (also called fields)	Yes	Examples of out-of-the-box attributes are Description, Alias, Release, and Name. Including attributes that are within relations such as Total Annual Cost. Attributes also includes any customer defined attributes. Depending on your SAP LeanIX package, you might have the ability via Self-Configuration or via SAP LeanIX	X	X	X	X

Functionality/Area of Interest	Is it possible to configure	Details/Example	Create	Read	Update	Delete
		Support to create attributes that do not exist in the standard meta model. If you have Virtual Workspaces, the "ACE" read and write can also be controlled in the same way as all other attributes.				
Relations	Yes		X	X	X	X
Attributes that exist within a Relation	Yes	For example, the standard Application to IT Component relation, includes a Total Annual Cost Attribute. This could be set to be READ only for the MEMBER role.	X	X	X	X
Quality Seal	Yes	It is possible to define for the property "update" only, this is due to the nature of the functionality. For more information on the Quality Seal, please see the Increase your Data Quality .			X	
Fact Sheet Tabs	No					

Administration Area Functionality

The entire "Administration" area is hardcoded to only be accessible to the ADMIN role. It is not possible for any other role to have access to the Administration area. Furthermore, it is not possible to define specific access to the Administration area even for the ADMIN role. Any user with this role will automatically have access to everything under the Administration area.

Please note that all users have access to Profile, Password, and Notifications. These three options are available under "My Settings".

Dashboards

At this time there aren't specific settings within the Authorization model that determine access to Dashboards. However, when users create Dashboards, they can define some access settings for the particular Dashboard that they are creating. For more information on dashboards, see [Dashboard Modeling](#).

i Note

At this time, all roles are able to create Public Dashboards even those with the role of VIEWER.

PII Change Audit Report

Export a record of changes to the personally identifiable information (PII) of individual users. Workspace administrators use this report to meet compliance and audit requirements.

Overview

The PII change audit report provides a downloadable log of updates to a users' personally identifiable information (PII). This report supports compliance requirements and audit processes by tracking changes to sensitive user attributes such as email addresses and names. It helps answer questions such as:

- When was a user's email address modified?
- Which administrator performed the change?

Report Content

The report exports as an XLSX spreadsheet containing user PII changes. It does not include audit logs for other areas of the LeanIX product.

Column	Description
Author	The administrator who made the change
Change Time	Date and time when the change was made
Column	The PII field that was modified
Old Value	The previous value before the change
New Value	The updated value after the change

Exporting PII Change Data

1. Go to the administration and navigate to [Users](#).
2. Select the user whose PII change history you want to review.
3. Go to the [PII Change Audit Report](#) tab.

← Edit User

User Details

User Subscriptions

PII Change Audit Report

Export a record of changes to the user's personally identifiable information (PII) for the specified time frame.

Start Date

dd.mm.yyyy

End Date

dd.mm.yyyy

The default time frame is last 30 days.

New Report

4. Select a **Start Date** and **End Date**.

If you don't select dates, the default time frame is the last 30 days.

5. Click **New Report**. Remain on the page until you see the success message that the report is completed. Then the file download starts automatically. Alternatively, you can leave the page to receive an email notification with the download link later.

→ Tip

If the link in the email doesn't work, verify whether Microsoft Outlook already opened the link in the background. The link works only once.

Result

You receive the report as an XLSX spreadsheet.

	A	B	C	D	E	
1	Author	Change Time	Column	Old Value	New Value	
2	aziz.ghorbani@sap.com	2026-01-23T15:53:32	Last name	Wilson	Meyer	
3	aziz.ghorbani@sap.com	2026-01-23T15:53:54	Email	jessica.wilson@sap.com	jessisa.meyer@sap.com	
4	aziz.ghorbani@sap.com	2026-01-23T15:53:54	User name	jessica.wilson@sap.com	jessisa.meyer@sap.com	
5						

Single Sign-On (SSO)

Learn how to configure single sign-on (SSO) with SAP LeanIX.

Overview

Single sign-on (SSO) is a centralized authentication process that allows users to access multiple applications with a single set of login credentials.

SAP LeanIX supports SSO through the following protocols:

- [Security Assertion Markup Language \(SAML\) 2.0](#) ➡

All SSO protocols securely exchange authentication and authorization data between an identity provider (IdP) and SAP LeanIX as service provider (SP).

We offer the following options for the SSO setup.

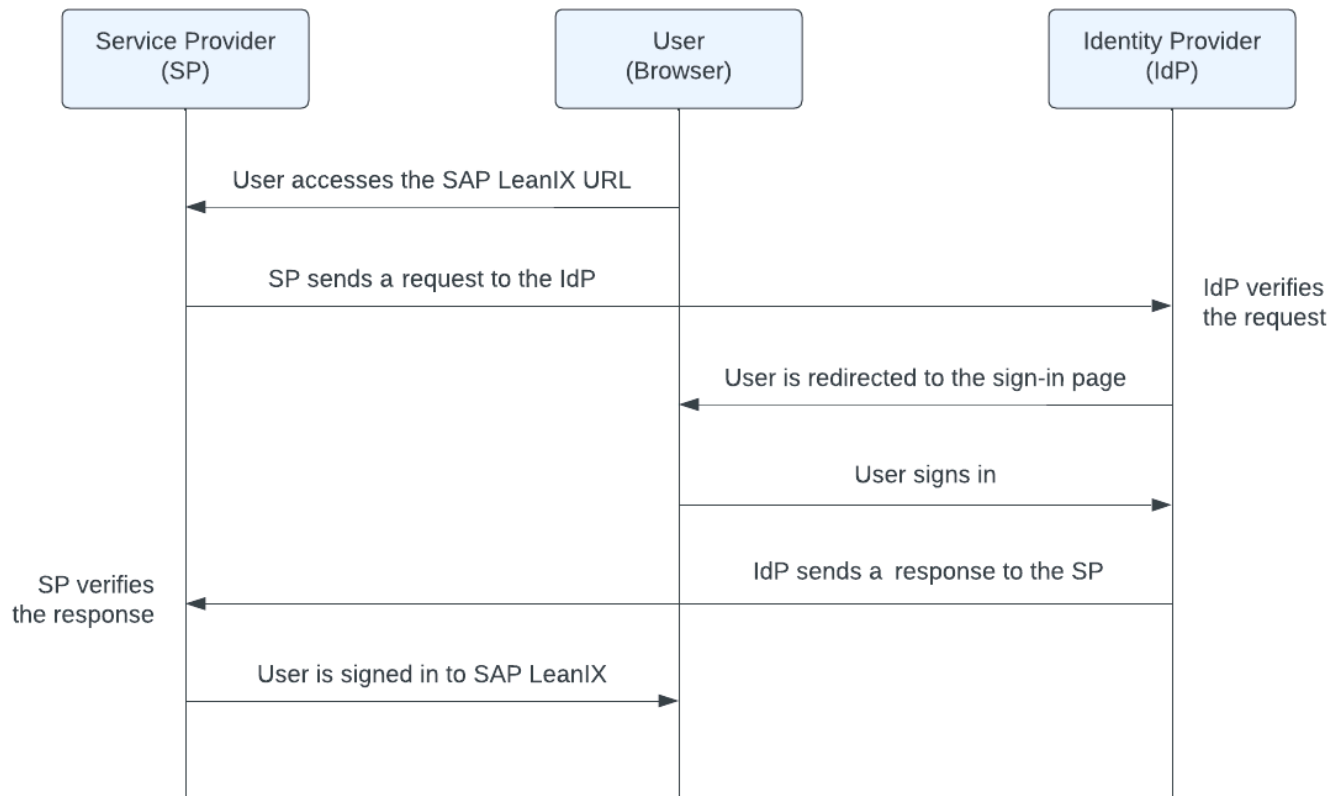
SSO Setup Options

	Authentication Only	Authentication and Authorization
How you manage authentication (system access)	Through SSO	Through SSO
Where you manage authorization (user roles)	In SAP LeanIX For details, see Managing User Roles Within SAP LeanIX .	In your IdP For details, see Managing User Roles Within the Identity Provider .
Additional options		• Custom user roles • Virtual workspaces

SAP LeanIX supports just-in-time (JIT) provisioning, a process in which user accounts are dynamically created when users sign in to SAP LeanIX through SSO for the first time.

Authentication Flow

SAP LeanIX supports SP-initiated SSO only. IdP-initiated login is not supported. The authentication flow must start from the SAP LeanIX login page or a configured service provider entry point. The following image illustrates the SAML 2.0 authentication flow.



Authentication Flow

When a user attempts to access the SAP LeanIX application through a browser, the SP first checks if the user is already authenticated. If not, the user is redirected to the identity provider system, which initiates the authentication process, typically through a username and password.

Once the user has been successfully authenticated within the IdP, the IdP sends a response to the SP. The response includes relevant attributes, such as the user's email address, that serve to identify the user within SAP LeanIX. Once the SP has verified the response, the user is granted access to the SAP LeanIX application.

Requirements for Identity Providers

When choosing an IdP for your SSO setup, ensure that:

- The IdP supports the SAML 2.0 protocol.
- The client browser has network access to the IdP.

Example Identity Providers

The following table lists example IdPs for which we've successfully tested SSO implementation.

Identity Provider	Instructions
Microsoft Entra ID	Configuring SSO with Microsoft Entra ID
Okta	Configuring SSO with Okta
SAP Cloud Identity Services - Identity Authentication Service (IAS)	Configuring SSO with SAP Cloud Identity Services - Identity Authentication Service (IAS)
PingOne	Configuring SSO with PingOne

Identity Provider	Instructions
OneLogin	Configuring SSO with OneLogin
Microsoft Active Directory Federation Services (AD FS)	Configuring SSO with Active Directory Federation Services
Shibboleth IdP	Refer to the Shibboleth IdP documentation ➡
CA Single Sign-On	Refer to the Broadcom documentation ➡
Google IdP	Refer to the Google documentation ➡
LemonLDAP::NG	Refer to the LemonLDAP::NG documentation ➡

Configuring SSO

With the self-service configuration you independently setup your SSO authentication for SAP LeanIX. To learn more, see [Getting Started With SSO](#).

Managing User Roles with SSO

With SSO, you can manage user roles internally within SAP LeanIX or externally within your IdP. Choose your preferred option when requesting to set up SSO.

Additionally, you can set up Transient users to grant them access to Self-service Portals. For more information, see [Transient Users with SSO](#).

Managing User Roles Within SAP LeanIX

If you choose to manage user roles within SAP LeanIX, the IdP manages authentication, while SAP LeanIX handles authorization.

You can assign roles to users when inviting them to SAP LeanIX. If needed, you can change user roles later in the **Users** section of the administration area.

You should decide whether you want to assign a default role in SAP LeanIX to all new users:

- If you set a default role, this role will be automatically assigned to all new users when they first sign in to a workspace. In this case, you don't need to manually invite users to your workspace.
- If you don't set a default role, you need to manually invite users to your workspace by selecting the **Invite User** option in user profile menu.

Managing User Roles Within the Identity Provider

If you choose to manage user roles externally within your IdP, then both authentication and authorization will be handled by the IdP.

When managing user roles within your IdP, you can define custom user roles in addition to standard roles. For more information, see [Custom User Roles](#).

The IdP determines the user role and maps it to the SAP LeanIX role during the sign-in process. To learn more about mapping user roles between your IdP and SAP LeanIX, see [SSO Attribute Overview](#). If you don't provide any role information, a default role is assigned to the user. If the default role is not set, the user loses access to SAP LeanIX.

Depending on the user roles provided in the IdP and SAP LeanIX, the following typical scenarios are possible:

- **Scenario 1:** You provide a user role in your IdP. If the user already has a role assigned in SAP LeanIX, it is overwritten by the role from the IdP.
- **Scenario 2:** You don't provide a user role in your IdP, and a default role exists in SAP LeanIX. In this case, the default role is assigned to the user. Their current role (if already assigned) is overwritten by the default role.
- **Scenario 3:** You don't provide a user role in your IdP, and a default role is not set in SAP LeanIX. In this case, the user loses access to a workspace. Their existing role is not overwritten but is not used.

Custom User Roles

i Note

You can create custom user roles only if you manage user roles externally within your IdP.

If you're managing user roles externally within your IdP, you can create custom user roles, which lets you assign permissions that are more granular than those provided by the standard roles. For more information, see [Custom User Roles](#).

User Account Deletion with SSO

If your workspace uses SSO, a user who deletes their own account can log in to the workspace again. You can prevent this in the following ways:

- Enable invitation-only mode. Without an explicit invitation, the user cannot log in again.
- Remove the access permission for the user in your identity provider.

Transient Users with SSO

SSO with an external IdP allows you to create transient user roles. A transient user is authenticated by SAP LeanIX based on their existence in your IdP, but is not assigned any role and therefore has no access to the workspace itself.

Transient users can access a simplified version of SAP LeanIX through self-service portals. You can embed these portals in your existing intranet, wiki, or any other system integrated with SSO. This setup allows you to directly share SAP LeanIX data without having to invite users to your workspace. For more information, see [Portal](#).

To create transient users, follow these steps:

1. Notify your support agent during the SSO setup or create a dedicated ticket.
2. To grant access to a self-service portal to transient users, in the portal configuration, enter **TRANSIENT** in the **Accessible for** field.

← Edit the configuration of this Portal

Basic
Status
Search
Detailed View
Branding
Localization

General

Unique name (will be part of the URL) *

☒ Enabled

Accessible for

ADMIN x
MEMBER x
POWERUSER x
VIEWER x
TRANSIENT x

No results

Modifying the Configuration of a Self-Service Portal to Grant Access to Transient Users

3. In your IdP, create a group for transient users if it doesn't already exist. Don't assign any permission roles (Admin, Viewer, or Member) to this group. If you're using Microsoft Entra ID, select the predefined value **Users**.

Users referenced in SAP LeanIX at any time, such as through an actual login or by being subscribed to a fact sheet as a contact, have a permanent user record. These users can't become transient users. Ensure they are granted proper role assignments, otherwise they'll lose access to the portal.

Upon signing in to SAP LeanIX, users are assigned the temporary transient role and are granted access to view the self-service portal data.

Getting Started With SSO

The self-service process of setting up SSO ensures a seamless integration with your chosen identity provider (IdP). You'll configure the SSO connection first, test the setup, and then activate. This ensures a flexible, zero-disruption deployment for different identity providers for different workspaces.

The SSO configuration options accommodate your existing infrastructure:

- **SAML** metadata provisioning

Before You Start

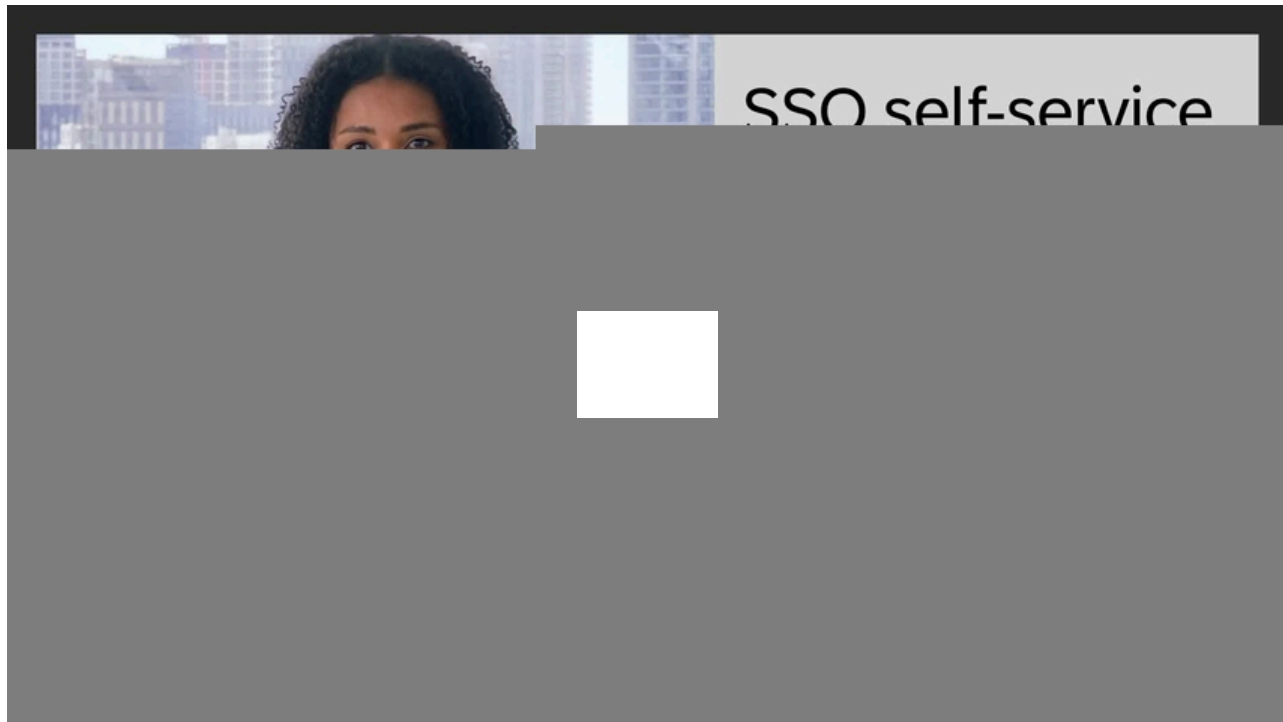
Flexible options allow you to customize your SSO setup to meet specific workspace needs and security requirements:

- **Multiple Identity Providers:** Choose one or more identity provider(s). You can connect to different identity providers, for example, region-specific identity providers. If you have different workspaces, for example a sandbox and a production workspace, you can connect each workspace to a different identity provider. For more information, see [Requirements for Identity Providers](#).
- **Authentication and Authorization:** Decide how you want to manage user roles. You can manage users internally within SAP LeanIX or externally within your identity provider. For more information, see [Managing User Roles with SSO](#).
- **Subdomain:** Decide whether you want a dedicated subdomain for your organization. You can choose any custom subdomain, such as your company's name, for example, `https://your-company.leanix.net`. You can also add domains later, as long as the workspace uses SSO.
- **Default Role:** Consider to set a default SAP LeanIX role for new users from your identity provider.

- **Invitations:** Find the suitable access setup for your workspace. The invite-only flow blocks users from accessing a workspace unless you have explicitly invited them, even if they have access through the identity provider.
- **Transient Users (Portal Viewers):** Enable transient users to potentially grant them access to self-service portals. To learn more, see [Transient Users with SSO](#).

After you set up SSO authentication, SSO becomes the sole access method for the platform. The self-service SSO configuration cannot revert to username and password authentication. If you need to revert, reach out to support.

To see the full SSO configuration process, watch the video that takes you through the SSO setup with Microsoft Entra ID as an example.



[Open this video in a new window](#)

Enable SSO Setup

If you haven't configured an SSO connection for your workspace before, you first enable the SSO setup in the administration area. If you have an SSO connection configured before November 2025 and want to migrate to the self-service SSO, the SSO setup should already be enabled and you can proceed with the next step to add a new identity provider.

1. In LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. Select one of the SSO authentication types.
3. Click **Save** to proceed with setting up the SSO configuration.

i Note

At this point, your current authentication method, including username and password, is still active and has not changed.

The active authentication type and the selection on the Authentication and SSO page might not match. Only once you activate the first SSO connection, the authentication is completely switched to SSO.

Search by item name

Subscription Roles

Tagging

Users

User Roles and Permissions

Dashboards

ADVANCED SETTINGS

Authentication and SSO

Automations

Calculations

Export

KPIs

Notifications Center

Optional Features and Early Access

Portals

Reports

To-Dos

Transformations

Workspace Views

Authentication

Authentication Type

- ☐ Username and Password (Default)
Managing user access and roles directly within the application is a straightforward way to get started. Users log in to the application with their SAP LeanIX email and password.
[Learn more](#)
- ☒ SSO - Authentication
Authentication (system access) is managed through SSO, while authorization (role management) is handled within SAP LeanIX. Users log in with their credentials for your identity provider.
[Learn more](#)
- ☐ SSO - Authentication and Authorization
Authentication (system access) and authorization (role management) are handled through SSO. This setup allows you to use extended authorization features, such as Virtual Workspaces and custom roles.
[Learn more](#)

Authentication Settings

Default Role

None

☐ Invitation Only
SSO users must be invited before first workspace access. When enabled, allowing transient users is not allowed.

☐ Allow Transient Users
Allows transient users to access the workspace. Transient users are users that are authenticated via SSO but unknown to LeanIX. Those users can only access the Self-Service Portal if they exist for the workspace.

Save

After saving, you can see a new **Identity Providers** section for setting up identity providers.

Search by item name

Subscription Roles

Tagging

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User Roles and Permissions

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ADVANCED SETTINGS

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Optional Features and Early Access

Portals

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Transformations

Workspace Views

DISCOVERY AND INTEGRATIONS

API Tokens

Authentication

Authentication Type

☐ **Username and Password (Default)**

Managing user access and roles directly within the application is a straightforward way to get started. Users log in to the application with their SAP LeanIX email and password.
[Learn more](#)

☒ **SSO - Authentication**

Authentication (system access) is managed through SSO, while authorization (role management) is handled within SAP LeanIX. Users log in with their credentials for your identity provider.
[Learn more](#)

☐ **SSO - Authentication and Authorization**

Authentication (system access) and authorization (role management) are handled through SSO. This setup allows you to use extended authorization features, such as Virtual Workspaces and custom roles.
[Learn more](#)

Identity Providers

[+ Add Identity Provider](#)

Authentication Settings

Default Role

None

☐ **Invitation Only**

SSO users must be invited before first workspace access. When enabled, allowing transient users is not allowed.

☐ **Allow Transient Users**

Allows transient users to access the workspace. Transient users are users that are authenticated via SSO but unknown to LeanIX. Those users can only access the Self-Service Portal if they exist for the workspace.

Save

Next Steps

Now you can start to configure SSO connections. See configuration instructions for different identity providers:

- [Configuring SSO with Microsoft Entra ID](#)
- [Configuring SSO with Okta](#)
- [Configuring SSO with SAP Cloud Identity Services - Identity Authentication Service \(IAS\)](#)
- [Configuring SSO with PingOne \(PingIdentity\)](#)
- [Configuring SSO with OneLogin \(OneIdentity\)](#)
- [Configuring SSO with Active Directory Federation Services](#)

FAQ

Can I keep roles managed within SAP LeanIX?

Yes, you can decide where you want to manage roles. SAP LeanIX offers the following authentication types:

- SSO - Authentication means roles are managed in SAP LeanIX
- SSO - Authentication and Authorization means roles are managed in the identity provider

Why can I not delete the identity provider?

Active identity providers cannot be deleted as this would remove access to the workspace for the users on this identity provider. Check if the identity provider is still active before deleting it.

If there is only a single identity provider configured, it cannot be deleted as otherwise you and your users would lose all access to SAP LeanIX.

Migration to the Self-Service SSO Configuration

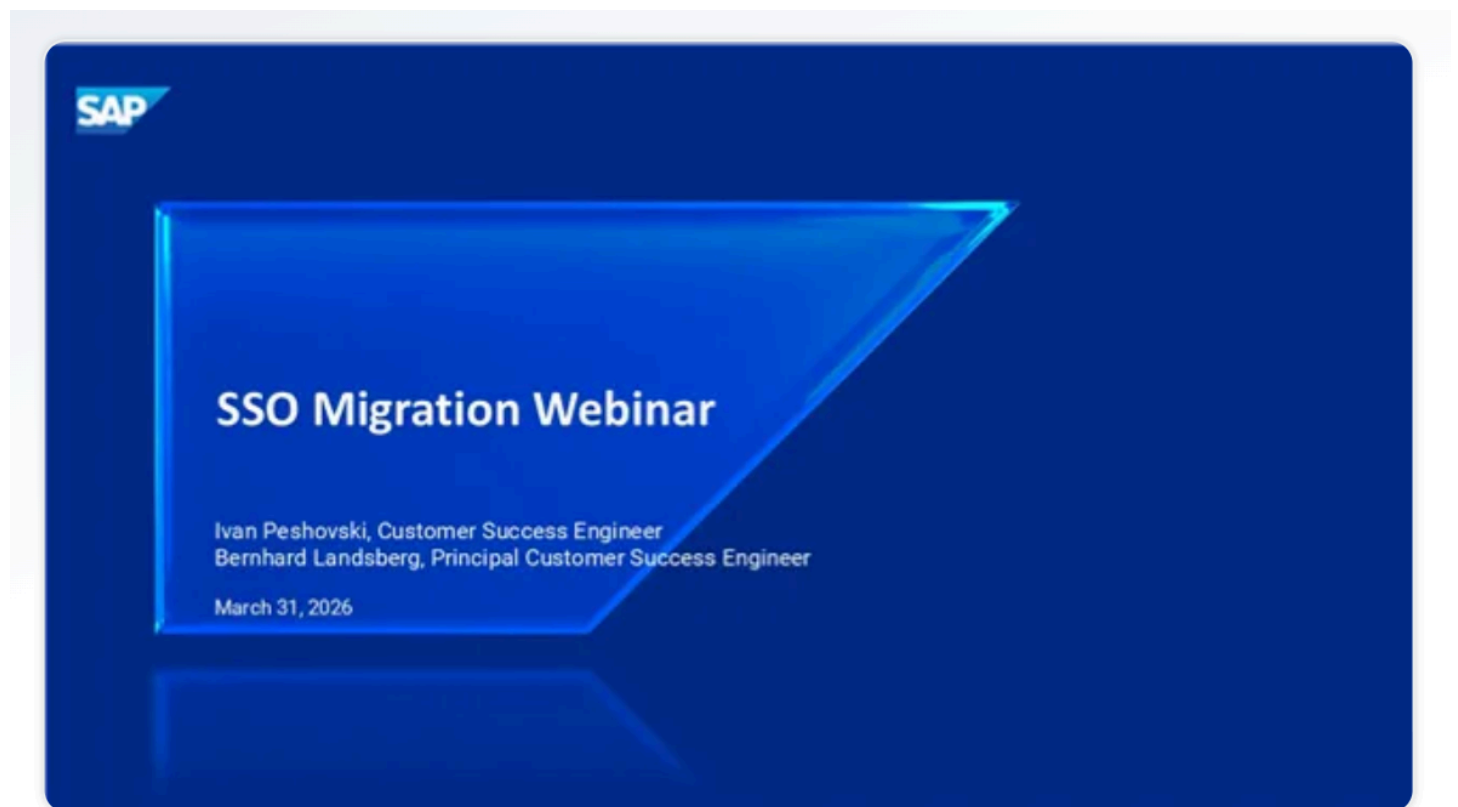
Migrate to the new SSO self-service for improved authentication and authorization until July 3, 2026.

Overview

SAP LeanIX is migrating all customers to a new single sign-on (SSO) service. This migration requires active changes in your identity provider (IdP) and workspace-specific SSO configurations in SAP LeanIX. During the migration, your users don't experience any disruptions and can always access your workspaces.

This guide covers the steps needed to migrate to the new SSO service.

View our expert-led SSO migration webinar to see step-by-step setup and learn about typical questions. To view the webinar, copy this URL to your browser: https://www.brighttalk.com/webcast/18723/664465?utm_source=SAPLeanIX&utm_medium=brighttalk&utm_campaign=664465.

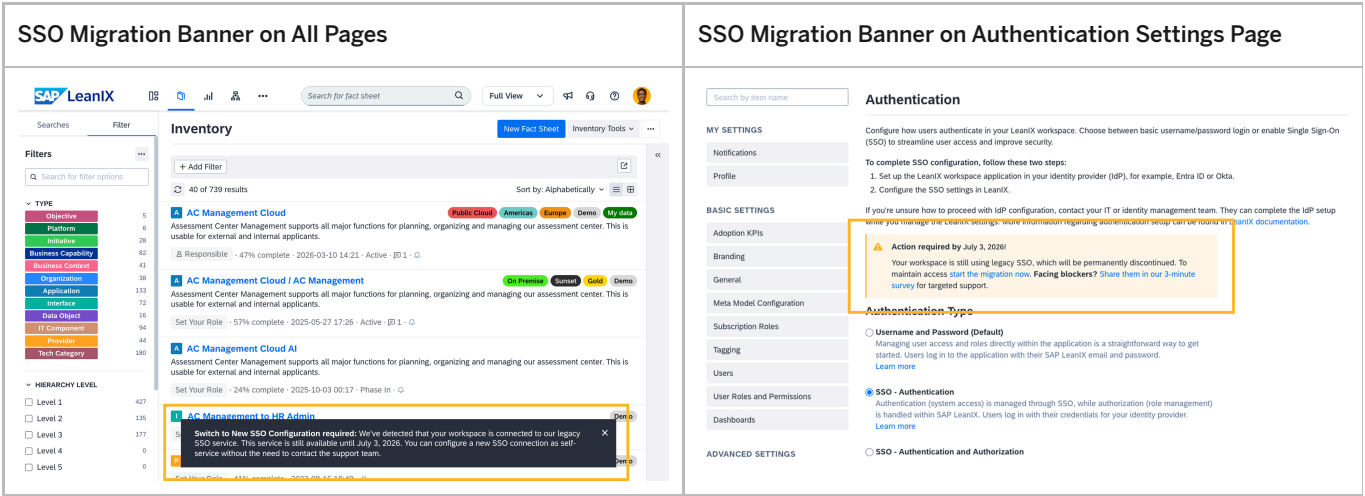


Who Needs to Migrate?

SAP LeanIX informs affected customers in different ways, for example, with emails and in-product banners. You can see if your workspace is affected directly in your workspace.

1. As admin, log in to your workspace.
2. Check for the informational SSO migration banner.

If you are an admin of a workspace on the legacy SSO configuration, you'll see an informational banner for the required SSO migration.



3. If you can see the banner, start the SSO migration.

If you miss the SSO migration deadline, you and your users may loose access to the workspace.

Important Changes

- **Each workspace now requires its own SSO configuration.** The legacy setup with one identity provider for all workspaces no longer works.

Each workspace must have its own SSO connection with a dedicated **Entity ID** and **Reply (ACS) URL** in SAP LeanIX. You do not necessarily need to set up a new SAML application in your identity provider for each workspace. It is possible to connect several workspace SSO connections to the same SAML application. This depends on what your identity provider supports and how your SSO setup is configured. For more details, see [Configuration Options](#).
- **Attribute names were updated.** Create a new SAML application in your identity provider and configure attributes from scratch. Make sure to check the new attributes.

Legacy Attribute Name	New Attribute Name
mail	email
customerRoles	customer_roles
entryACI	ace

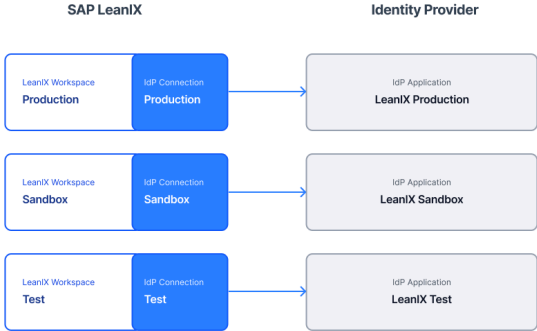
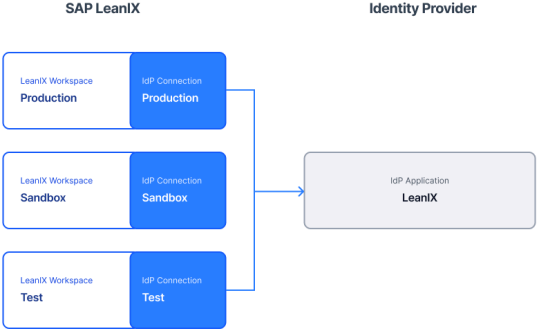
The configuration guides explain how to configure attributes for authentication and authorization for specific identity providers. For an overview of available attributes, see [SSO Attribute Overview](#).

Prerequisites

- You are starting the SSO migration before July 3, 2026.
- You have admin access to your identity provider. If you don't have access, coordinate with your IT or identity management team to support you during the migration.

- You have admin permissions in SAP LeanIX.
- Once you start the SSO configuration, keep your admin session active. Don't close the browser.

Configuration Options

(Recommended) Option 1: Dedicated SAML Applications for each Workspace	Option 2: Shared SAML Application Across Multiple Workspaces
 Each SAP LeanIX workspace connects to its own identity provider application.	 Multiple SAP LeanIX workspaces connect to the same identity provider application. Note that some identity providers may not support multiple entity IDs and reply (ACS) URLs.

Migration Process

The migration process consists of the same steps as starting a new SSO configuration and you repeat it for each workspace. Start the SSO migration before July 3, 2026.

1. Learn about the new SSO configuration options, see [Getting Started With SSO](#).
2. Configure a new SSO connection in SAP LeanIX and a new SAML application in your identity provider.

Create new SAML applications rather than copying existing ones. If you keep your existing application unchanged, users will keep access until you fully migrate to the new SSO setup. The new application already uses the correct attribute names and can be tested without interfering with existing SSO connections.

For detailed instructions, see the configuration guides for your identity provider:

- [Configuring SSO with Okta](#)
- [Configuring SSO with Microsoft Entra ID](#)
- [Configuring SSO with SAP Cloud Identity Services - Identity Authentication Service \(IAS\)](#)
- [Configuring SSO with PingOne \(PingIdentity\)](#)
- [Configuring SSO with OneLogin \(Onedidentity\)](#)
- [Configuring SSO with Active Directory Federation Services](#)

Each SSO connection that you configure for a workspace is added to the **Authentication and SSO** page in the administration. You can activate and deactivate the SSO connections separately.

3. Repeat step 2 for each workspace.

Troubleshooting

- Check attribute mappings using a SAML tracer.
- If the problems persist, submit a ticket to support.

Reverting the SSO Migration

If something goes wrong during the self-service SSO configuration migration, contact support to revert the changes.

1. Go to [SAP for Me](#).
2. Enter the ticket title: "Revert self-service SSO configuration migration for workspace [your workspace name]".
3. Provide your workspace details and reason for reverting:
 - Workspace name
 - Subdomain, for example, "customername.leanix.net" or "https://eu.leanix.net"
 - Migration completion date (if known)
 - Reason for reverting
4. Provide the migration steps already completed, for example:
 - Created an app in the identity provider
 - Created an identity provider in SAP LeanIX
 - Activated the identity provider in SAP LeanIX
5. Choose SAP LeanIX as the system, then populate the other required fields and submit the ticket.

The SAP LeanIX support team assists with rolling back the changes to your previous SSO configuration.

FAQs

Why Is the SSO migration necessary?

The SSO service that SAP LeanIX uses is expiring on July 3, 2026. As there is no extension possible, **SAP LeanIX** is offering a new SSO service. This switch to the new SSO service includes the modernized authentication process to make user login more convenient and support additional SSO configurations.

For the migration, do we need to select the authentication type **SSO - Authentication** even if we currently have **SSO - Authentication and Authorization**?

You can use your preferred authentication type as all provided SSO options work for the new SSO services and for migration.

SSO Authentication Option	Legacy SSO Service	New SSO Service	SSO Migration
SSO - Authentication	yes	yes	yes
SSO - Authentication and Authorization	yes	yes	yes

Do you support IAS (SAP Cloud Identity Services - Identity Authentication Service)?

Yes, IAS is supported. For more details, see [Configuring SSO with SAP Cloud Identity Services - Identity Authentication Service \(IAS\)](#).

We already have an existing SAML application. Can we clone it?

Most identity providers don't allow cloning SAML applications. Cloning or copy+pasting also imports the wrong attribute configuration. Start from scratch with a new SAML application in your identity provider.

Where do I find Entity ID and Reply URL?

1. In SAP LeanIX, go to **Administration** > **Authentication and SSO**.
2. When you start adding a new identity provider, the **Entity ID** and **Reply URL** are generated and displayed.

How can I test the SSO connection first on a sandbox instance?

It's not possible to test an SSO connection on a different instance. The identity provider and the workspace establish a trusted connection that only works for this specific workspace. The SSO connection for the sandbox won't work for the production workspace.

To test your SSO connection, the new SSO self-service process includes a test step. You can repeat the test with different setups and test users. So before activating the SSO connection, you can be sure that everything works as expected.

Related Resources

- [Getting Started With SSO](#)
- [Contact Support](#)

Configuring SSO with Microsoft Entra ID

Instructions for configuring single sign-on (SSO) with Microsoft Entra ID as an identity provider.

Prerequisites

- You have enabled SSO configuration in SAP LeanIX, see [Getting Started With SSO](#)
- You have admin access to Microsoft Entra ID with permissions to create enterprise applications and define attributes. If you don't have access, coordinate with your IT or identity management team to support you during the SSO configuration.

Add New Identity Provider in SAP LeanIX

You can create up to 5 connections to identity providers (IdP).

1. In SAP LeanIX, navigate to **Administration** > **Authentication and SSO** and go to the **Identity Providers** section.
2. To add a new identity provider, select the authentication type and click **Add Identity Provider**.
3. Provide the identity provider details.
4. Choose **Next**.

To proceed with the setup, you need to set up a SAML application in your identity provider.

→ Tip

To change the settings of an SSO connection in SAP LeanIX, add a new identity provider. Then delete the old configuration.

Create SAML Application in Microsoft Entra ID

1. In Microsoft Entra ID, create a new SAML application (non-gallery) with SAML as sign-in method.
2. In the new SAML application, navigate to the ► [Single sign-on](#) ► [SAML](#) ► [Basic SAML Configuration](#) ►.

Configure Metadata

Exchanging the metadata between SAP LeanIX as service provider and Microsoft Entra ID as identity provider establishes a trusted SSO connection.

1. In Microsoft Entra ID, navigate to ► [Applications](#) ► [{Your new SAML application}](#) ► [Single sign-on](#) ► [SAML](#) ► [Basic SAML Configuration](#) ►.

Edit the [Basic SAML Configuration](#). Copy the following IDs from the SAP LeanIX configuration:

- [Entity ID](#) as the [Identifier \(Entity ID\)](#)
- [Reply URL](#) as the [Reply URL \(Assertion Consumer Service URL\)](#)

2. Save and close the SAML configuration of the new SAML application.
3. In Microsoft Entra ID, navigate to ► [Single sign-on](#) ► [SAML](#) ► [SAML Certificates](#) ►.
4. Download the [Federation Metadata XML](#) file.
5. In SAP LeanIX, upload the metadata file.

Configure Service Provider-Initiated Authentication Flow

SAP LeanIX supports sp-initiated SSO only. IdP-initiated login is not supported. Users must login directly to the SAP LeanIX workspace to start the authentication flow. Provide your workspace URL during SSO configuration to ensure proper authentication flow between your identity provider and workspace.

1. In Microsoft Entra ID, navigate to ► [Applications](#) ► [{Your new SAML application}](#) ► [Single sign-on](#) ► [SAML](#) ► [Basic SAML Configuration](#) ►.
2. For [Sign on URL](#) add the workspace URL, for example, `https://{region}.leanix.net/{workspacename}/`.

If you want to setup your workspace with a subdomain, you request the subdomain after the SSO configuration and update the [Sign on URL](#) later.

Configure Authentication Attributes

Configure the required attributes to authenticate users through Microsoft Entra ID.

1. In Microsoft Entra ID, navigate to ► [Applications](#) ► [{Your new SAML application}](#) ► [Single sign-on](#) ► [Attributes & Claims](#) ►.
- Configure the required claims (attributes):

Claim Name	Type	Value
firstname	SAML	user.givenname
lastname	SAML	user.surname
email	SAML	user.mail
uid	SAML	Use a unique ID that is not the email address. uid must be in email format. Recommendation: Use user.userPrincipalName or user.objectID in email format. Do not use the user's actual email address, as email addresses can change.

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App registrations

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Attributes & Claims

+ Add new claim + Add a group claim Columns Got feedback?

Required claim

Claim name	Type	Value
Unique User Identifier (Name ID)	SAML	user.userprincipalname [...]

Additional claims

Claim name	Type	Value	
email	SAML	user.mail	...
firstname	SAML	user.givenname	...
lastname	SAML	user.surname	...
role	SAML	user.assignedroles	...
uid	SAML	user.userprincipalname	...

2. For each claim, delete the **Namespace** value in the configuration.

Manage claim

×

Save Discard changes Got feedback?

Name *

firstname ✓

Namespace

Enter a namespace URI ✓

Choose name format

Source *

Attribute

Transformation

Directory schema extension (Preview)

Source attribute *

user.givenname

Claim conditions

Advanced SAML claims options

Optional: Configure Authorization Attributes

If you want to manage user roles within Microsoft Entra ID, configure the corresponding roles.

Authorization Scope

The necessary attributes and their configuration depend on the authorization scope:

- [Configure Role Attributes in Microsoft Entra ID](#)

Role attributes are used for the default user roles: VIEWER, MEMBER, ADMIN

- [Configure Custom Role Attributes in Microsoft Entra ID](#)

Custom roles can be additional roles such as ARCHITECT or AUDITOR that you define for your organization. Custom roles require the default user role attributes to be configured first.

- [Configure Access Control Entity Attributes in Microsoft Entra ID](#)

Access control entities are used for virtual workspaces and require the default user role attributes to be configured first.

Assign Users to Your SAML Application

1. In Microsoft Entra ID, navigate to ► **Enterprise apps** ► **{Your new SAML application}** ► **Single Sign-on** ► **Manage** ► **Add Assignment** ► **Users and groups** ►.
2. Assign users or user groups to the application.

Test the SSO Connection

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. On the identity provider tile, select **Test Configuration**.



3. Follow the test instructions. Open the provided URL in a private/incognito browser window and log in with the credentials for your identity provider.

If you cannot use a private or incognito browser window, copy the provided URL, log out of SAP LeanIX, and then you use the provided URL in your regular browser.

If the test is successful, you are redirected to your identity provider for login. After logging in, you're in your workspace.

You can re-run the test and try different setups, for example, different role assignments for your test user.

If you encounter any issues during SSO configuration, see our troubleshooting guide at [Troubleshooting SSO](#).

Activate the SSO Connection

⚠ Caution

You and your users can lose all access to SAP LeanIX if the identity provider configuration is not working!

Only proceed if the configuration test was successful.

! Restriction

You cannot switch back to authentication with username and password. Contact [Support](#) if you have questions or need help.

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. On the identity provider tile, toggle to activate.

Confirm in the dialog that opens. Now the users managed in this identity provider can only log in to the associated workspace through their SSO account.

Optional: Request to Add Identity Provider Connection to a Custom Subdomain

With the new SSO activated the users already have access to the workspace with the direct workspace URL, but the custom subdomain won't work yet.

If your SAP LeanIX workspace uses a custom subdomain, the subdomain must be updated to include the new identity provider connection. SAP LeanIX manages subdomains, so this process is not available through self-service. After a successful migration, submit a ticket to support. Provide the direct URLs of the relevant workspaces.

Related Information

- [Microsoft Entra ID documentation](#) ➡
- [Single Sign-On \(SSO\)](#)
- [Getting Started With SSO](#)
- [SCIM Provisioning](#)

Configure Role Attributes in Microsoft Entra ID

Manage who can view, edit, or administer SAP LeanIX by configuring roles for user authorization in Microsoft Entra ID.

If you want to authorize users through SSO, you configure the role attributes. Role attribute configuration consists of the following settings:

- User group for each role
- Claim (attribute) definition
- Claim conditions
- App role

1. In Microsoft Entra ID, navigate to ► **Groups** ► **All groups** ►. Ensure you have the required user groups or otherwise add:
 - Group for the VIEWER role
 - Group for the MEMBER role
 - Group for the ADMIN role

2. Add the claim (attribute): Navigate to **{Your new SAML application} > Single Sign-on > Attributes and Claims > Add new claim**. Configure the (attribute) for user roles:

Name	Namespace	Source	Value
role	blank	Attribute	user.assignedroles

3. Add the claim conditions: In **Attributes and Claims** click the **role** claim. Open the dropdown for **Claim conditions**. Configure the claim conditions for user roles:

i Note

Make sure you add the claim conditions in exactly this order, as conditions are processed sequentially from the first entry to the last.

If a user belongs to both the **VIEWER** and **MEMBER** groups, they receive the **MEMBER** permission because the last matching condition takes precedence.

User type	Scoped Groups	Source	Value
Any	Group for VIEWER role	Attribute	VIEWER
Any	Group for MEMBER role	Attribute	MEMBER
Any	Group for ADMIN role	Attribute	ADMIN

4. Add the app roles: Navigate to **App registrations > App roles > Create app role**.

Display Name	Description	Value
VIEWER	LeanIX Viewer Role	VIEWER
MEMBER	LeanIX Member Role	MEMBER
ADMIN	LeanIX Admin Role	ADMIN

Configure Custom Role Attributes in Microsoft Entra ID

Configure custom roles like ARCHITECT or AUDITOR in Microsoft Entra ID to send tailored permissions to SAP LeanIX through SSO.

If you want to authorize custom roles for your workspace, you need to add an additional custom role attribute. Custom role attribute configuration consists of the following settings:

- User group for each role
- Claim (attribute) definition
- Claim conditions
- App role

i Note

Custom roles also require the default user role configuration, see [Configure Role Attributes in Microsoft Entra ID](#).

1. Add the group: In Microsoft Entra ID, navigate to **Groups > All groups**. Ensure you have a user groups or otherwise add:
 - o Group for the custom role
2. Add the claim (attribute): Navigate to **{Your new SAML application} > Single Sign-on > Attributes and Claims > Add new claim**. Configure the (attribute) for the custom roles:

Name	Namespace	Source	Value
customer_roles	blank	Attribute	user.assignedroles

3. Adjust the claim conditions: In **Attributes and Claims** click the **role** claim. Open the dropdown for **Claim conditions**. To the claim conditions you previously configured for the default user roles, add the custom roles:

User type	Scoped Groups	Source	Value
Any	- o Group for VIEWER role - o Group for custom role	Attribute	VIEWER

Custom roles only need to be added to the VIEWER claim condition. This ensures custom roles get access to SAP LeanIX.

4. Add the app role: Navigate to **App registrations > App roles > Create app role**.

i Note

Create the app role value using **uppercase** letters. This is equivalent to the role attribute in SAP LeanIX, for example, ARCHITECT or AUDITOR. Ensure the spelling is correct.

Display name	Description	Value
{Custom Role}	LeanIX Custom Role	{custom role name, for example, ARCHITECT}

Configure Access Control Entity Attributes in Microsoft Entra ID

Enable virtual workspace access by adding the ace attribute to your Entra ID SAML application.

To start using virtual workspace, you configure the attribute for access control entities.

i Note

Custom roles also require the default user role configuration, see [Configure Role Attributes in Microsoft Entra ID](#).

1. In Microsoft Entra ID, add the claim (attribute): Navigate to **{Your new SAML application} > Single Sign-on > Attributes and Claims > Add new claim**. Configure the (attribute) for access control entities:

Name	Namespace	Source	Value
ace	blank	Attribute	user.assignedroles

Related Information

- [Virtual Workspaces](#)

Configuring SSO with Okta

Instructions for configuring single sign-on (SSO) with Okta as an identity provider.

Prerequisites

- You have enabled SSO configuration in SAP LeanIX, see [Getting Started With SSO](#)
- You have admin access to Okta with permissions to create SAML applications and define profiles and attributes. If you don't have access, coordinate with your IT or identity management team to support you during the SSO configuration.

Add New Identity Provider in SAP LeanIX

You can create up to 5 connections to identity providers (IdP).

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ► and go to the **Identity Providers** section.
2. To add a new identity provider, select the authentication type and click **Add Identity Provider**.
3. Provide the identity provider details.
4. Choose **Next**.

To proceed with the setup, you need to set up a SAML application in your identity provider.

→ Tip

To change the settings of an SSO connection in SAP LeanIX, add a new identity provider. Then delete the old configuration.

Create SAML Application in Okta

1. In the Okta Admin Console, navigate to **Applications**. Click **Create App Integration**.
2. In the overlay that is displayed, select **SAML 2.0** as the sign-in method and click **Next**.

Create a new app integration

Sign-in method

[Learn More](#) 

- ☐ **OIDC - OpenID Connect**
Token-based OAuth 2.0 authentication for Single Sign-On (SSO) through API endpoints. Recommended if you intend to build a custom app integration with the Okta Sign-In Widget.
- ☒ **SAML 2.0**
XML-based open standard for SSO. Use if the Identity Provider for your application only supports SAML.
- ☐ **SWA - Secure Web Authentication**
Okta-specific SSO method. Use if your application doesn't support OIDC or SAML.
- ☐ **API Services**
Interact with Okta APIs using the scoped OAuth 2.0 access tokens for machine-to-machine authentication.

[Cancel](#)

[Next](#)

3. Enter a name for your application, then click [Next](#).

4. Activate the checkmark for [App Visibility](#).

Configure Metadata

Exchanging the metadata between SAP LeanIX as service provider and Okta as identity provider establishes a trusted SSO connection.

1. In the new SAML application, go to **Sign On** **SAML Settings** and click [Generate new certificate](#). Copy the following IDs from the SAP LeanIX configuration:

- [Entity ID](#) as the [Audience URI \(SP Entity ID\)](#)
- [Reply URL](#) as the [ACS URL \(Single Sign On URL\)](#)

2. Configure the application settings:

- [Name ID format](#): Select [EmailAddress](#).
- [Application username](#): This parameter depends on your Okta implementation. If the Okta username matches the SAP LeanIX email, select [Okta username](#).

3. In the [Feedback](#) section, specify that the app is internal.

3 Help Okta Support understand how you configured this application



The optional questions below assist Okta Support in understanding your app integration.

App type 



This is an internal app that we have created

[Previous](#)

[Finish](#)

4. Click **Finish** to save the IDs and settings.
5. In the SAML application, go to ► **Sign On** ► **SAML Signing Certificates** ►. Either copy the metadata URL or download the metadata XML file.
6. In SAP LeanIX, provide the metadata.

Configure Service Provider-Initiated Authentication Flow

SAP LeanIX supports sp-initiated SSO only. IdP-initiated login is not supported. Users must login directly to the SAP LeanIX workspace to start the authentication flow. Provide your workspace URL during SSO configuration to ensure proper authentication flow between your identity provider and workspace.

To configure the sp-initiated SSO for Okta, you configure a bookmark application. This workaround is also documented in the Okta documentation. To view it, copy this URL to your browser `https://help.okta.com/en-us/content/topics/apps/apps_bookmark_app.htm`.

1. In the Okta Admin Console, navigate to ► **Applications** ► **Applications** ► **Browse App Catalog** ►.
2. Search **Bookmark App**, then choose **Bookmark App**.
3. Choose **Add Integration**.
4. Enter **Application label**.
5. For URL, add the workspace URL, for example, `https://{region}.leanix.net/{workspacename}/`.

If you want to setup your workspace with a subdomain, you request the subdomain after the SSO configuration and update the **URL** later.

6. Ensure both checkboxes are clear, **Request Integration** and **Application Visibility**.
7. Choose **Done**. The bookmark application is created.
8. Go to the **Assignments** tab. Assign all relevant user groups to this bookmark application.

Configure Authentication Attributes

Configure the required attributes to authenticate users through Okta.

1. In Okta, navigate to ► **Applications** ► **{Your new SAML application}** ► **Sign On** ►.
2. Click **Show Legacy configuration** and go to ► **SAML Settings** ► **Attributes** ►.
3. Configure the required attributes:

Name	Name Format	Value	Description
firstname	Basic	user.firstName	The first name of the user.
lastname	Basic	user.lastName	The last name of the user.
uid	Basic	[unique ID]	Use a unique ID that is not the email address. uid must be in email format. Recommendation: Use <code>user.id</code> . Do not use the user's actual email address,

Name	Name Format	Value	Description
			as email addresses can change.
email	Basic	user.email	The email address of the user.

Optional: Configure Authorization Attributes

If you want to manage user roles within Okta, configure the corresponding roles.

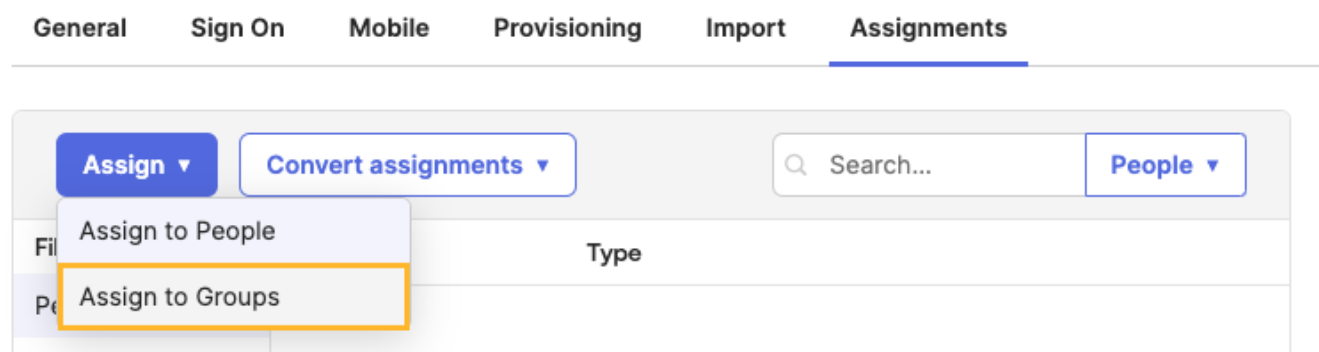
Authorization Scope

The necessary attributes and their configuration depend on the authorization scope:

- [Configure Role Attributes in Okta](#)
- [Configure Custom Role Attributes in Okta](#)
- [Configure Access Control Entity Attributes in Okta](#)

Assign Users to the SAML Application

1. In Okta, navigate to **Applications** > **{Your new SAML application}**. On the **Assignments** tab of the application, click **Assign** > **Assign to Groups**.



2. In the overlay that is displayed, select a user group.
3. If you have authorization configured: For this group, select the according authorization attributes.
 - For the **Role** attribute, select the authorization level that this group has. From the list select VIEWER, MEMBER, or ADMIN.
 - For the **Custom Role** attribute, select from the list of your defined custom roles.
 - For the **ACE** attribute, select from the list of your defined access control entities.
4. Save the configuration.
5. Repeat the group selection and attribute assignment for all relevant groups.
6. Adjust the **priority** of the group assignments with drag-and-drop.

Group priority is important when a user is a member of more than one group. The highest group priority determines which **Role** the user gets assigned.

Test the SSO Connection

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. On the identity provider tile, select **Test Configuration**.



3. Follow the test instructions. Open the provided URL in a private/incognito browser window and log in with the credentials for your identity provider.

If you cannot use a private or incognito browser window, copy the provided URL, log out of SAP LeanIX, and then you use the provided URL in your regular browser.

If the test is successful, you are redirected to your identity provider for login. After logging in, you're in your workspace.

You can re-run the test and try different setups, for example, different role assignments for your test user.

If you encounter any issues during SSO configuration, see our troubleshooting guide at [Troubleshooting SSO](#).

Activate the SSO Connection

⚠ Caution

You and your users can lose all access to SAP LeanIX if the identity provider configuration is not working!

Only proceed if the configuration test was successful.

! Restriction

You cannot switch back to authentication with username and password. Contact [Support](#) if you have questions or need help.

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. On the identity provider tile, toggle to activate.

Confirm in the dialog that opens. Now the users managed in this identity provider can only log in to the associated workspace through their SSO account.

Optional: Request to Add Identity Provider Connection to a Custom Subdomain

With the new SSO activated the users already have access to the workspace with the direct workspace URL, but the custom subdomain won't work yet.

If your SAP LeanIX workspace uses a custom subdomain, the subdomain must be updated to include the new identity provider connection. SAP LeanIX manages subdomains, so this process is not available through self-service. After a successful migration, submit a ticket to support. Provide the direct URLs of the relevant workspaces.

Related Information

- [Okta website](#) ➦ .
- [Single Sign-On \(SSO\)](#).
- [Getting Started With SSO](#)
- [SCIM Provisioning](#)

Configure Role Attributes in Okta

Manage who can view, edit, or administer SAP LeanIX by configuring roles for user authorization in Okta.

If you want to authorize users through SSO, you configure the role attributes. Role attribute configuration in Okta consists of the following settings:

- Set up attribute in the SAML application
- Profile mapping to define attribute details

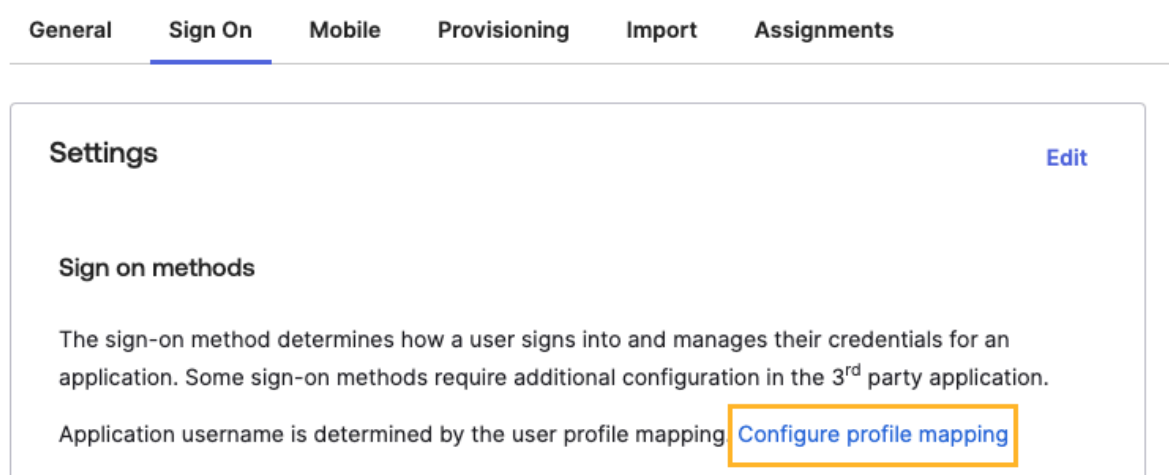
Set Up Attribute in the SAML Application

1. In Okta, navigate to ► **Applications** ► {Your new SAML application} ► **Sign On** ►.
2. Click **Show Legacy configuration** and go to ► **SAML Settings** ► **Attributes** ►.
3. Configure the role attribute:

Name	Name Format	Value
role	Basic	appuser.role

Profile Mapping to Define Attribute Details

1. In Okta, navigate to ► **Applications** ► {Your new SAML application} ► **Sign On** ►.
2. Select **Configure profile mapping**.



3. Close the overlay with user profile mappings that is displayed.

You are now in the **Profile Editor** for {Your new SAML application}.

4. In the **Attributes** section, click **Add Attribute**.
5. Specify the attribute details:
 - **Data type**: Select **string**.
 - **Display name**: Role
 - **Variable name** and **External name**: role, matches the attribute that you added to the SAML application
 - **Enum**: Select **Define enumerated list of values**, this adds the attribute member configuration
 - **Attribute type**: Select **Group**.
6. Configure the **Attribute Members** of the **Enum** with the role values:

Display Name	Value
ADMIN	ADMIN
MEMBER	MEMBER
VIEWER	VIEWER

7. Save the changes.

Configure Custom Role Attributes in Okta

Configure custom roles like ARCHITECT or AUDITOR in Microsoft Entra ID to send tailored permissions to SAP LeanIX through SSO.

If you want to authorize custom roles for your workspace, you need to add an additional custom role attribute. Custom role attribute configuration consists of the following settings:

- Setup attribute in the SAML application
- Profile mapping to define attribute details

i Note

Custom roles also require the default user role configuration, see [Configure Role Attributes in Okta](#).

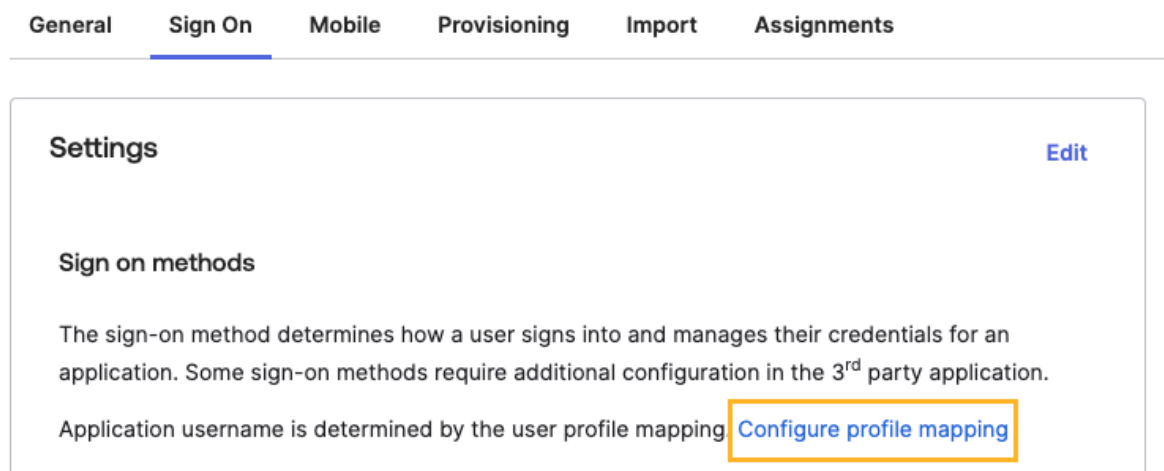
Setup Attribute in SAML Application

1. In Okta, navigate to **Applications** > **{Your new SAML application}** > **Sign On**.
2. Click **Show Legacy configuration** and go to **SAML Settings** > **Attributes**.
3. Configure the custom role attribute:

Name	Name Format	Value
customer_roles	Basic	appuser.customRole

Profile Mapping to Define Attribute Details

1. In Okta, navigate to **Applications** > **{Your new SAML application}** > **Sign On**.
2. Select **Configure profile mapping**.



3. Close the overlay with user profile mappings that is displayed.

You are now in the **Profile Editor** for **{Your new SAML application}**.

4. In the **Attributes** section, click **Add Attribute**.
5. Specify the attribute details:
 - **Data type**: Select **string**.
 - **Display name**: is Custom Role
 - **Variable name** and **External name**: is customRole, matches the attribute that you added to the SAML application
 - **Enum**: Select **Define enumerated list of values**, this adds the attribute member configuration
 - **Attribute type**: Select **Group**.
6. Configure the **Attribute Members** of the **Enum** with the role values, adding a new row for each custom role:

i Note

Create the role value using **uppercase** letters. This is equivalent to the role attribute in SAP LeanIX, for example, ARCHITECT or AUDITOR. Ensure the spelling is correct.

Display Name	Value
{CUSTOM ROLE, for example, ARCHITECT}	{CUSTOM ROLE, for example, ARCHITECT}

7. Save the changes.

Configure Access Control Entity Attributes in Okta

Enable virtual workspace access by adding the ace attribute to your Entra ID SAML application.

To start using virtual workspace, you configure the attribute for access control entities.

i Note

Access control entities also require the default user role configuration, see [Configure Role Attributes in Okta](#).

1. In Okta, navigate to ► **Applications** ► **{Your new SAML application}** ► **Sign On** ►.
2. Click **Show Legacy configuration** and go to ► **SAML Settings** ► **Attributes** ►.
3. Configure the access control entity attribute:

Name	Name Format	Value
ace	Basic	appuser.ace

Configuring SSO with SAP Cloud Identity Services - Identity Authentication Service (IAS)

Configure single sign-on (SSO) with SAP Cloud Identity Services - Identity Authentication service (IAS) as identity provider (IdP) to enhance security and simplify user authentication.

Prerequisites

- You have enabled SSO configuration in SAP LeanIX, see [Getting Started With SSO](#)
- You have admin access to SAP Cloud Identity Services with permissions to create SAML applications and define attributes. If you don't have access, coordinate with your IT or identity management team to support you during the SSO configuration.

Add New Identity Provider in SAP LeanIX

You can create up to 5 connections to identity providers (IdP).

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ► and go to the **Identity Providers** section.
2. To add a new identity provider, select the authentication type and click **Add Identity Provider**.
3. Provide the identity provider details.
4. Choose **Next**.

To proceed with the setup, you need to set up a SAML application in your identity provider.

→ Tip

To change the settings of an SSO connection in SAP LeanIX, add a new identity provider. Then delete the old configuration.

Create a SAML Application in IAS

1. In SAP Cloud Identity Services, navigate to ► **Applications & Resources** ► **Applications** ►.
2. Choose **Create** to add a new application.

SAP Cloud Identity Services Europe (Frankfurt / Amsterdam) a4qbvq1ui

Home Users & Authorizations Identity Provisioning Applications & Resources Identity Providers Monitoring & Reporting

Applications

Applications (9) Search

Charged Applications

- Demo DE Signin OIDC
- Demo DE Signin SAML
- DE Debugging invalid certificate

Demo DE Signin OIDC

Application type: Charged Application ID:
 Name Type: Unknown URL: Home URL not configured
 Organization ID: global Protocol Type: OpenID Connect

Trust Authentication and Access Provisioning Branding and Layout

Single Sign-On

OpenID Connect Configuration

Configure trust with a service provider for web-based authentication.

Subject Name Identifier

Configure the attribute which the application uses to identify the users. Identity Authentication sends the attribute to the application as subject in OpenID Connect tokens.

Apply Function to Subject Name Identifier

Convert the subject name identifier to uppercase or lowercase.

3. Complete the following fields:

- Type: SAP LeanIX
- Protocol Type: SAML 2.0

Create Application

Display Name: *

Home URL:

Type:

Parent application:

Organization ID: *

Protocol Type:

☒ SAML 2.0

☐ OpenID Connect

Create Application

Display Name: *

Home URL:

Type: **SAP LeanIX**

Parent application: **None**

Organization ID: * **global**

Protocol Type:

☒ SAML 2.0

☐ OpenID Connect

+ Create **Cancel**

4. Choose **Create**.

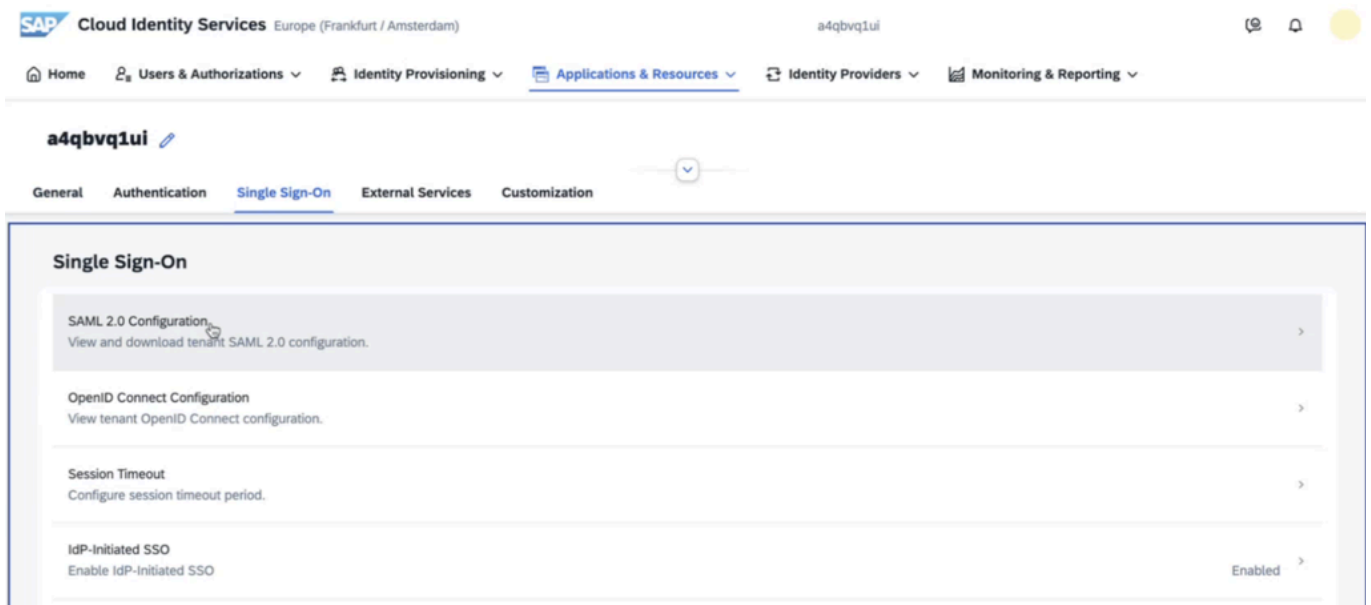
The application is added to the list of applications.

The screenshot shows the SAP Cloud Identity Services console for the 'Europe (Frankfurt / Amsterdam)' region. The 'Applications & Resources' tab is selected, displaying a list of applications on the left and the configuration for 'Docudemo' on the right. The 'Docudemo' application is a 'Charged' application with 'SAML 2.0' protocol type and 'global' organization ID. The 'Single Sign-On' configuration tab is active, showing the 'SAML 2.0 Configuration' section. A notification bubble states 'Application "Docudemo" created'. The 'URL' field is marked as 'Home URL not configured'.

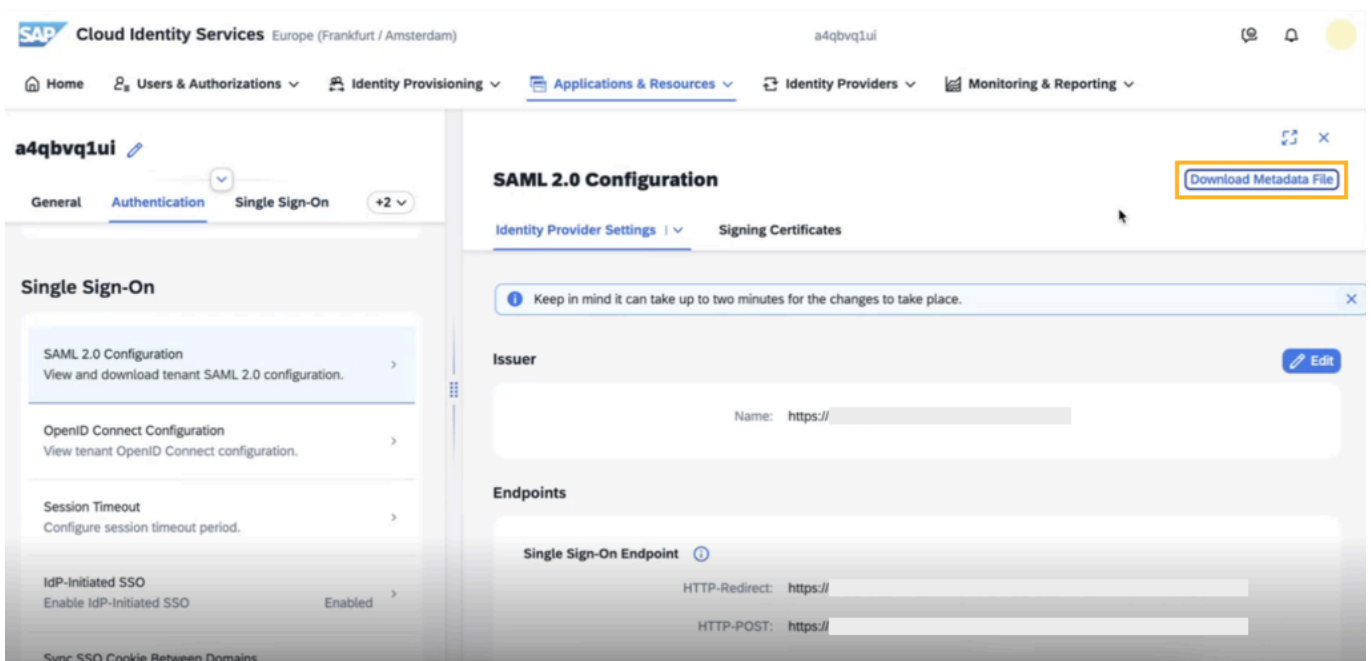
Configure Metadata

Exchanging the metadata between SAP LeanIX as service provider and IAS as identity provider establishes a trusted SSO connection.

1. In SAP Cloud Identity Services, navigate to **Applications & Resources** > **Tenant Settings**.
2. On the **Single Sign-On** tab, select **SAML 2.0 Configuration**.



3. Choose [Download Metadata File](#).



4. In SAP LeanIX, upload the metadata file in the identity provider configuration.

After uploading the IAS metadata, the SAP LeanIX metadata URL is generated.

Configuration

SAP LeanIX Metadata

Metadata

https://eu-signin.leanix.net/realms/service-provider/broker



Identity Provider reply URL

https://eu-signin.leanix.net/realms/service-provider/broker

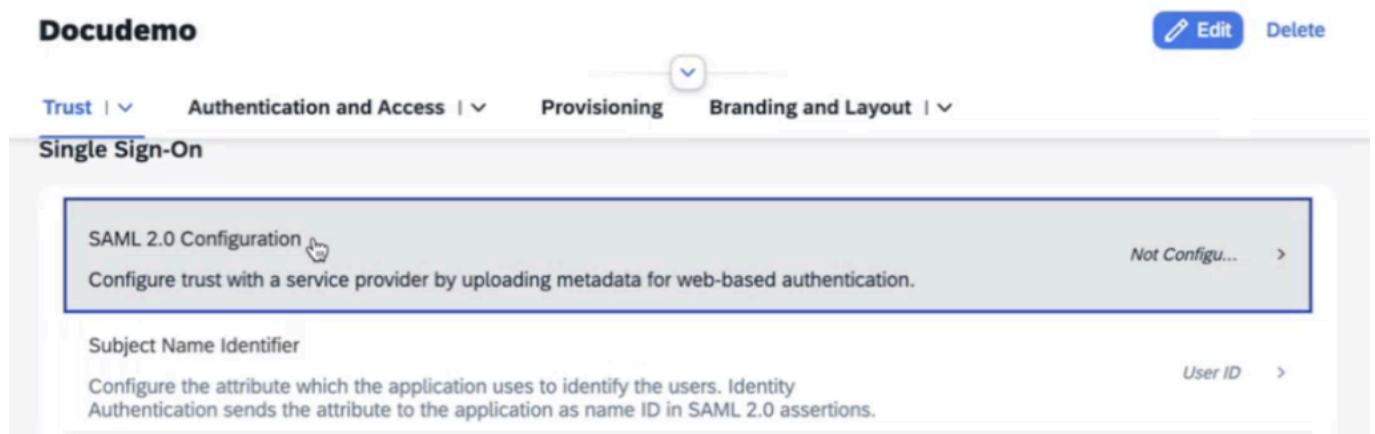


Entity ID

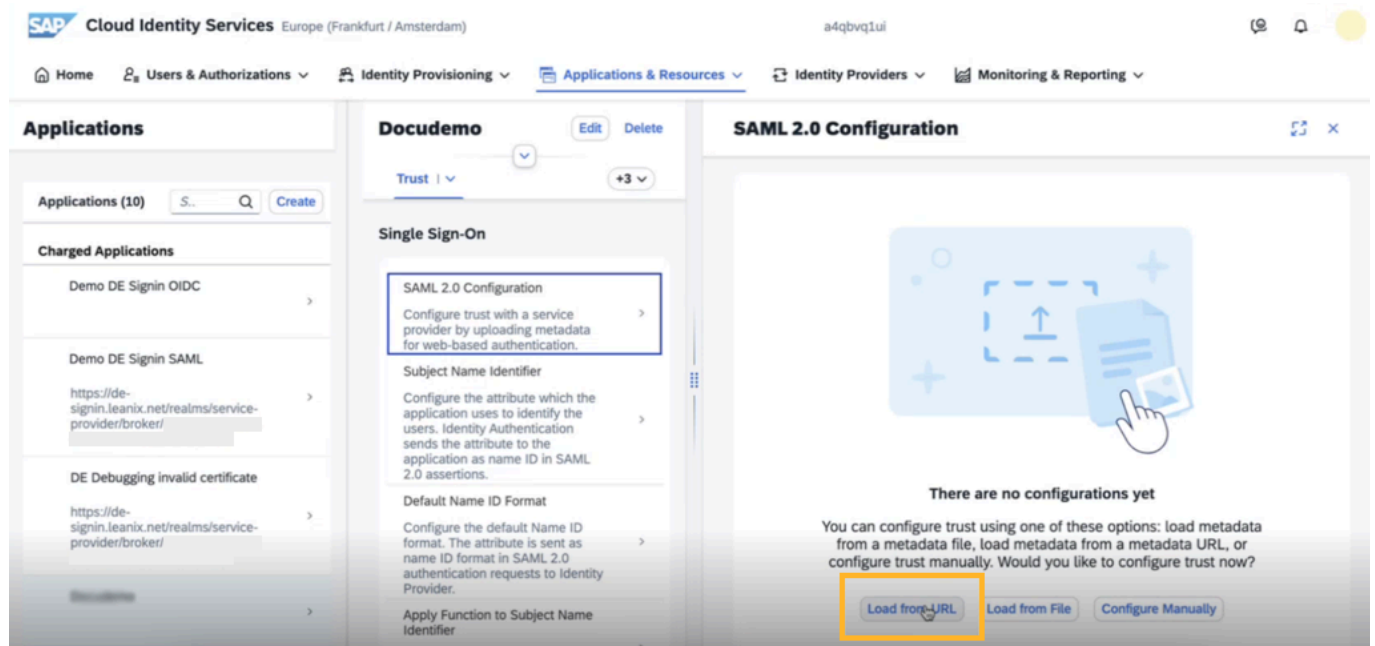
https://eu-signin.leanix.net/realms/service-provider/broker



5. In IAS, navigate to **Applications & Resources > Applications > {Your Application} > SAML 2.0 Configuration**.

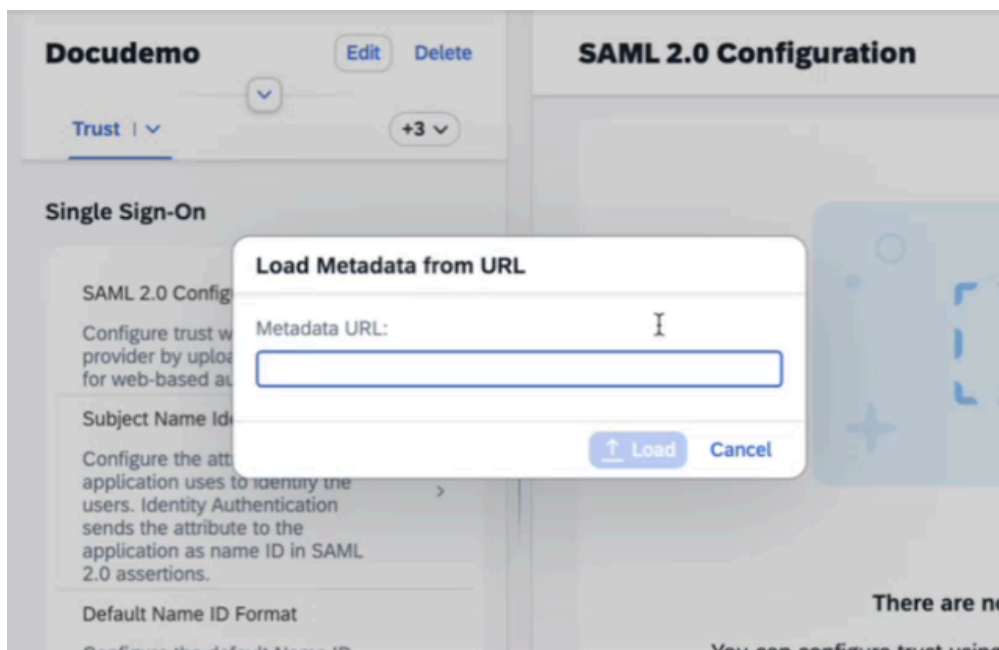


6. Choose **Load from URL**.



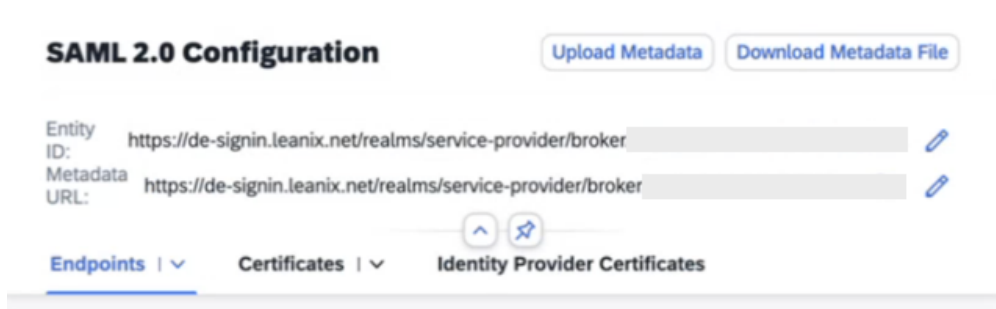
7. In SAP LeanIX, copy the previously generated **Metadata URL**.

8. Go back to Cloud Identity Services and paste the LeanIX metadata URL.



9. Choose **Load**.

Using the metadata URL automatically populates all relevant fields in the SAML application configuration.



Configure Service Provider-Initiated Authentication Flow

SAP LeanIX supports sp-initiated SSO only. IdP-initiated login is not supported. Users must login directly to the SAP LeanIX workspace to start the authentication flow. Provide your workspace URL during SSO configuration to ensure proper authentication flow between your identity provider and workspace.

1. In IAS, navigate to ► **Applications & Resources** ► **Applications** ► **[Your Application]** ►.
2. In the application overview, choose **Edit**.

Home Users & Authorizations Identity Provisioning Applications & Resources

Applications

Applications (20)

LeanIX Prod

LeanIX Sandbox

Docudemo

Docudemo

Edit **Delete**

Application type: Charged
 Name Type: Unknown
 Organization ID: global
 Application ID: [REDACTED]
 URL: Home URL not configured
 Protocol Type: SAML 2.0

Trust | v +3 v

Single Sign-On

SAML 2.0 Configuration
 Configure trust with a service provider by uploading metadata for web-based authentication.

Subject Name Identifier
 Configure the attribute which the application uses to identify the users. Identity

3. For **Home URL** add the workspace URL, for example, `https://{region}.leanix.net/{workspaceName}/`.

If you want to setup your workspace with a subdomain, you request the subdomain after the SSO configuration and update the **Home URL** later.

Configure Authentication Attributes

Configure the required attributes to authenticate users through IAS.

1. In IAS, navigate to **Applications & Resources > Applications > [Your Application] > Attributes**.

SAP Cloud Identity Services Europe (Frankfurt / Amsterdam) a4qbv1ui

Home Users & Authorizations Identity Provisioning Applications & Resources Identity Providers Monitoring & Reporting

Applications

Applications (10) S... Q Create

Charged Applications

Demo DE Signin OIDC

Demo DE Signin SAML
<https://de-signin.leanix.net/realms/service-provider/broker/>

DE Debugging invalid certificate
<https://de-signin.leanix.net/realms/service-provider/broker/>

Docudemo

Docudemo

Edit **Delete**

Trust | v +3 v

Configure the default Name ID format. The attribute is sent as name ID format in SAML 2.0 authentication requests to Identity Provider.

Apply Function to Subject Name Identifier
 Convert the subject name identifier to uppercase or lowercase.

Attributes
 Configure user attributes sent to the application.

Application APIs

Client Authentication
 Configure how clients authenticate with the application.

Attributes

Self-defined Attributes Collapse All Expand All Add

Name	Source	Value	
> first_name			+
> last_name			+
> locale			+
> mail			+
> user_uid			+

For more information, please see [Identity Authentication Documentation](#).

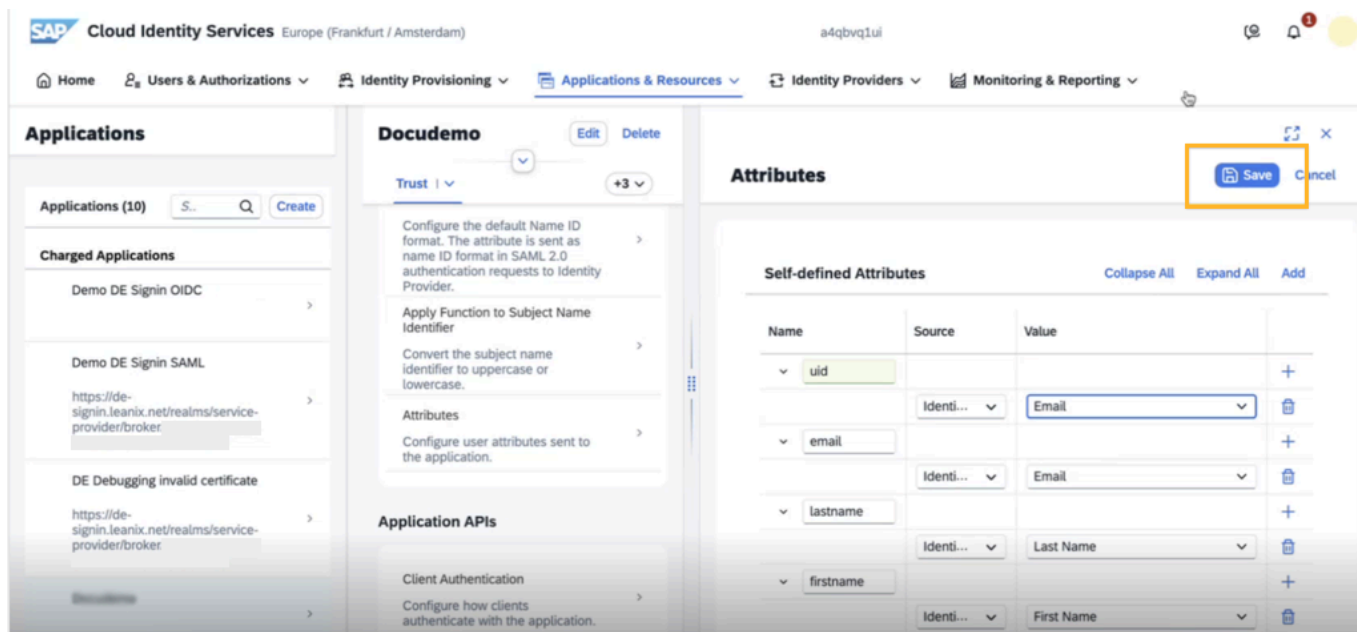
2. Remove the provided default attributes: Choose **Expand All** and then delete all entries. The attributes are now empty.

3. Configure the required attributes. All fields are case-sensitive.

Required Attributes

Name	Source	Value
firstname	Identity Directory	First Name
lastname	Identity Directory	Last Name
email	Identity Directory	Email
uid	Identity Directory	Email

4. Choose **Save**.



Optional: Configure Authorization Attributes

If you want to manage user roles within IAS, configure the corresponding roles.

Authorization Scope

The necessary attributes and their configuration depend on the authorization scope:

- [Configure Role Attributes in IAS](#)

Role attributes are used for the default user roles: VIEWER, MEMBER, ADMIN

- [Configure Custom Role Attributes in IAS](#)

Custom roles can be additional roles like ARCHITECT or AUDITOR that you define for your use cases. Custom roles require the default user role attributes to be configured first.

- [Configure Access Control Entity Attributes in IAS](#)

Access control entities are used for virtual workspaces and require the default user role attributes to be configured first.

Assign Users to the SAML Application

Unlike other identity providers, IAS does not require you to explicitly assign users to the SAML application. All users managed in IAS User Management are automatically permitted to sign in to SAP LeanIX once the SSO connection is active.

IAS also offers options for advanced access management. For more details, refer to the [IAS documentation](#).

Test the SSO Connection

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. On the identity provider tile, select **Test Configuration**.



3. Follow the test instructions. Open the provided URL in a private/incognito browser window and log in with the credentials for your identity provider.

If you cannot use a private or incognito browser window, copy the provided URL, log out of SAP LeanIX, and then you use the provided URL in your regular browser.

If the test is successful, you are redirected to your identity provider for login. After logging in, you're in your workspace.

You can re-run the test and try different setups, for example, different role assignments for your test user.

If you encounter any issues during SSO configuration, see our troubleshooting guide at [Troubleshooting SSO](#).

Activate the SSO Connection

⚠ Caution

You and your users can lose all access to SAP LeanIX if the identity provider configuration is not working!

Only proceed if the configuration test was successful.

! Restriction

You cannot switch back to authentication with username and password. Contact [Support](#) if you have questions or need help.

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. On the identity provider tile, toggle to activate.

Confirm in the dialog that opens. Now the users managed in this identity provider can only log in to the associated workspace through their SSO account.

Related Information

- [IAS documentation](#)
- [Single Sign-On \(SSO\)](#)
- [Getting Started With SSO](#)
- [SCIM Provisioning](#)

Configure Role Attributes in IAS

Manage who can view, edit, or administer SAP LeanIX by configuring roles for user authorization in IAS.

If you want to authorize users through SSO, you configure the role attributes. Role attribute configuration in IAS consists of the following settings:

- User group for each role
- Add users to user groups
- Map role attribute to the application

Create Groups and Link them to the SAML Application

The default role attribute can have different values and you create exactly one group for each value:

- VIEWER
- MEMBER
- ADMIN

i Note

You cannot change group settings in IAS. To edit a group, delete it and create a new one.

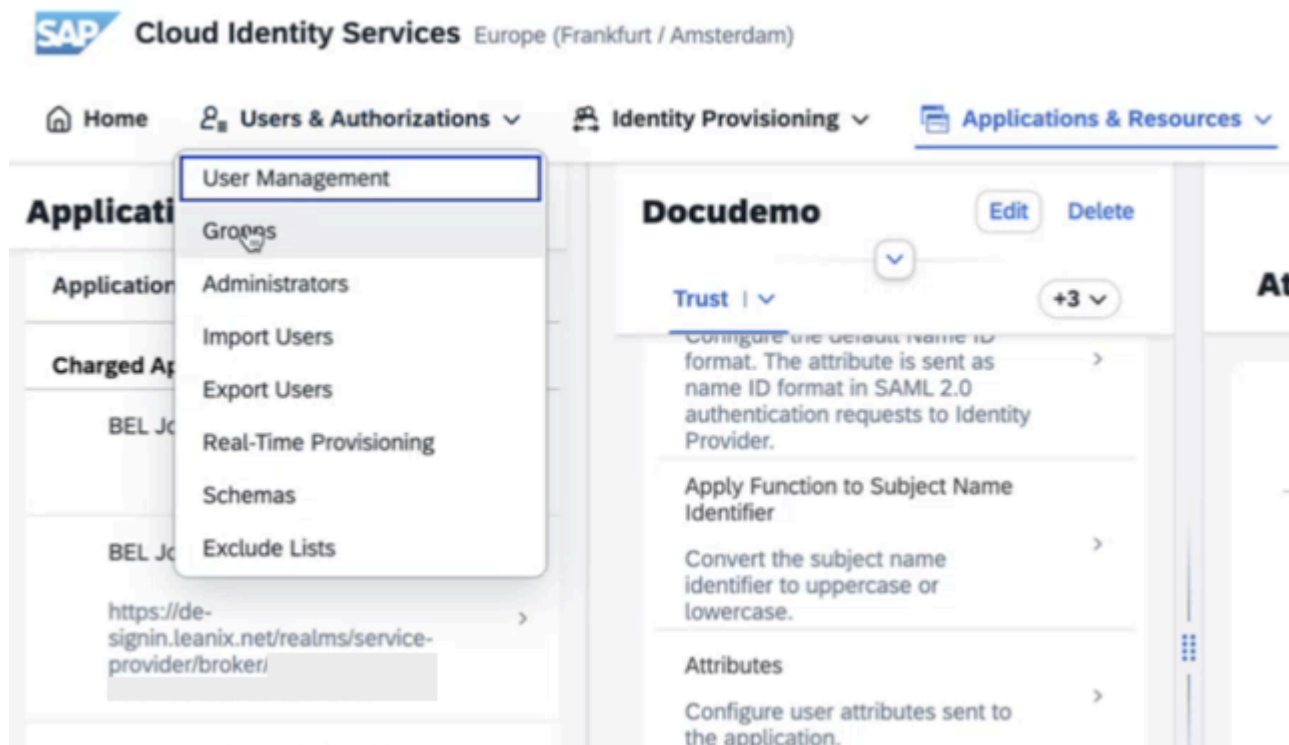
! Restriction

You can only create one group for each SAP LeanIX role attribute due to limitations in IAS.

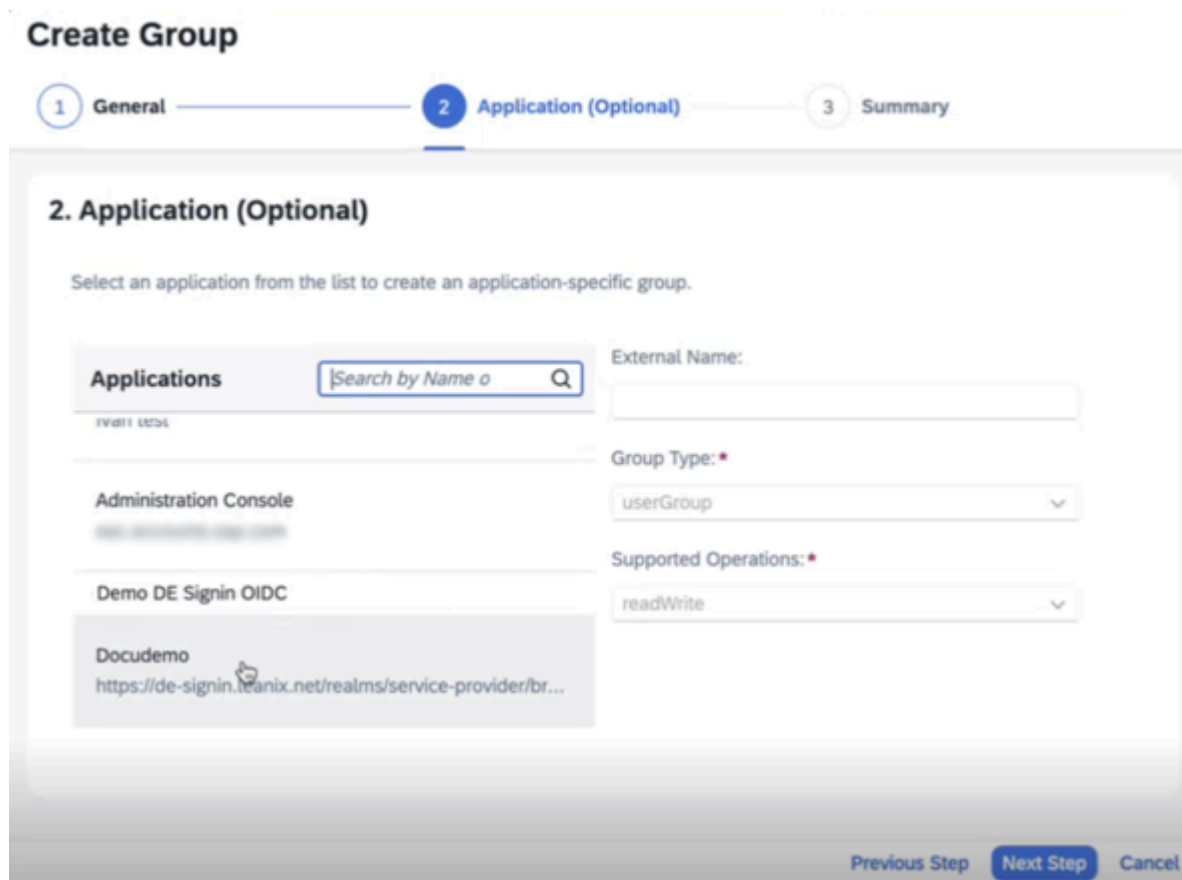
After you've created, for example, an ADMIN group, this is the only group that works for giving the default admin permissions in SAP LeanIX. All users that require the default admin permissions must be assigned to this user group.

Start with creating the first group for the VIEWER role.

1. In IAS, navigate to [Users & Authorizations](#) > [Groups](#).



2. Choose **Create**.
3. Enter general information for the application, then choose **Next Step**.
4. Select the application you created.



5. Provide the **External Name VIEWER**.

⚠ Caution

This is equivalent to the role attribute in SAP LeanIX, for example, ADMIN or VIEWER. Ensure the spelling and capitalization are correct.

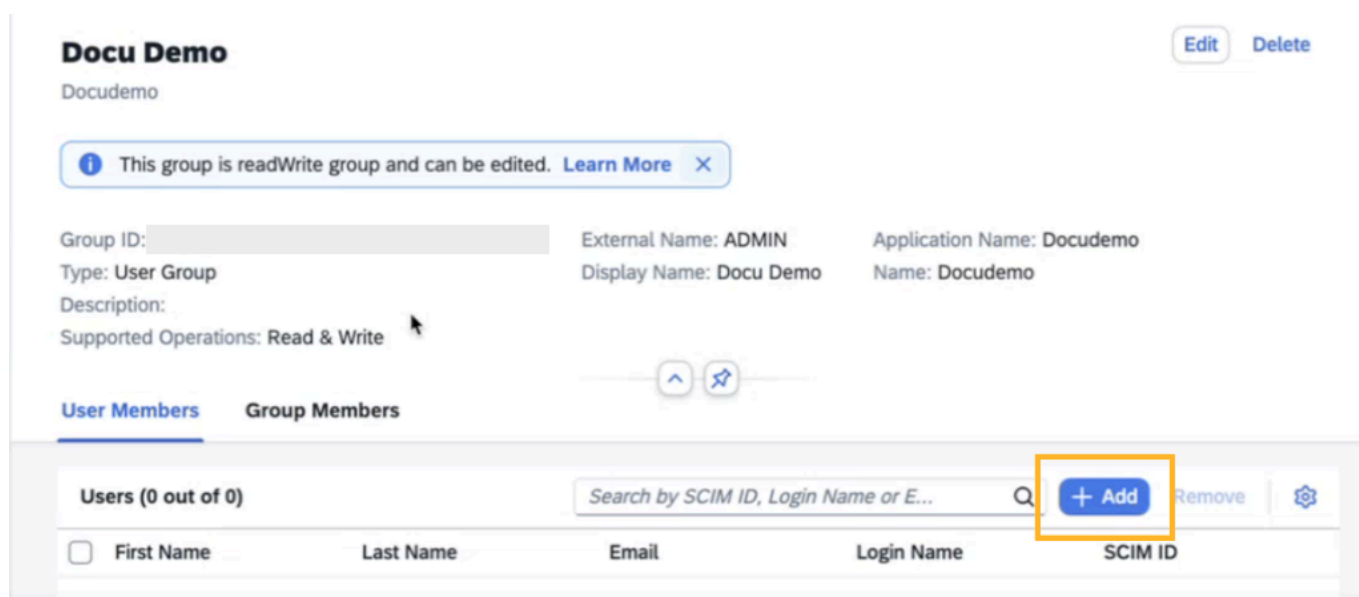
6. For **Group Type**, select **userGroup**.
7. Choose **Next Step**.
8. Check the configuration summary, then choose **Finish**.
9. Repeat the steps to create each a group for the MEMBER and ADMIN role.

Overview of Groups for Roles

External Name	Group Type
VIEWER	userGroup
MEMBER	userGroup
ADMIN	userGroup

Assign Users to Attribute-Related User Groups

1. In IAS, navigate to **Users & Authorization > User Management > Groups**.
2. Select the group you created.
3. Choose **Add** to add users to the group.



4. Select user groups.
5. Choose **Add** to save.

Assign Attribute to the SAML Application

1. In IAS, navigate to **Applications & Resources > Applications > [Your Application] > Attributes**.
2. Choose **Add**.
3. Configure the attribute

Name	Source	Value
role	Identity Directory	Application Groups - External name

4. Choose [Save](#).

Configure Custom Role Attributes in IAS

Configure custom roles like ARCHITECT or AUDITOR in IAS to send tailored permissions to SAP LeanIX through SSO.

If you want to authorize custom roles for your workspace, you need to add an additional custom role attribute. Custom role attribute configuration consists of the following settings:

- User group for each role
- Add users to user groups
- Map role attribute to the application

i Note

Custom roles also require the default user role configuration, see [Configure Role Attributes in IAS](#).

Create Groups and Link them to the SAML Application

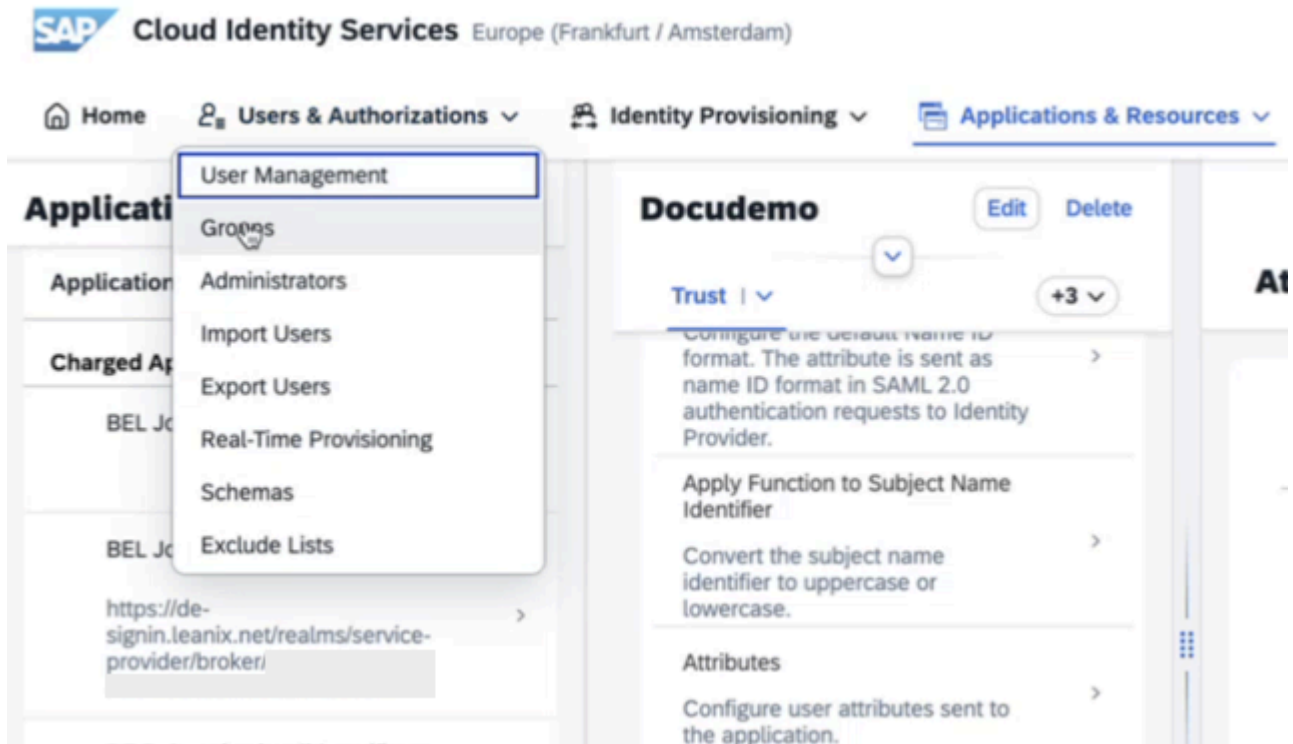
i Note

You cannot change group settings in IAS. To edit a group, delete it and create a new one.

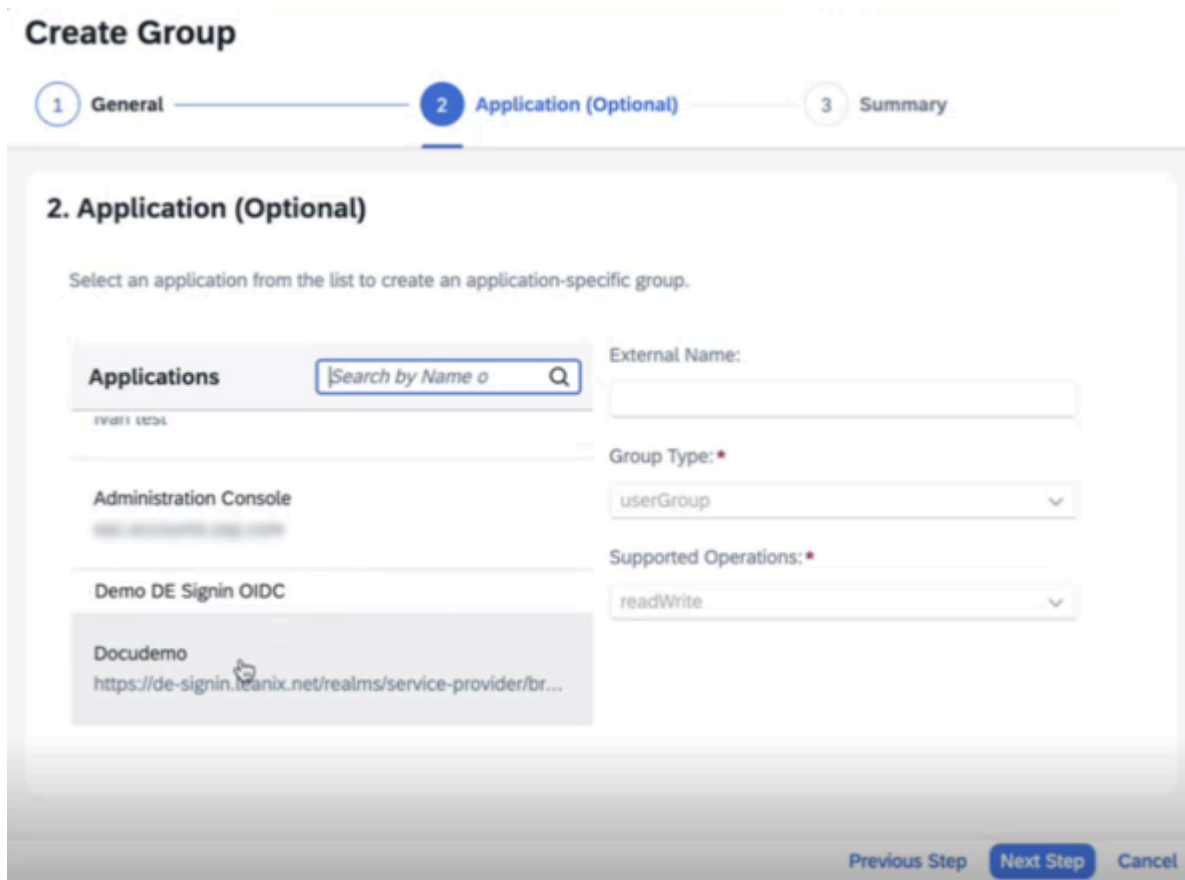
The custom role attribute can have different values and you create a group for each value, for example:

- ARCHITECT
- AUDITOR
- {CUSTOM ROLE}

1. In IAS, navigate to [Users & Authorizations](#) > [Groups](#).



2. Choose **Create**.
3. Enter general information for the application, then choose **Next Step**.
4. Select the application you created.



5. Provide the **External Name** {CUSTOM_ROLE}.

i Note

Create the external name using **uppercase** letters. This is equivalent to the role attribute in SAP LeanIX, for example, ARCHITECT or AUDITOR. Ensure the spelling is correct.

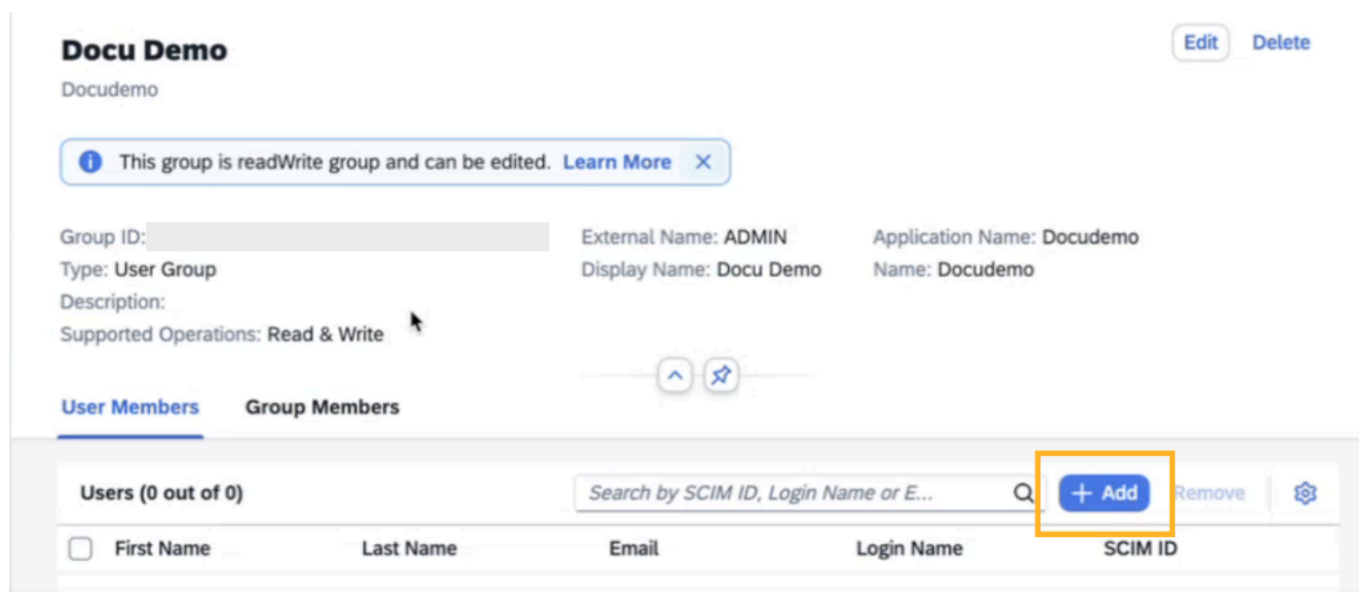
6. For **Group Type**, select **userGroup**.
7. Choose **Next Step**.
8. Check the configuration summary, then choose **Finish**.
9. Repeat the steps to create a group for each of your custom roles.

Overview of Groups for Custom Roles

External Name	Group Type
{CUSTOM ROLE A}	userGroup
{CUSTOM ROLE B}	userGroup
{CUSTOM ROLE ...}	userGroup

Assign Users to Attribute-Related User Groups

1. In IAS, navigate to **Users & Authorization > User Management > Groups**.
2. Select the group you created.
3. Choose **Add** to add users to the group.



4. Select user groups.
5. Choose **Add** to save.

Assign Attribute to the SAML Application

1. In IAS, navigate to **Applications & Resources > Applications > [Your Application] > Attributes**.
2. Choose **Add**.
3. Configure the attribute

Name	Source	Value
customer_roles	Identity Directory	Application Groups - External name

4. Choose [Save](#).

Configure Access Control Entity Attributes in IAS

Enable virtual workspace access by adding the ace attribute to your IAS SAML application.

To start using virtual workspace, you configure the attribute for access control entities with the following settings:

- User group for each role
- Add user to user groups
- Map role attribute to the application

i Note

Access control entities also require the default user role configuration, see [Configure Role Attributes in IAS](#).

Create Groups and Link them to the SAML Application

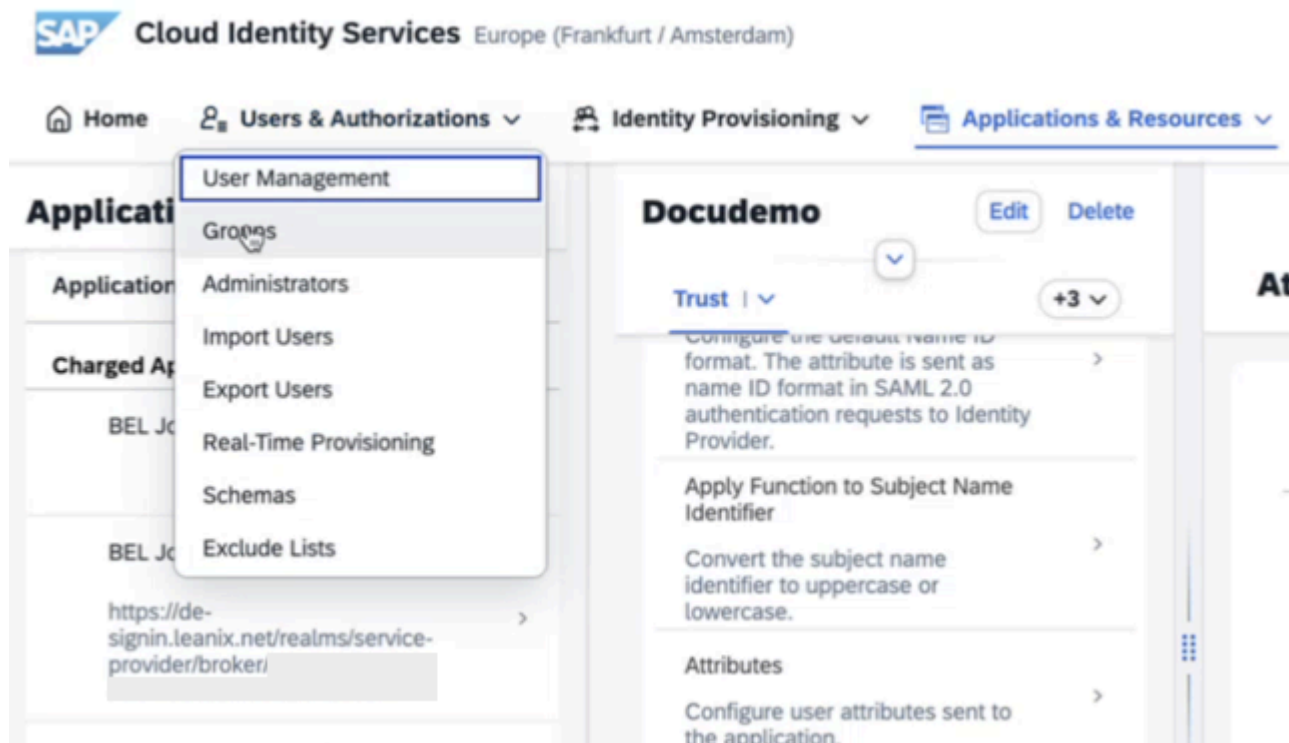
i Note

You cannot change group settings in IAS. To edit a group, delete it and create a new one.

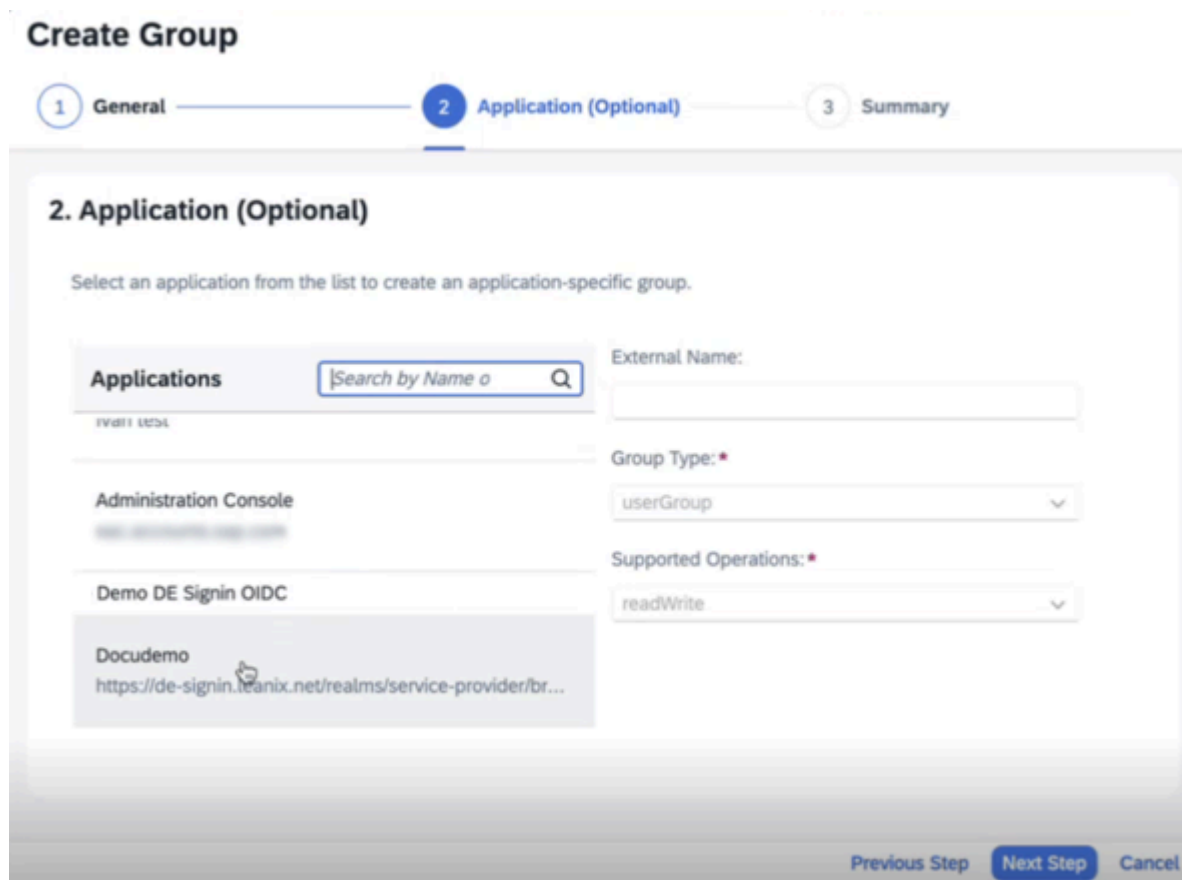
The access control attribute can have different values and you create a group for each value, for example:

- FINANCE-DEPARTMENT
- HEADQUARTER
- {ACE}

1. In IAS, navigate to [Users & Authorizations](#) > [Groups](#).



2. Choose **Create**.
3. Enter general information for the application, then choose **Next Step**.
4. Select the application you created.



5. Provide the **External Name** {ACE VALUE}.

⚠ Caution

This is equivalent to the access control entity in SAP LeanIX, for example, FINANCE-DEPARTMENT or HEADQUARTER. Ensure the spelling and capitalization are correct.

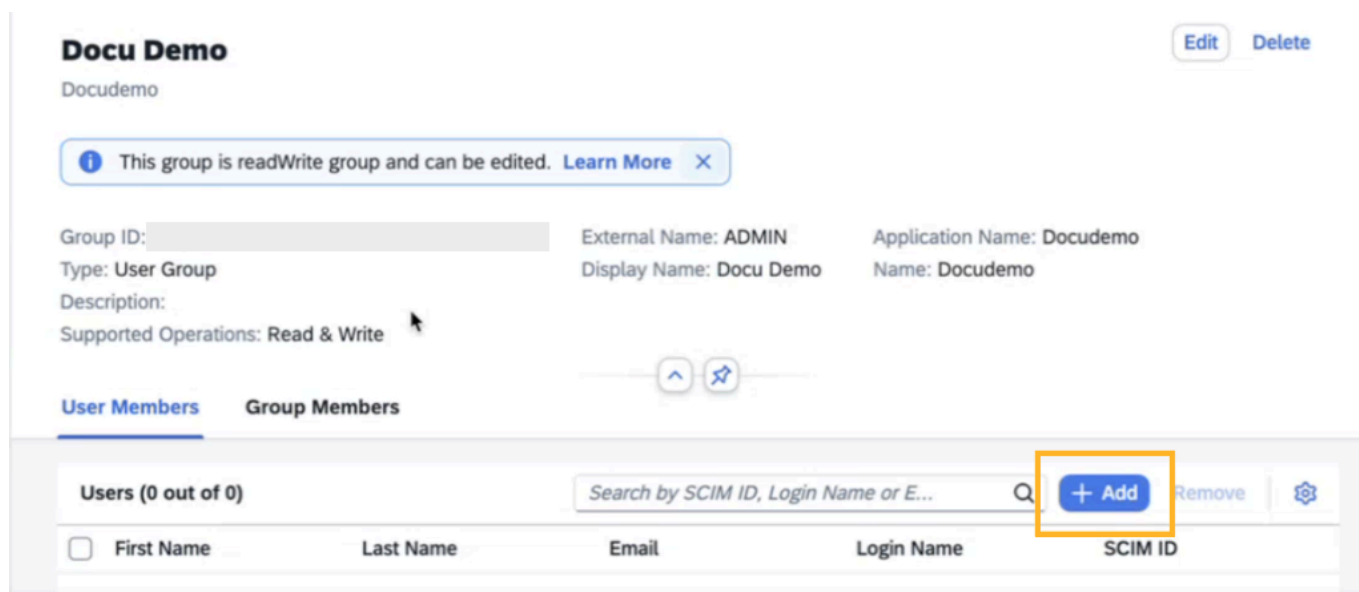
6. For **Group Type**, select **Authorization**.
7. Choose **Next Step**.
8. Check the configuration summary, then choose **Finish**.
9. Repeat the steps to create a group for each of your access control entities.

Overview of Groups for Custom Roles

External Name	Group Type
{ACE A}	Authorization
{ACE B}	Authorization
{ACE ...}	Authorization

Assign Users to Attribute-Related User Groups

1. In IAS, navigate to **Users & Authorization > User Management > Groups**.
2. Select the group you created.
3. Choose **Add** to add users to the group.



4. Select user groups.
5. Choose **Add** to save.

Assign Attribute to the SAML Application

1. In IAS, navigate to **Applications & Resources > Applications > [Your Application] > Attributes**.
2. Choose **Add**.
3. Configure the attribute

Name	Source	Value
ace	Identity Directory	Application Groups - External name

4. Choose [Save](#).

Configuring SSO with PingOne (PingIdentity)

Instructions for configuring single sign-on (SSO) with PingOne from PingIdentity as an identity provider.

Prerequisites

- You have enabled SSO configuration in SAP LeanIX, see [Getting Started With SSO](#)
- You have admin access to PingOne with permissions to create SAML applications and define attributes. If you don't have access, coordinate with your IT or identity management team to support you during the SSO configuration.

Add New Identity Provider in SAP LeanIX

You can create up to 5 connections to identity providers (IdP).

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ► and go to the **Identity Providers** section.
2. To add a new identity provider, select the authentication type and click **Add Identity Provider**.
3. Provide the identity provider details.
4. Choose **Next**.

To proceed with the setup, you need to set up a SAML application in your identity provider.

→ Tip

To change the settings of an SSO connection in SAP LeanIX, add a new identity provider. Then delete the old configuration.

Create SAML Application in PingOne

Follow these steps:

1. In the PingOne admin dashboard, in the **Applications** section, click **Add Application**.
2. Enter a name for your application.
3. Select **SAML Application** as the application type, then click **Configure**.

The screenshot shows the PingOne Admin Dashboard. On the left, the 'Applications' section lists three existing applications: 'PingOne Admin Console', 'PingOne Application Portal', and 'PingOne Self-Service - MyAccount'. On the right, the 'Add Application' form is open. The 'Application Name' field is filled with 'LeanIX'. The 'Description' field is empty. The 'Icon' field shows a default icon. The 'Application Type' section has several options: 'SAML Application' (selected), 'OIDC Web App', 'Native', 'Single-Page', 'Worker', and 'Device Authorization'. The 'Configure' button is visible at the bottom of the form.

Creating a SAML Application in the PingOne Admin Dashboard

Configure Metadata

Exchanging the metadata between SAP LeanIX as service provider and PingOne as identity provider establishes a trusted SSO connection.

1. In PingOne, navigate to **SAML Configuration > Provide Application Metadata**. Select **Manually Enter**. Copy the following IDs from the SAP LeanIX configuration:

- **Entity ID**
- **Reply URL** as the **ACS URL**

2. Save the changes.

3. Configure the destination attribute.

Known issue: To correctly configure a SAML application in PingOne, you must add the destination attribute manually. The known issue is documented in the PingOne documentation. To view it, copy the URL to your browser:

<https://support.pingidentity.com/s/article/Destination-Attribute-not-included-in-the-SAML-Response>.

- Navigate to **Applications > Integration > SP Connections** and access the relevant connection.
- Go to **Protocol Settings** and click **Signature Policy**.
- Enable **Sign Response As Required**.
- Click **Save**.

4. In PingOne, navigate back to the metadata and download the metadata file.

5. In SAP LeanIX, upload the metadata file.

Configure Service Provider-Initiated Authentication Flow

SAP LeanIX supports sp-initiated SSO only. IdP-initiated login is not supported. Users must login directly to the SAP LeanIX workspace to start the authentication flow. Provide your workspace URL during SSO configuration to ensure proper authentication flow between your identity provider and workspace.

1. In the PingOne, navigate to **Applications > Applications > [Your Application]**.
2. Edit the application settings.
3. For the **Signon URL (Initiate Single Sign-On URL)** add the workspace URL, for example, `https://{region}.leanix.net/{workspacename}/`.

If you want to setup your workspace with a subdomain, you request the subdomain after the SSO configuration and update the **Signon URL** later.

Configure Authentication Attributes

Configure the required attributes to authenticate users through PingOne.

In the **Attribute Statements** section, specify attributes to be added to the SAML assertion as shown in the following tables. Set all attributes as required. All fields are case-sensitive.

For the following required attributes, the corresponding values already exist in PingOne Mapping.

Attribute	Required	PingOne Mapping
firstname	Required	Given Name
lastname	Required	Family Name
email	Required	Email Address
uid	Required	The unique ID of the user in the email format.

Optional: Configure Authorization Attributes

If you want to manage user roles within PingOne and not within SAP LeanIX, configure optional attributes.

Authorization Scope

The necessary attributes and their configuration depend on the authorization scope:

- Role attributes are used for the default user roles: VIEWER, MEMBER, ADMIN
- Custom roles can be additional roles like ARCHITECT or AUDITOR that you define for your use cases. Custom roles require the default user role attributes to be configured first.
- Access control entities are used for virtual workspaces and require the default user role attributes to be configured first.

i Note

Create the authorization and role attributes using **uppercase** letters. This is equivalent to the role attribute in SAP LeanIX, for example, VIEWER or AUDITOR. Ensure the spelling is correct.

Specify the attributes using expressions. To implement the expressions correctly, you must use accurate group names. PingOne maps group membership to a role by extracting a segment from the group name. For example, for the group name LEANIX

MEMBER, MEMBER is sent to SAP LeanIX as the user role. User roles in the group names must exactly match the roles listed in the [User Roles and Permissions](#) section of your workspace's admin area in SAP LeanIX.

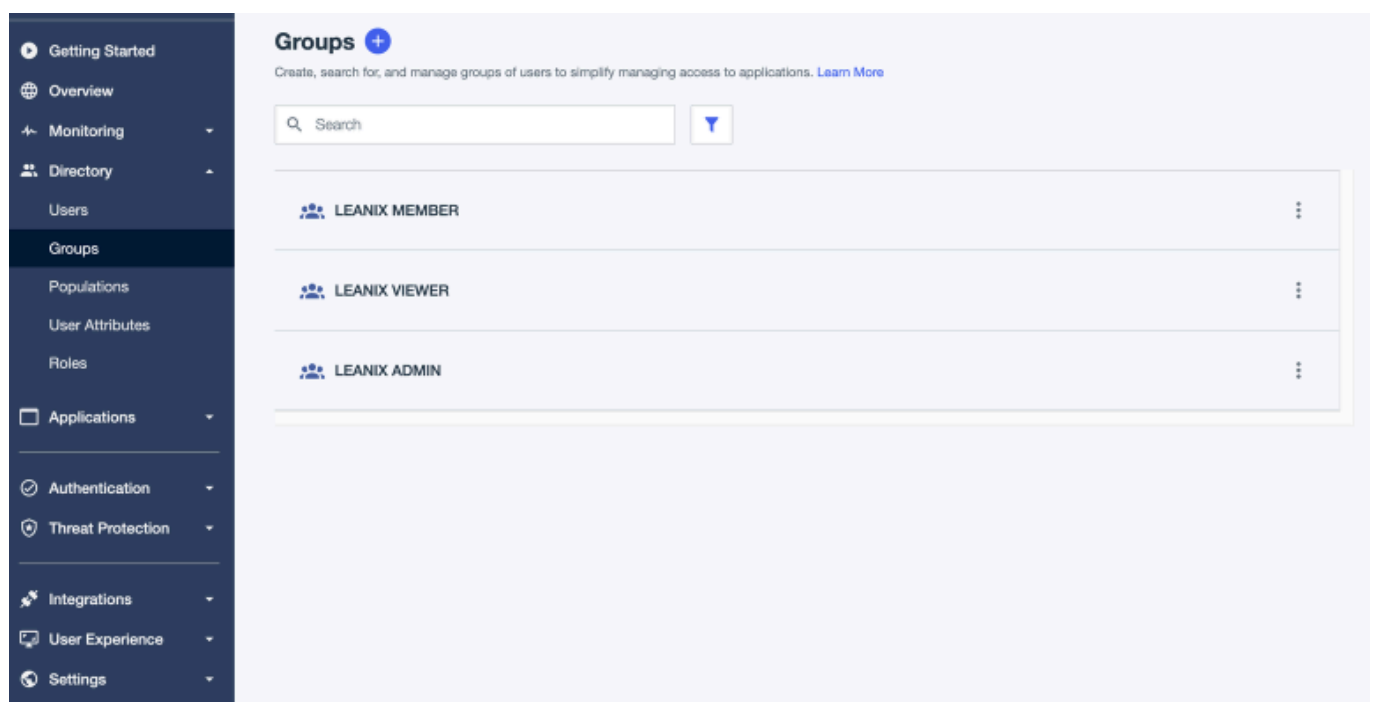
You can also map specific user attributes instead of group membership. This approach is less common but applies in scenarios such as virtual workspaces.

Attribute	Required	PingOne Mapping
role	Required only if you want to manage user roles within PingOne	Example expression: #string.replace(user.memberOfGroupNames.? [#string.startsWith(#this, 'LEANIX ')], "LEANIX ", "", -1)
customer_roles	Required only if you want to manage user roles within PingOne	Example expression: #string.replace(user.memberOfGroupNames.? [#string.startsWith(#this, 'LEANIX ')], "LEANIX ", "", -1)
ace	Required only if you want to manage user roles within PingOne and want to use virtual workspaces	Example expression: #string.replace(user.memberOfGroupNames.? [#string.startsWith(#this, 'LEANIX ')], "LEANIX ", "", -1)

Assign Users to the SAML Application

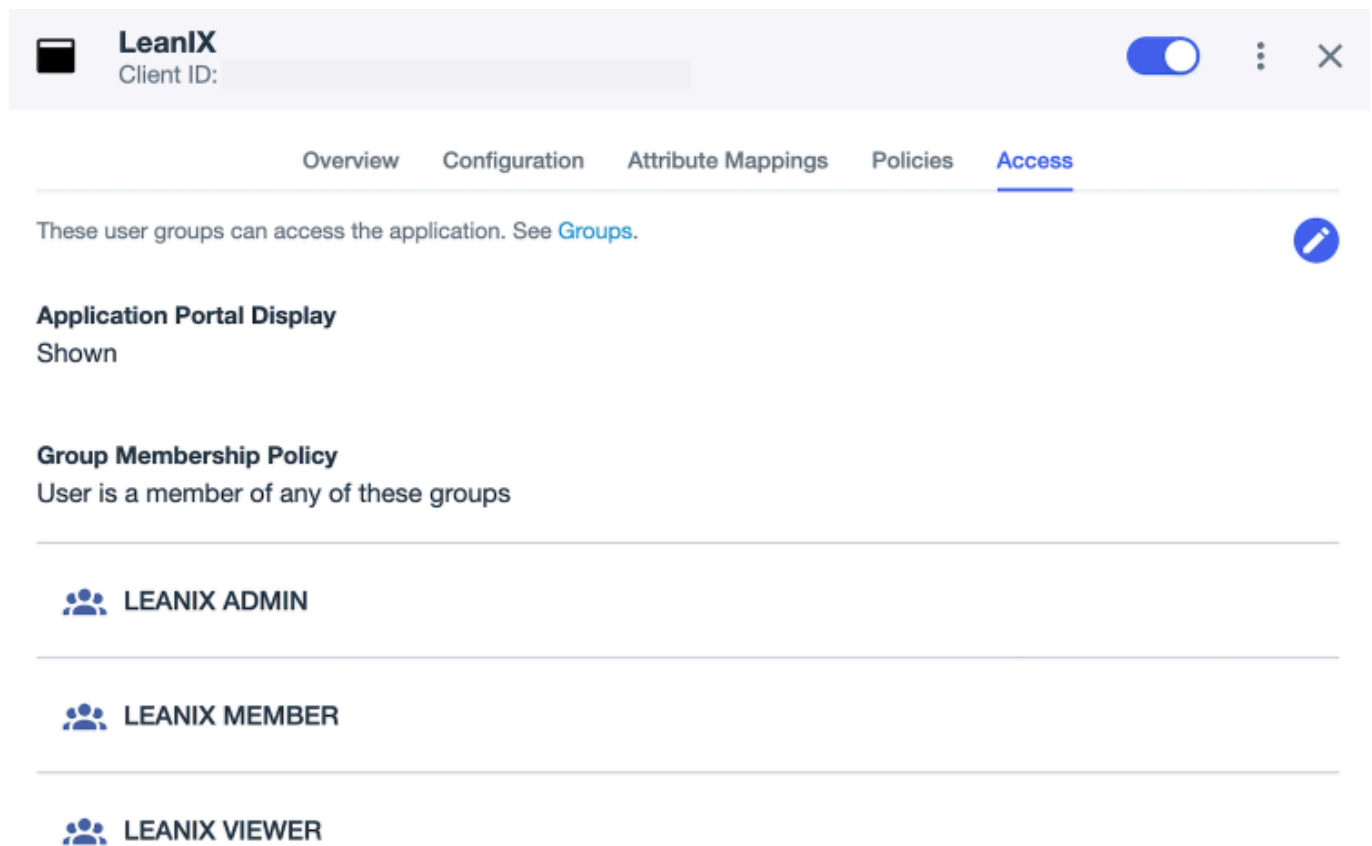
Follow these steps:

1. In the PingOne admin dashboard, navigate to [Directory](#) > [Groups](#). Review your user groups and verify that they're already populated with users. In the following screenshot, the example user groups are LEANIX MEMBER, LEANIX VIEWER, and LEANIX ADMIN.



Creating User Groups in PingOne

2. Navigate to the SAML application page. On the **Access** tab, grant access to the application to the user groups. If you don't need to manage authorization, assigning fewer groups will suffice.

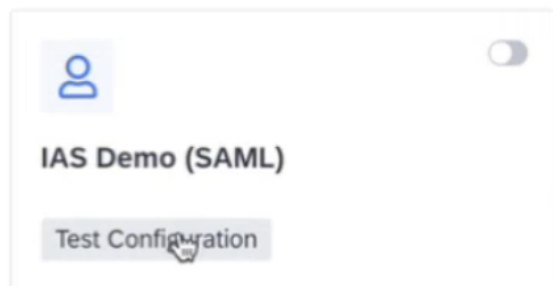


The screenshot shows the SAP LeanIX interface. At the top, there's a header with the LeanIX logo, a Client ID field, and a toggle switch. Below the header is a navigation bar with tabs: Overview, Configuration, Attribute Mappings, Policies, and Access (which is selected). The main content area shows a message: "These user groups can access the application. See [Groups](#)." Below this, there's a section titled "Application Portal Display" with the status "Shown". Another section titled "Group Membership Policy" states "User is a member of any of these groups". A list of user groups is shown below, each with a group icon and name: LEANIX ADMIN, LEANIX MEMBER, and LEANIX VIEWER.

Assigning User Groups to a SAML Application

Test the SSO Connection

1. In SAP LeanIX, navigate to **Administration** > **Authentication and SSO**.
2. On the identity provider tile, select **Test Configuration**.



3. Follow the test instructions. Open the provided URL in a private/incognito browser window and log in with the credentials for your identity provider.

If you cannot use a private or incognito browser window, copy the provided URL, log out of SAP LeanIX, and then you use the provided URL in your regular browser.

If the test is successful, you are redirected to your identity provider for login. After logging in, you're in your workspace.

You can re-run the test and try different setups, for example, different role assignments for your test user.

If you encounter any issues during SSO configuration, see our troubleshooting guide at [Troubleshooting SSO](#).

Activate the SSO Connection

⚠ Caution

You and your users can lose all access to SAP LeanIX if the identity provider configuration is not working!

Only proceed if the configuration test was successful.

! Restriction

You cannot switch back to authentication with username and password. Contact [Support](#) if you have questions or need help.

1. In SAP LeanIX, navigate to ► [Administration](#) ► [Authentication and SSO](#) ►.
2. On the identity provider tile, toggle to activate.

Confirm in the dialog that opens. Now the users managed in this identity provider can only log in to the associated workspace through their SSO account.

Optional: Request to Add Identity Provider Connection to a Custom Subdomain

With the new SSO activated the users already have access to the workspace with the direct workspace URL, but the custom subdomain won't work yet.

If your SAP LeanIX workspace uses a custom subdomain, the subdomain must be updated to include the new identity provider connection. SAP LeanIX manages subdomains, so this process is not available through self-service. After a successful migration, submit a ticket to support. Provide the direct URLs of the relevant workspaces.

Related Information

- [PingOne documentation](#) ➡
- [Single Sign-On \(SSO\)](#)
- [Getting Started With SSO](#)
- [SCIM Provisioning](#)

Configuring SSO with OneLogin (OneIdentity)

Instructions for configuring single sign-on (SSO) with OneLogin as an identity provider.

Prerequisites

- You have enabled SSO configuration in SAP LeanIX, see [Getting Started With SSO](#)
- You have admin access to OneLogin with permissions to create SAML applications and define attributes. If you don't have access, coordinate with your IT or identity management team to support you during the SSO configuration.

Add New Identity Provider in SAP LeanIX

You can create up to 5 connections to identity providers (IdP).

1. In SAP LeanIX, navigate to ► [Administration](#) ► [Authentication and SSO](#) ► and go to the [Identity Providers](#) section.
2. To add a new identity provider, select the authentication type and click [Add Identity Provider](#).

3. Provide the identity provider details.

4. Choose **Next**.

To proceed with the setup, you need to set up a SAML application in your identity provider.

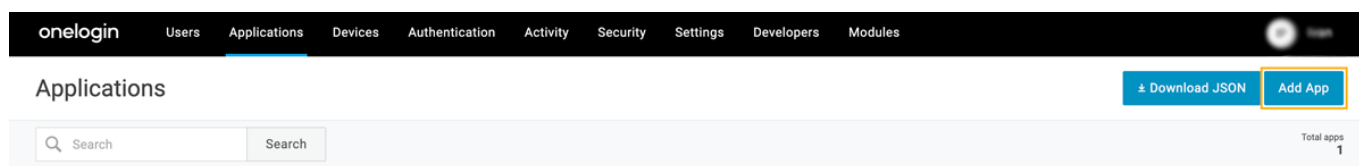
→ Tip

To change the settings of an SSO connection in SAP LeanIX, add a new identity provider. Then delete the old configuration.

Create a SAML Application in OneLogin

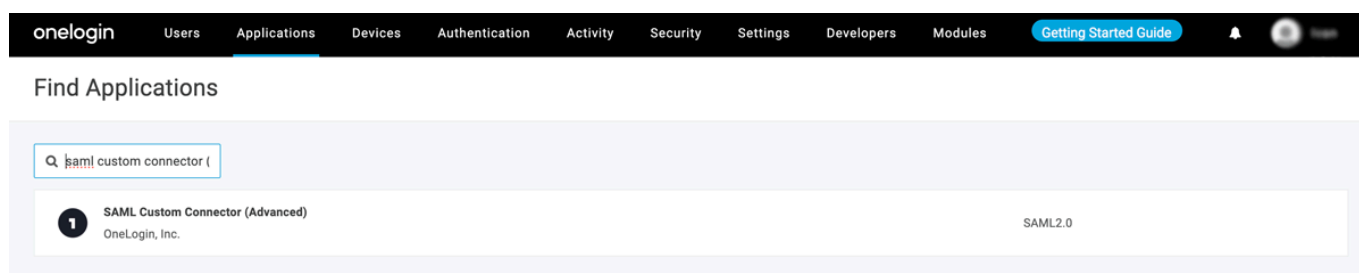
Follow these steps:

1. In the OneLogin admin dashboard, in the **Applications** section, click **Add App**.



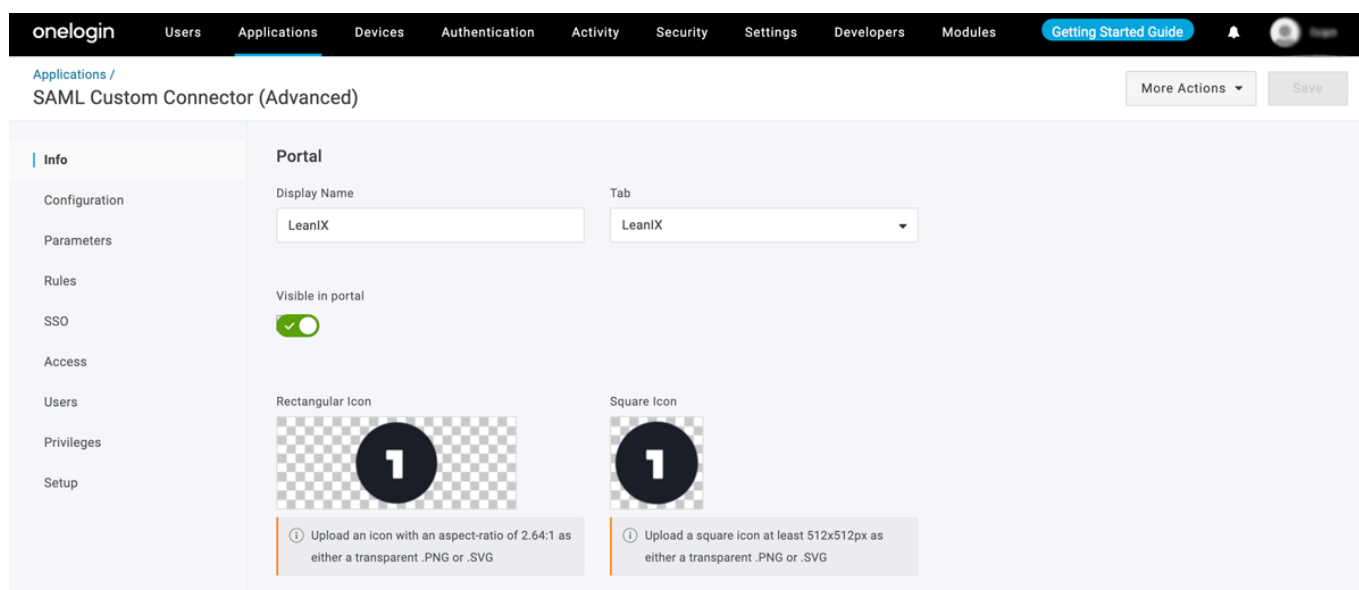
Adding an Application in OneLogin

2. On the page that appears, search for and select **SAML Custom Connector (Advanced)**.



Searching for SAML Custom Connector (Advanced)

3. Enter a name for your application and adjust other settings as needed.



Entering Basic Details for a SAML Application

Configure Metadata

Exchanging the metadata between SAP LeanIX as service provider and OneLogin as identity provider establishes a trusted SSO connection.

In the **Configuration** section of the application settings, enter the metadata URLs:

- **Audience** (Entity ID): `https://{REGION}-signin.leanix.net/realms/service-provider/broker/{UUID}`
- **Recipient**: `https://{REGION}-signin.leanix.net/realms/service-provider/broker/{UUID}/endpoint`
- **ACS (Consumer) URL Validator**: `^https:\\/{REGION}-signin\\.leanix\\.net\\/realms\\/service-provider\\/broker\\/\\{UUID}\\endpoint$`
- **ACS (Consumer) URL**: `https://{REGION}-signin.leanix.net/realms/service-provider/broker/{UUID}/endpoint`

Configure Service Provider-Initiated Authentication Flow

SAP LeanIX supports sp-initiated SSO only. IdP-initiated login is not supported. Users must login directly to the SAP LeanIX workspace to start the authentication flow. Provide your workspace URL during SSO configuration to ensure proper authentication flow between your identity provider and workspace.

1. Navigate to the **Configuration** section of the application settings.
2. For the **Login URL** add the workspace URL, for example, `https://{region}.leanix.net/{workspacename}/`.

If you want to setup your workspace with a subdomain, you request the subdomain after the SSO configuration and update the **Login URL** later.

Configure Authentication Attributes

Configure the required attributes to authenticate users through OneLogin.

In the **Parameters** section of the application settings, specify attributes to be added to the SAML assertion. Set all attributes as required. All fields are case-sensitive.

Attribute	Required	OneLogin Mapping
firstname	Required	First Name
lastname	Required	Last Name
email	Required	Email
uid	Required	Email

Optional: Configure Authorization Attributes

If you want to manage user roles within OneLogin and not within SAP LeanIX, configure additional role attributes.

Authorization Scope

The necessary attributes and their configuration depend on the authorization scope:

- Role attributes are used for the default user roles: VIEWER, MEMBER, ADMIN
- Custom roles can be additional roles like ARCHITECT or AUDITOR that you define for your use cases. Custom roles require the default user role attributes to be configured first.

- Access control entities are used for virtual workspaces and require the default user role attributes to be configured first.

i Note

Create the authorization and role attributes using **uppercase** letters. This is equivalent to the role attribute in SAP LeanIX, for example, ARCHITECT or AUDITOR. Ensure the spelling is correct.

Attribute	Required	OneLogin Mapping
role	Required only if you manage user roles within OneLogin	User Roles
customer_roles	Required only if you manage user roles within OneLogin	User Roles

- In the OneLogin admin dashboard, navigate to the **Users** section.
- In the **Roles** section, create application roles.

onelogin Users Applications Devices Authentication Activity Security Settings Developers Modules

Roles

Q search roles... Search

Role	Apps
ADMIN	LeanIX
Default	LeanIX
MEMBER	LeanIX
VIEWER	LeanIX

Creating Application Roles

- In the **Mappings** section, map the application roles that you created to user groups.

onelogin Users Applications Devices Authentication Activity Security Settings Developers Modules

Mappings / MEMBER

Conditions

Group is LeanIX_MEMBER

+

Actions

Set role MEMBER

+

Mapping an Application Role to a User Group

Test the SSO Connection

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. On the identity provider tile, select **Test Configuration**.



3. Follow the test instructions. Open the provided URL in a private/incognito browser window and log in with the credentials for your identity provider.

If you cannot use a private or incognito browser window, copy the provided URL, log out of SAP LeanIX, and then you use the provided URL in your regular browser.

If the test is successful, you are redirected to your identity provider for login. After logging in, you're in your workspace.

You can re-run the test and try different setups, for example, different role assignments for your test user.

If you encounter any issues during SSO configuration, see our troubleshooting guide at [Troubleshooting SSO](#).

Activate the SSO Connection

⚠ Caution

You and your users can lose all access to SAP LeanIX if the identity provider configuration is not working!

Only proceed if the configuration test was successful.

! Restriction

You cannot switch back to authentication with username and password. Contact [Support](#) if you have questions or need help.

1. In SAP LeanIX, navigate to ► **Administration** ► **Authentication and SSO** ►.
2. On the identity provider tile, toggle to activate.

Confirm in the dialog that opens. Now the users managed in this identity provider can only log in to the associated workspace through their SSO account.

Optional: Request to Add Identity Provider Connection to a Custom Subdomain

With the new SSO activated the users already have access to the workspace with the direct workspace URL, but the custom subdomain won't work yet.

If your SAP LeanIX workspace uses a custom subdomain, the subdomain must be updated to include the new identity provider connection. SAP LeanIX manages subdomains, so this process is not available through self-service. After a successful migration, submit a ticket to support. Provide the direct URLs of the relevant workspaces.

Related Information

- [OneLogin website](#) ➡
- [Single Sign-On \(SSO\)](#)
- [Getting Started With SSO](#)
- [SCIM Provisioning](#)

Configuring SSO with Active Directory Federation Services

Instructions for configuring single sign-on (SSO) with Active Directory Federation Services (AD FS) as an identity provider.

Prerequisites

Before you start, do the following:

- Learn about the SSO configuration process in [Getting Started With SSO](#).

Configuring SSO

You can configure SSO for SAP LeanIX through a self-hosted Active Directory Federation Services (AD FS) server. AD FS is a service provided by Microsoft as a standard role for Windows Server that provides a web login using existing Active Directory credentials.

The SAP LeanIX SAML service provider is based on SIGNIN. To learn how to set up AD FS SAML federation with SIGNIN, please refer to the [Microsoft documentation](#) ➡ .

Usually, the AD FS metadata information is available at the following URL:

`https://FederationMetadata/2007-06/FederationMetadata.xml`

SAP LeanIX supports sp-initiated SSO only. IdP-initiated login is not supported. Users must login directly to the SAP LeanIX workspace to start the authentication flow. Provide your workspace URL during SSO configuration to ensure proper authentication flow between your identity provider and workspace.

Configuring Attribute Mapping

To learn how to configure attribute mapping, see [SSO Attribute Overview](#).

When using AD FS for SSO, you can pass values in addition to the authentication values. These values are defined as Claim Rules in the Relying Party Trust. To edit Claim Rules, select the **Relying Party Trusts** folder in AD FS Management, then select **Edit Claim Rules** in the **Actions** sidebar.

You should also configure custom claim rules. Please see example AD FS claim rules below.

```
@RuleName = "Get tokenGroups"
c:[Type == "http://schemas.microsoft.com/ws/2008/06/identity/claims/windowsaccountname", Issuer ==
=> add(store = "Active Directory", types = ("http://claims.contoso.com/tokenGroups"), query = ";tokenGroups")

@RuleName = "Add Given-Name from AD"
c:[Type == "http://schemas.microsoft.com/ws/2008/06/identity/claims/windowsaccountname", Issuer ==
=> add(store = "Active Directory", types = ("http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname"))
```

```

@RuleName = "Add Surname from AD"
c:[Type == "http://schemas.microsoft.com/ws/2008/06/identity/claims/windowsaccountname", Issuer ==
=> add(store = "Active Directory", types = ("http://schemas.xmlsoap.org/ws/2005/05/identity/claims/

@RuleName = "Add UPN from AD"
c:[Type == "http://schemas.microsoft.com/ws/2008/06/identity/claims/windowsaccountname", Issuer ==
=> add(store = "Active Directory", types = ("http://schemas.xmlsoap.org/ws/2005/05/identity/claims/

@RuleName = "Add E-Mail-Address from AD"
c:[Type == "http://schemas.microsoft.com/ws/2008/06/identity/claims/windowsaccountname", Issuer ==
=> add(store = "Active Directory", types = ("http://schemas.xmlsoap.org/ws/2005/05/identity/claims/

@RuleName = "Send Given-Name as firstname"
c:[Type == "http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname"]
=> issue(Type = "firstname", Value = c.Value, Issuer = c.Issuer, Properties["http://schemas.xmlsoap.org/

@RuleName = "Send Surname as lastname"
c:[Type == "http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname"]
=> issue(Type = "lastname", Value = c.Value, Issuer = c.Issuer, Properties["http://schemas.xmlsoap.org/

@RuleName = "Send ADUPN as uid"
c:[Type == "http://schemas.xmlsoap.org/ws/2005/05/identity/claims/upn"]
=> issue(Type = "uid", Value = c.Value, Issuer = c.Issuer, Properties["http://schemas.xmlsoap.org/

@RuleName = "Send E-Mail-Address as email"
c:[Type == "http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress"]
=> issue(Type = "email", Value = c.Value, Issuer = c.Issuer, Properties["http://schemas.xmlsoap.org/

@RuleName = "Send role ADMIN"
EXISTS([Type == "http://schemas.xmlsoap.org/ws/2005/05/identity/claims/tokenGroups", Value == "EAM-ADMIN"])
=> issue(Type = "role", Value = "ADMIN", Properties["http://schemas.xmlsoap.org/ws/2005/05/identity/claims/

@RuleName = "Send role MEMBER"
EXISTS([Type == "http://schemas.xmlsoap.org/ws/2005/05/identity/claims/tokenGroups", Value == "EAM-MEMBER"])
=> issue(Type = "role", Value = "MEMBER", Properties["http://schemas.xmlsoap.org/ws/2005/05/identity/claims/

@RuleName = "Send role VIEWER"
EXISTS([Type == "http://schemas.xmlsoap.org/ws/2005/05/identity/claims/tokenGroups", Value == "EAM-VIEWER"])
=> issue(Type = "role", Value = "VIEWER", Properties["http://schemas.xmlsoap.org/ws/2005/05/identity/claims/

@RuleName = "Send customer_roles CUSTOMERROLE1"
EXISTS([Type == "http://schemas.xmlsoap.org/ws/2005/05/identity/claims/tokenGroups", Value == "EAM-CUSTOMERROLE1"])
=> issue(Type = "customer_roles", Value = "CUSTOMERROLE1", Properties["http://schemas.xmlsoap.org/

```

Verify your SSO Configuration

To verify your SSO configuration, first, access your workspace at <https://SUBDOMAIN.leanix.net> and log in, then open a SAML tracer browser extension or desktop application. In the SAML tracing you can see a list of required user attributes.

Troubleshooting SSO

Use the SSO connection test and a SAML tracer to inspect your SSO connection and attribute mappings. Understand what the support team needs to help you in case of persistent problems.

If you encounter issues during SSO configuration, try the following to troubleshoot.

SAML Tracer

A SAML tracer helps you understand most SSO issues. Use it to inspect SAML requests and check which SSO attributes your identity provider sends. You can use browser extensions or desktop applications, such as SAML Tracer, to inspect the SAML request. These tools show which SSO attributes are being sent.

1. Install the SAML tracer of your choice.
2. Access your workspace at `https://{SUBDOMAIN}.leanix.net` and log in.
3. Open the SAML tracer browser extension or desktop application.
4. In the SAML tracer, check attribute mappings.
5. Optionally, take a screenshot or export details from the SAML tracer for future reference, for example, to reach out to support.

Contact Support

If something goes wrong during the self-service SSO configuration that you cannot fix, contact support:

1. Go to [SAP for Me](#) .
2. Enter the ticket title: "Problem with self-service SSO configuration - [your workspace name]".
3. Provide your workspace details and reason for reverting:
 - Workspace name
 - Subdomain, for example, "customername.leanix.net" or "http://eu.leanix.net"
 - SSO configuration date (if known)
4. Provide the configuration steps that you already completed, for example:
 - Created a SAML application in the identity provider (IdP)
 - Created an identity provider in SAP LeanIX
 - Activated the identity provider in SAP LeanIX
5. Provide a screenshot or export from your SAML tracer.
6. Choose SAP LeanIX as the system, then populate the other required fields and submit the ticket.

The SAP LeanIX support team assists solving your SSO issue.

SSO Attribute Overview

Reference the SAML 2.0 attribute overview for required authentication attributes and optional authorization attributes for roles and access control entities.

Attribute Mapping

Configure the SAML 2.0 attribute mapping in your identity provider.

Use a browser extension or desktop application such as [SAML Tracer](#) to inspect SAML requests and verify which attributes your identity provider sends.

Attribute Name	Required	Format	Example	Description
firstname	Required	urn:oasis:names:tc:SAML:2.0:attrname-format:basic	Peter	The first name of the user.
lastname	Required	urn:oasis:names:tc:SAML:2.0:attrname-format:basic	Schmidt	The last name of the user.
uid	Required	urn:oasis:names:tc:SAML:2.0:attrname-format:basic	01586285682568@customer.com	The unique identifier of the user. The user needs to provide an email for authentication. Use an ID that is different from the user's email address.
email	Required	urn:oasis:names:tc:SAML:2.0:attrname-format:basic	peter.schmidt@customer.com	The email address of the user.
role	Optional	urn:oasis:names:tc:SAML:2.0:attrname-format:basic	MEMBER	The role assigned to the user. Required values: ADMIN, MEMBER, VIEWER. If you submit multiple values, separate them with commas. SAP LeanIX assigns the role with the highest privilege level. If you're managing roles with SAP LeanIX, you can configure this attribute. To learn more, see Managing Roles .

Attribute Name	Required	Format	Example	Descripti
				User Role with SSO
customer_roles	Optional	urn:oasis:names:tc:SAML:2.0:attrname-format:basic	MANAGER	The custo role to be assigned the user. this attrik only for custom r otherwise omit it. To learn r see Custo User Role
ace	Optional	urn:oasis:names:tc:SAML:2.0:attrname-format:basic	member-orgunit1	The ID of access control er (ACE) of i virtual workspac Use this attribute when configuri access to virtual workspac otherwise omit it. To learn r see Virtu Workspa Configur

Example SAML Message with User Attributes

The following example SAML message contains required user attributes.

≡ Sample Code

```
<md:AttributeConsumingService isDefault="true" index="1">
<md:ServiceName xml:lang="en">service-provider</md:ServiceName>
<md:RequestedAttribute Name="customer_roles" FriendlyName="customer_roles" NameFormat="urn:oasis
<md:RequestedAttribute Name="blacklisted" FriendlyName="blacklisted" NameFormat="urn:oasis:names
<md:RequestedAttribute Name="email" FriendlyName="email" NameFormat="urn:oasis:names:tc:SAML:2.0
<md:RequestedAttribute Name="role" FriendlyName="role" NameFormat="urn:oasis:names:tc:SAML:2.0:a
<md:RequestedAttribute Name="transient" FriendlyName="transient" NameFormat="urn:oasis:names:tc:
<md:RequestedAttribute Name="uid" FriendlyName="uid" NameFormat="urn:oasis:names:tc:SAML:2.0:att
<md:RequestedAttribute Name="ace" FriendlyName="ace" NameFormat="urn:oasis:names:tc:SAML:2.0:att
<md:RequestedAttribute Name="firstname" FriendlyName="firstname" NameFormat="urn:oasis:names:tc:
```

```
<md:RequestedAttribute Name="lastname" FriendlyName="lastname" NameFormat="urn:oasis:names:tc:SA
</md:AttributeConsumingService>
```

Related Information

Reference the attribute configuration guides for the following identity providers for more details:

- Microsoft Entra ID
 - [Configure Authentication Attributes](#)
 - [Optional: Configure Authorization Attributes](#)
- Okta
 - [Configure Authentication Attributes](#)
 - [Optional: Configure Authorization Attributes](#)
- SAP Cloud Identity Services - Identity Authentication Service (IAS)
 - [Configure Authentication Attributes](#)
 - [Optional: Configure Authorization Attributes](#)
- PingOne (PingIdentity)
 - [Configure Authentication Attributes](#)
 - [Optional: Configure Authorization Attributes](#)
- OneLogin
 - [Configure Authentication Attributes](#)
 - [Optional: Configure Authorization Attributes](#)
- ADFS
 - [Configuring Attribute Mapping](#)

SCIM Provisioning

Learn how to configure System for Cross-Domain Identity Management (SCIM) for seamless user state synchronization.

Introduction

System for Cross-domain Identity Management (SCIM) is a standard protocol designed to automate the synchronization of user states across multiple systems. SCIM facilitates the transfer of user information from a source system, such as an external identity provider (IdP), to a target system, such as SAP LeanIX.

In SAP LeanIX, SCIM works together with single sign-on (SSO). SCIM is used to automate the process of user provisioning and deprovisioning across multiple systems, while SSO is used for authentication and authorization. During the sign-in process, SSO checks if a user has the right to access the system (known as authentication) and determines the user's permissions within the system (known as authorization). For more information, see [Single Sign-On \(SSO\)](#).

SCIM is used for the following purposes:

This is custom documentation. For more information, please visit [SAP Help Portal](#).

- **User provisioning:** Provisioning means creating a user account. If a user exists in the source system, such as an external IdP, but not in the target system (SAP LeanIX), then a corresponding user object is created in the target system.
- **User deprovisioning:** Deprovisioning means archiving a user account. If a user exists in the target system but not in the source system, then the user object in the target system gets deprovisioned.
- **User updating:** If the user data in the source system differs from the data in the target system, then this data is transferred to the target system.

i Note

You can choose a SCIM provider that is not the same as your SSO provider.

Synchronized Attributes

During periodic synchronizations, the following user details are updated:

- First name
- Last name
- Email address
- Username
- Role, custom roles, and Access Control Entities (ACEs), if applicable
- Department, if applicable

Limitations

- You can enable SCIM for a single workspace. SCIM is not supported across multiple workspaces.
- User state synchronization only works for the workspace where you configure SCIM.
- If you enabled the **Invite Only** flow with SSO for your workspace, user permissions are not created with SCIM.

Configuring SCIM

To configure SCIM between your IdP and SAP LeanIX, follow these steps:

1. In SAP LeanIX, create a technical user with the admin permission role. Save the API token that appears. For instructions, see [Technical Users](#).
2. Request the ACCOUNTADMIN role for the technical user by [submitting a ticket](#) 📩 to support. If you're an SAP customer, submit a request from the [SAP for Me](#) 🏠 portal. In the request, provide the name of the technical user.
3. Get an access token required for the SCIM integration:
 - a. Using the API token of the technical user, get a short-lived access token. For instructions, see [Get a Short-Lived Access Token](#).
 - b. Using the short-lived access token, get a long-lived access token. For instructions, see [Get a Long-Lived Access Token](#).
4. In your IdP, configure user provisioning. For instructions, refer to the documentation of your IdP. Use the following details:
 - **SCIM endpoint:** `https://{SUBDOMAIN}.leanix.net/services/mtm/v1/scim/v2`

- **SCIM access token:** Long-lived access token that you obtained.

5. In your IdP, configure attribute mapping. For more information, see [SCIM Attribute Mapping](#).
6. Depending on the configuration of your IdP, you may need to enable the synchronization of user states.

After you've set up SCIM, user states are synchronized between your IdP and SAP LeanIX.

SCIM Attribute Mapping

The following table lists attributes that are supported in the SCIM integration. Your IdP may require other attributes that are not listed in the table.

Attribute	Required	Description
userName	Required	As configured in SSO in the uid claim
givenName	Required	User's given name
familyName	Required	User's family name
emails or email	Required	User's work email address
active	Required (in Microsoft Entra ID)	Controls provisioning and deprovisioning

i Note

To learn how to synchronize authorization, refer to the instructions for a specific identity provider.

The username and email address are unique user identifiers. The following scenarios are possible:

- **The username and email address in SAP LeanIX match with the provisioned information:** The user can be matched. No changes to the user in SAP LeanIX are applied.
- **The username or email address in SAP LeanIX matches with the provisioned information:** The user can be matched. The user in SAP LeanIX gets updated with the provisioned information.
- **Neither the username nor email address matches with the provisioned information:** The user can't be matched. A new user is created in SAP LeanIX.

Access Tokens Required for SCIM

Learn how to obtain and manage long-lived access tokens required for SCIM.

To configure SCIM, you need a long-lived access token. In this guide, you'll learn how to obtain, retrieve, and deactivate access tokens through the MTM REST API. To view reference documentation for the MTM API, navigate to the [OpenAPI Explorer](#) ➦ .

Get a Short-Lived Access Token

To obtain a long-lived access token required for the SCIM integration, you need a short-lived access token. The long-lived token inherits the user role from the short-lived token. The Technical User through which you request a short-lived token must have the ACCOUNTADMIN role.

To obtain a short-lived access token, follow these steps:

1. Create a technical user with the admin permission role. Save the API token that appears. For instructions, see [Technical Users](#).
2. Request the ACCOUNTADMIN role for the Technical User by [submitting a ticket](#) to SAP LeanIX Support. If you're an SAP customer, submit a request from the [SAP for Me](#) portal. In the request, provide the name of the Technical User.
3. Using the API token of the Technical User, obtain a short-lived access token. Replace the following placeholders with your values:
 - o {SUBDOMAIN}: Your SAP LeanIX subdomain. You can copy the subdomain value from the URL of your workspace.
 - o {API_TOKEN}: The API token that you obtained by creating a Technical User.

Example cURL request:

≡ Sample Code

```
curl --request POST https://{SUBDOMAIN}.leanix.net/services/mtm/v1/oauth2/token \
  -u apitoken:{API_TOKEN} \
  --data grant_type=client_credentials
```

A short-lived access token is returned in the access_token attribute in the response. The token is valid for 3600 seconds.

Save the token. You need it to obtain a long-lived access token.

Example JSON response:

```
{
  "scope": "",
  "expired": false,
  "access_token": "eyJhbGciOiJIUzI1NiJ9.eyJz ssqaPSA",
  "token_type": "bearer",
  "expires_in": 3599
}
```

Get a Long-Lived Access Token

A long-lived access token is required for the SCIM integration. This token does not have an expiration date. It's valid until deleted or deactivated.

Before proceeding, obtain a short-lived access token. For instructions, see [Get a Short-Lived Access Token](#).

To obtain a long-lived access token, make a POST request to the following endpoint:

```
https://{SUBDOMAIN}.leanix.net/services/mtm/v1/longlivedBearerTokens
```

The following table lists parameters that you should pass in the request body.

Parameter	Parameter Type	Data Type	Required	Description
description	Body	String	Optional	The description for the long-lived access token.

Parameter	Parameter Type	Data Type	Required	Description
scimConfiguration.workspaceId	Body	String	Required	The ID of the workspace to configure SCIM for. To get your workspace ID, in the administration area, navigate to API Tokens , then copy the WorkspaceId value.
scimConfiguration.permissionRole	Body	String	Required	The default SAP LeanIX role to be assigned to the user. Possible values: • MEMBER • ADMIN • VIEWER

The following example request contains placeholders that you should replace with your values.

- {SUBDOMAIN}: Your SAP LeanIX subdomain. You can copy the subdomain value from the URL of your workspace.
- {SHORT_LIVED_ACCESS_TOKEN}: Short-lived access token that you retrieved in the previous step.
- {WORKSPACE_ID}: The ID of your workspace. To get your workspace ID, in the administration area, navigate to the [API Tokens](#) section, then copy the WorkspaceId value.
- {DEFAULT_USER_ROLE}: The default SAP LeanIX role to be assigned to the user. Possible values: VIEWER, MEMBER, or ADMIN.

Example cURL request:

Sample Code

```
curl --request POST \
  --url https://{SUBDOMAIN}.leanix.net/services/mtm/v1/longlivedBearerTokens \
  --header 'Authorization: Bearer {SHORT_LIVED_ACCESS_TOKEN}' \
  --header 'Content-Type: application/json' \
  --data '{"description":"My first long-lived bearer token","scope":"","scimConfiguration":{"wor
```

A long-lived access token is returned in the accessToken attribute in the response.

Caution

The access token is shown only once in the response. Save the token and make sure that you store it securely.

Example JSON response:

```
{
  "id": "47394a58-4165-45e4-6ge4-f9a706dad4fb",
```

```

    "accountId": "fb526496-751b-44cd-31h3-369f233fa543",
    "accessTokenId": "b25f9c17-5fab-41a5-96a5-fef3bc1943e1",
    "valid": true,
    "accessToken": "eyJraWQiOiI0MDJjODg3NTBjZmJhOGQzZTQ0Nj...LlRK-8-W7cg",
    "creatorId": "8c263138-afab-4823-34gh-3e32628d4dc6",
    "description": "My first long-lived bearer token",
    "createdAt": "2024-04-19T12:45:23.461Z"
  }

```

Get All Long-Lived Access Tokens

Before proceeding, obtain a short-lived access token. The associated Technical User must have the ACCOUNTADMIN role. For instructions, see [Get a Short-Lived Access Token](#).

To retrieve existing long-lived access tokens, make a GET request to the following endpoint:

```
https://{SUBDOMAIN}.leanix.net/services/mtm/v1/longlivedBearerTokens
```

Replace {SUBDOMAIN} and {SHORT_LIVED_ACCESS_TOKEN} with your values.

Example cURL request:

```

curl --request GET \
  --url https://{SUBDOMAIN}.leanix.net/services/mtm/v1/longlivedBearerTokens \
  --header 'Authorization: Bearer {SHORT_LIVED_ACCESS_TOKEN}'

```

Example JSON response:

```

[
  {
    "id": "47394a58-4165-45e4-6ge4-f9a706dad4fb",
    "accountId": "fb526496-751b-44cd-31h3-369f233fa543",
    "accessTokenId": "b25f9c17-5fab-41a5-96a5-fef3bc1943e1",
    "valid": true,
    "creatorId": "8c263138-afab-4823-34gh-3e32628d4dc6",
    "description": "My first long-lived bearer token",
    "createdAt": "2024-04-19T12:45:23.461Z"
  },
  ...
]

```

Deactivate a Long-Lived Access Token

Before proceeding, do the following:

- Obtain a short-lived access token. The associated Technical User must have the ACCOUNTADMIN role. For instructions, see [Get a Short-Lived Access Token](#).
- Get the id of the token that you want to deactivate. To do that, retrieve all long-lived access tokens and copy the token id from the response. For instructions, see [Get All Long-Lived Access Tokens](#).

To deactivate a long-lived access token, make a POST request to the following endpoint:

`https://{SUBDOMAIN}.leanix.net/services/mtm/v1/longlivedBearerTokens/{id}/invalidate`

Replace `{SUBDOMAIN}` and `{SHORT_LIVED_ACCESS_TOKEN}` with your values.

Example cURL request:

```
curl --request POST \
  --url https://{SUBDOMAIN}.leanix.net/services/mtm/v1/longlivedBearerTokens/47394a58-4165-45e4-6ge4-f9a706dad4fb \
  --header 'Authorization: Bearer {SHORT_LIVED_ACCESS_TOKEN}'
```

Example JSON response:

```
{
  "id": "47394a58-4165-45e4-6ge4-f9a706dad4fb",
  "accountId": "fb526496-751b-44cd-31h3-369f233fa543",
  "accessTokenId": "b25f9c17-5fab-41a5-96a5-fef3bc1943e1",
  "valid": false,
  "creatorId": "8c263138-afab-4823-34gh-3e32628d4dc6",
  "description": "My first long-lived bearer token",
  "createdAt": "2024-04-19T12:45:23.461Z"
}
```

The `valid` attribute returned in the response is set to `false`, which means that the token is deactivated and can no longer be used to perform any authenticated operations.

Configuring SCIM in Okta

Instructions for configuring SCIM provisioning in Okta. SCIM allows you to enable seamless and automated user management across your applications.

Prerequisites

Before you start, do the following:

- Learn about the SCIM configuration process in [Configuring SCIM](#).
- Configure single sign-on (SSO) with Okta as an identity provider (IdP). For instructions, see [Configuring SSO with Okta](#).
- Get a long-lived access token through the MTM API. For instructions, see [Get a Long-Lived Access Token](#).

Configuring User Provisioning

Follow these steps:

1. In the Okta admin dashboard, in the **Applications** section, select the SSO application for SAP LeanIX.
2. On the **General** tab, in the **App Settings** section, click **Edit**, then select **SCIM** against **Provisioning**. Save the changes.

 **LeanIX**

Active ▾

 View Logs

 Monitor Imports

General

Sign On

Mobile

Provisioning

Import

Assignments

App Settings

Cancel

Application label

LeanIX

This label displays under the app on your home page

Application visibility

☐ Do not display application icon to users

☐ Do not display application icon in the Okta Mobile app

Provisioning

☐ None

☐ On-Premises Provisioning

☒ SCIM

Selecting the SCIM Option for Provisioning in Okta

3. If you've previously created the role attribute, in the [Profile Editor](#) section, delete the attribute and then recreate it with the following [External namespace](#) value. Save the changes. The update might take some time. To verify that the property update has been completed, check a sample user assignment for their role inheritance.

External namespace

```
urn:ietf:params:scim:schemas:extension:workspacePermission:2.0:User
```

Add Attribute

* Local app attributes are only stored on Okta and not created in LeanIX. Use local attributes if you plan to add the attribute to LeanIX or only want to store the mapped value in Okta.

Data type

string

Display name ?

role

Variable name ?

role

External name ?

role

External namespace ?

urn:ietf:params:scim:schemas:extension:workspace

Description

LeanIX permission role

Enum

☒ Define enumerated list of values

Attribute members

	Display name	Value	
⋮	ADMIN	ADMIN	×
⋮	MEMBER	MEMBER	×
⋮	VIEWER	VIEWER	×

+ Add Another

Creating the "role" Attribute

4. On the SSO application page, navigate to the **Provisioning** tab. In the **Integration** section, click **Edit** against **SCIM Connection** and specify the following details. Save the changes.

Parameter	Value
SCIM connector base URL	https://{SUBDOMAIN}.leanix.net/services/mtm/v1/scim/v2
Unique identifier field for users	userName
Supported provisioning actions	Select the following checkboxes: Import New Users and Profile Updates: Any potential changes only apply to users assigned to the SAML application in Okta. If you leave this checkbox unselected, new users assigned to the SAML application won't be created in SAP LeanIX, and any potential changes made to user attributes won't be synchronized.

Parameter	Value
	◦ Push New Users ◦ Push Profile Updates
Authentication Mode	HTTP Header
Authorization	Long-lived access token that you obtained

General
Sign On
Mobile
Provisioning
Import
Assignments

Settings
To App
To Okta
Integration

SCIM Connection Cancel

SCIM version 2.0

SCIM connector base URL

Unique identifier field for users

Supported provisioning actions

- ☒ Import New Users and Profile Updates
- ☒ Push New Users
- ☒ Push Profile Updates
- ☐ Push Groups
- ☐ Import Groups

Authentication Mode

HTTP Header

Authorization

[Test Connector Configuration](#)

[Save](#) [Cancel](#)

Configuring SCIM Connection Settings

5. On the **Provisioning** tab, in the **To App** section, enable the following options:

- [Create Users](#)
- [Update User Attributes](#)
- [Deactivate Users](#)

Save the changes.

Settings
To App
To Okta
Integration

okta

→

LeanIX

Provisioning to App

Cancel

Create Users

☒ Enable

Creates or links a user in LeanIX when assigning the app to a user in Okta.
The [default username](#) used to create accounts is set to **Okta username**.

Update User Attributes

☒ Enable

Okta updates a user's attributes in LeanIX when the app is assigned. Future attribute changes made to the Okta user profile will automatically overwrite the corresponding attribute value in LeanIX.

Deactivate Users

☒ Enable

Deactivates a user's LeanIX account when it is unassigned in Okta or their Okta account is deactivated. Accounts can be reactivated if the app is reassigned to a user in Okta.

Sync Password

☐ Enable

Creates a LeanIX password for each assigned user and pushes it to LeanIX.

Save

Configuring Provisioning to App Settings

6. To configure attribute mappings, scroll down to the **Attribute Mappings** section. Update attribute mapping to include only the following attributes and delete all other attributes:

- Username
- Given name
- Family name
- Email
- Primary email type
- role: This attribute is only relevant for customers using an external IdP for SSO.

LeanIX Attribute Mappings

Select a(n) LeanIX attribute to set its value based on values stored in Okta.

[Go to Profile Editor](#)

[Force Sync](#)

Attribute	Attribute Type	Value	Apply on		
Username userName	Personal	Configured in Sign On settings			
Given name givenName	Personal	user.firstName	Create and update		
Family name familyName	Personal	user.lastName	Create and update		
Email email	Personal	user.email	Create and update		
Primary email type emailType	Personal	(user.email != null && user.email != "") ? 'work' : "	Create and update		

Configuring Attribute Mappings for SCIM

SCIM with Okta is set up. User states should be synchronized between the systems.

Configuring SCIM in Microsoft Entra ID

Instructions for configuring SCIM provisioning in Microsoft Entra ID. SCIM allows you to enable seamless and automated user management across your applications.

Prerequisites

Before you start, do the following:

- Learn about the SCIM configuration process in [Configuring SCIM](#).
- Configure single sign-on (SSO) with Microsoft Entra ID as an identity provider. For instructions, see [Configuring SSO with Entra ID](#).
- Get a long-lived access token through the MTM API. For instructions, see [Get a Long-Lived Access Token](#).

Configuring User Provisioning

Follow these steps:

1. In Microsoft Entra ID, navigate to the SSO application for SAP LeanIX.
2. On the [Provisioning](#) page, configure provisioning settings.

- a. In the **Provisioning Mode** list, select **Automatic**.
- b. Under **Admin Credentials**, in the **Tenant URL** field, enter the following URL:

`https://{SUBDOMAIN}.leanix.net/services/mtm/v1/scim/v2`

- c. In the **Secret Token** field, enter the long-lived access token that you obtained.

Provisioning Mode: Automatic

Use Microsoft Entra to manage the creation and synchronization of user accounts in Emma SSO based on user and group assignment.

Admin Credentials

Admin Credentials
Microsoft Entra needs the following information to connect to Emma SSO's API and synchronize user data.

Tenant URL * ⓘ
`https://{SUBDOMAIN}.leanix.net/services/mtm/v1/scim/v2` ✓

Secret Token
[Empty field]

Test Connection

Mappings

Configuring Provisioning Settings in Microsoft Entra ID

- d. Under **Mappings**, ensure that provisioning for Active Directory Groups is disabled.

Secret Token
[Field with asterisk]

Test Connection

Mappings

Mappings
Mappings allow you to define how data should flow between Microsoft Entra ID and customappsso.

Name	Enabled
Provision Azure Active Directory Groups	No
Provision Azure Active Directory Users	Yes

☐ Restore default mappings

Mapping Settings on the Provisioning Page

3. On the **Attribute Mapping** page, update attribute mapping to include only the following attributes and delete all other attributes.
 - **userPrincipalName**: Please ensure that the property you're passing is identical to your SSO configuration. Change the SCIM property as needed. This property must be in the email format.
 - **mail**

- Switch
- givenName
- surname
- department

Attribute Mapping ... ×

Save Discard

Enabled

YesNo

Source Object

User

Source Object Scope

All records

Target Object

urn:ietf:params:scim:schemas:extension:enterprise:2.0:User

Target Object Actions

☒ Create

☒ Update

☒ Delete

Attribute Mappings

Attribute mappings define how attributes are synchronized between Microsoft Entra ID and customappsso

customappsso Attribute	Microsoft Entra ID Attribute	Matching precedence	Edit	Remove
emails[type eq "work"].value	mail	2	Edit	Delete
active	Switch([IsSoftDeleted], , "False", "True", "True", "False")		Edit	Delete
userName	userPrincipalName	1	Edit	Delete
name.givenName	givenName		Edit	Delete
name.familyName	surname		Edit	Delete

[Add New Mapping](#)

☐ Show advanced options

Required Attribute Mappings for SCIM Provisioning Between Microsoft Entra ID and SAP LeanIX

To make Entra ID compatible with the SCIM implementation, configure the predefined email attribute as a matching attribute with the Matching precedence set to 2.

Edit Attribute ...

A mapping lets you define how the attributes in one class of Microsoft Entra object (e.g. Users) should flow to and from this application.

Mapping type ⓘ

Direct

Source attribute * ⓘ

mail

Default value if null (optional) ⓘ

Target attribute * ⓘ

emails[type eq "work"].value

Match objects using this attribute

Yes

Matching precedence * ⓘ

2

Apply this mapping ⓘ

Always

Configuring the Email Attribute

i Note

To learn more about attribute mapping and SCIM provisioning, refer to the Microsoft Entra ID documentation:

- [Customize attribute mappings](#)
- [Plan provisioning for a SCIM endpoint](#)

4. To test provisioning, select **Provision on demand** and enable provisioning for a test user.

Overview

Provision on demand

Manage

Provisioning

Users and groups

Expression builder

Monitor

Provisioning logs

Audit logs

Insights

Troubleshoot

New support request

Start provisioning

Stop provisioning

Restart provisioning

Edit provisioning

Provision on demand

Refresh

Current cycle status

Incremental cycle completed.

100% complete

Users

1

View provisioning logs

Statistics to date

View provisioning details

View technical information

Manage provisioning

Update credentials

Edit attribute mappings

Add scoping filters

Provision on demand

Selecting the Provision on demand Option for Testing the Configuration

5. After you've tested provisioning for a test user, click **Start provisioning**.

Configuring Externally Managed User Roles

If your organization manages user roles externally within Microsoft Entra ID and not within SAP LeanIX, create an attribute for the role property. To learn more about managing user roles, see [Managing User Roles with SSO](#).

Follow these steps:

1. On the [Attribute Mapping](#) page, select the [Show advanced options](#) checkbox, then select [Edit attribute list](#).

Attribute Mapping

Save

Discard

Enabled

Yes

No

Source Object

User

Source Object Scope

All records

Target Object

urn:ietf:params:scim:schemas:extension:enterprise:2.0:User

Target Object Actions

Create

Update

Delete

Attribute Mappings

Attribute mappings define how attributes are synchronized between Microsoft Entra ID and customappsso

customappsso Attribute	Microsoft Entra ID Attribute	Matching precedence	Edit	Remove
emails[type eq "work"].value	mail	2	Edit	Delete
active	Switch([IsSoftDeleted], , "False", "True", "True", "False")		Edit	Delete
userName	userPrincipalName	1	Edit	Delete
name.givenName	givenName		Edit	Delete
name.familyName	surname		Edit	Delete

Add New Mapping

Show advanced options

Navigating to Advanced Options from the Attribute Mapping Page

2. On the [Edit Attribute List](#) page, create a new attribute for the role property using the following string:

urn:ietf:params:scim:schemas:extension:workspacePermission:2.0:User:role

Edit Attribute List ...

Save

Discard

addresses[type eq "other"].region	String	☐	☐	☐	☐
addresses[type eq "other"].postalCode	String	☐	☐	☐	☐
addresses[type eq "other"].country	String	☐	☐	☐	☐
roles[primary eq "True"].display	String	☐	☐	☐	☐
roles[primary eq "True"].type	String	☐	☐	☐	☐
roles[primary eq "True"].value	String	☐	☐	☐	☐
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:employeeNumber	String	☐	☐	☐	☐
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:costCenter	String	☐	☐	☐	☐
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:organization	String	☐	☐	☐	☐
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:division	String	☐	☐	☐	☐
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:department	String	☐	☐	☐	☐
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:manager	Reference	☐	☐	☐	☐
urn:ietf:params:scim:schemas:extension:workspacePermission:2.0:User:role	String	☐	☐	☐	☐
	String	☐	☐	☐	☐

Tips

Creating an Attribute for the Role Property

3. On the **Edit Attribute** page, configure mapping for the attribute, then save the changes.
- a. In the **Mapping type** list, select **Expression**.

b. In the **Expression** field, enter the following:

```
IIF(Instr([appRoleAssignments], "ADMIN", "", "")>"0", "ADMIN", IIF(Instr([appRoleAssignr
```

Edit Attribute ...

A mapping lets you define how the attributes in one class of Azure AD object (e.g. Users) should flow to and from this application.

Mapping type ⓘ

Expression

▼

Expression ⓘ

IIF(Instr([appRoleAssignments], "ADMIN", "", "")>"0", "ADMIN", IIF(Instr([appRoleAssignments], "MEMBER", "", "")>"0", "MEMBER", IIF(Instr([appRoleAssignments], "VIEWER", "", "")>"0", "VIEWER", "")))

The expression was correctly parsed.

Default value if null (optional) ⓘ

[Use the expression builder](#)

Target attribute * ⓘ

urn:ietf:params:scim:schemas:extension:workspacePermission:2.0:User:role

▼

Match objects using this attribute

No

▼

Matching precedence ⓘ

0

Apply this mapping ⓘ

Always

▼

Ok

Configuring Mapping for the Role Attribute

User role mapping between Microsoft Entra ID and SAP LeanIX is enabled.

Virtual Workspaces

Manage access to fact sheets for custom user groups

Overview

With virtual workspaces you manage access control to fact sheets. Organize users into groups and assign or restrict access to fact sheets for each group. Allow users to only access specific fact sheets, for example:

- Fact sheets of the legal entity they belong to
- Fact sheets of their team

- Fact sheets of the region they are working in

Virtual workspaces are an additional option within your existing workspace. Although they are called workspaces, you can think of them as filters that hide or show fact sheets. User role permissions, such as viewer, member, or admin within SAP LeanIX, remain unchanged.

Essentials

- **Fact sheets only**

Virtual workspaces manage access to fact sheets. Diagrams, reports, and update notifications for fact sheets from all workspaces are accessible to all users. Fact sheets in diagrams are visible, even if the virtual workspace restricts access for this user. Clicking on a restricted fact sheet will result in an error for this user.

- **Activation required**

Reach out to the support team to enable the feature. If in doubt, check your contract to determine if you already have the virtual workspaces feature. If not, it is always possible to add it through your CSM.

- **Single sign-on (SSO) required**

You need to authenticate your users via SSO and also manage roles via your identity provider (IdP).

Access Control for Virtual Workspaces

The following components work together to achieve effective access control management:

- **Virtual workspaces:**

Virtual workspaces represent logical or organizational units within your organization. Such a unit could be a region, for example Americas or APAC. You define virtual workspaces based on the user groups you want to manage access for.

- **Access control entities:**

Access control entities represent virtual workspaces. To create a workspace, you add a new access control entity. For example, the access control entity APAC creates the virtual workspace for the region APAC.

Access control entities are key to managing the access. You list all your access control entities in the ACL and assign them to users and to fact sheets.

- **User groups:**

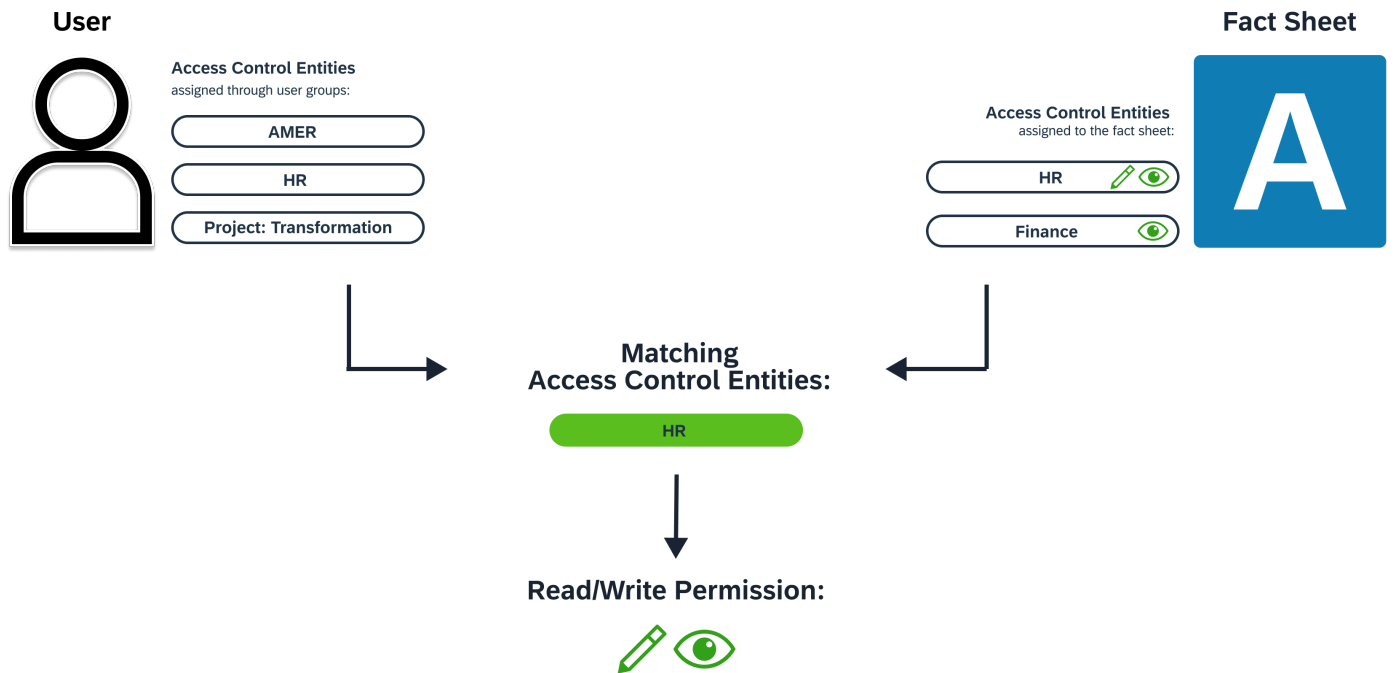
You assign users to access control entities. A user can belong to one or more entities, or none at all. You can combine several different access control entities to represent where the user needs access. You perform the assignment in your IdP through Active Directory (AD) user groups.

- **Fact sheets:**

To apply access control, you assign access control entities to fact sheets. Once assigned, fact sheets belong to virtual workspaces. For every access control entity you specify read or write permissions for a fact sheet.

- **Access control list (ACL):**

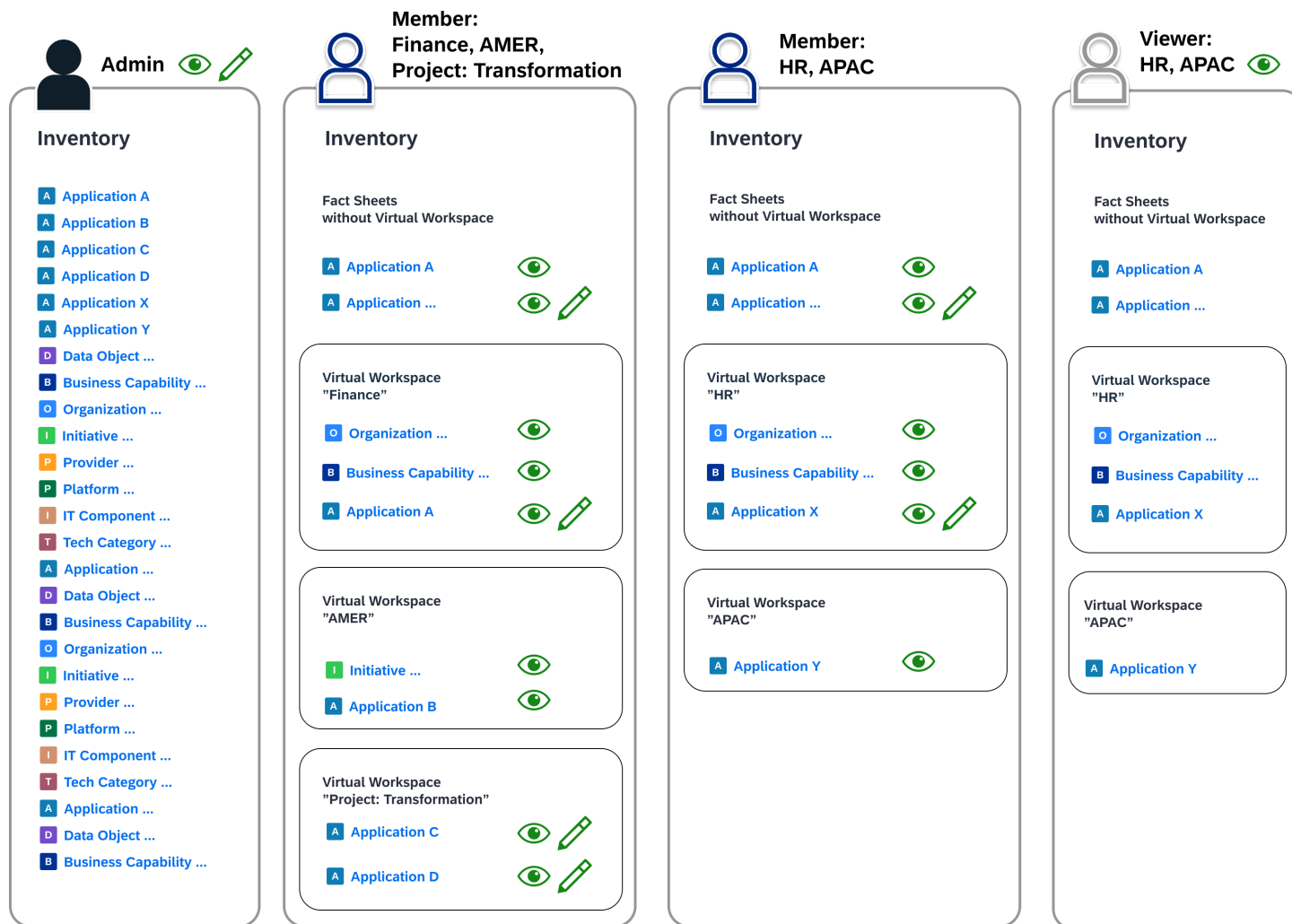
The ACL brings together all the access rights you configure for users and fact sheets. When a user logs in to SAP LeanIX, their details are checked including access permissions from the ACL. Only users that match the access control entities of the fact sheet are given read/write access to a fact sheet.



Checking for Matching Access Control Entities

Together, the access control entity assignment for each user and for each fact sheet create the unique virtual workspaces for each user. The user will see all fact sheets of each access control entity. The more access control entities a user is assigned to, the more fact sheets the user can access. Additionally, all fact sheets without any restrictions are accessible for all users in all virtual workspaces. By limiting access to fact sheets, you give your users focus.

To start configuring virtual workspaces see See [Virtual Workspace Configuration](#).

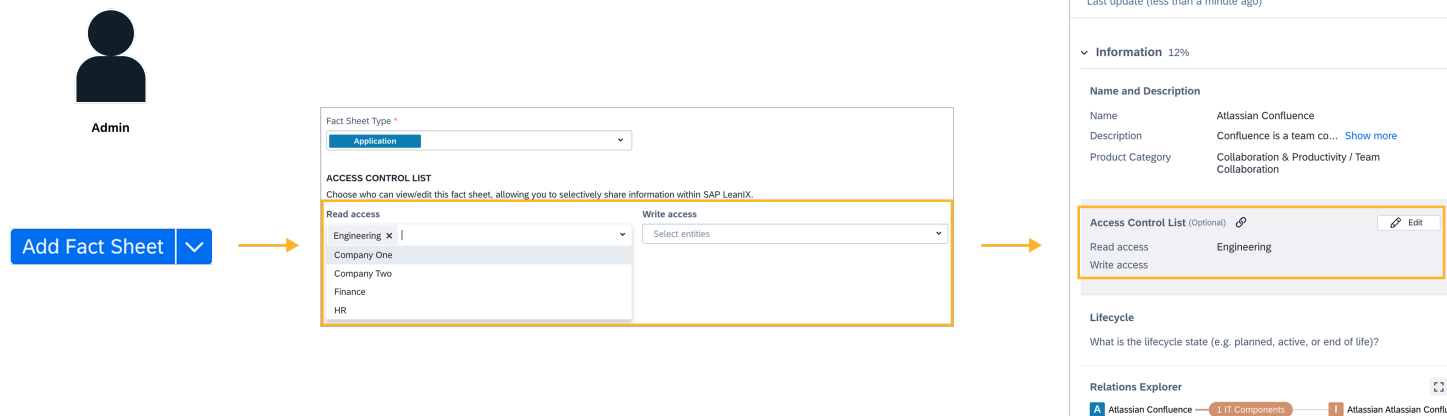


Access Control Behavior for Different User Roles

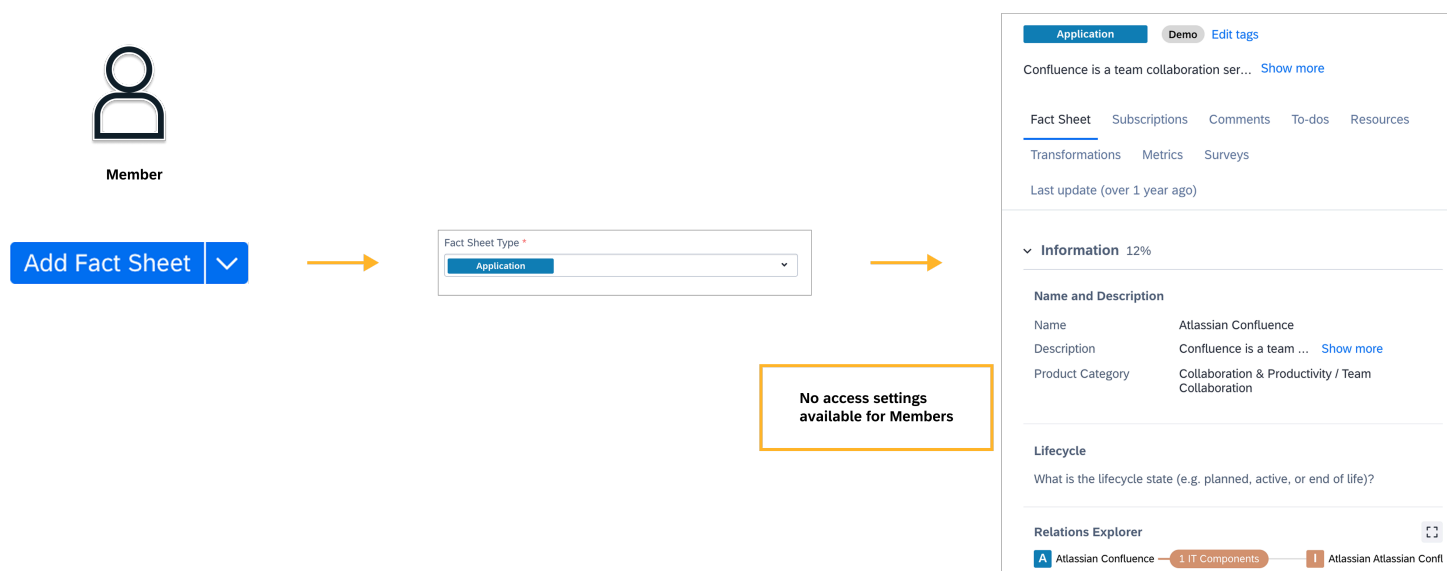
User role permissions, such as viewer, member, or admin take precedence over the virtual workspaces. Depending on the user role, the read/write settings apply differently.

User Role	Read/Write Access Behavior
Viewer	Viewers cannot edit anything, but they can see fact sheets for their virtual workspaces.
Member	Members can see and edit fact sheets in their virtual workspace. When a member creates a new fact sheet, it is created with the member's access settings. Members cannot see the access control configuration and therefore cannot change these settings. See User Flow for a Member Creating a New Fact Sheet
Admin	Admins can see and edit all fact sheets in all virtual workspaces. Admins can change the access control settings for the meta model and the fact sheet. See User Flow for an Admin Creating a New Fact Sheet

User Flow for an Admin Creating a New Fact Sheet



User Flow for a Member Creating a New Fact Sheet



Virtual Workspaces Configuration

Set up virtual workspaces to manage access for custom user groups

Prerequisites

Virtual workspaces only work with SSO configured for your SAP LeanIX:

- Make sure that users are authenticated via SSO from your identity provider, meaning the access to SAP LeanIX is managed by your identity provider.
- Make sure that users are authorized via SSO, meaning your identity provider sends the user roles (viewer, member, admin).
- Make sure you can add attributes to the SSO configuration. You need to be able to do this in later steps of the configuration. If you are not able to do this, reach out to the administrator of your identity provider for the configuration.

To learn more, see [Single Sign-On](#).

Step 1: Define User Groups for Access Control

This is custom documentation. For more information, please visit [SAP Help Portal](#).

It is important to have a solid concept of your user groups. Keep it as simple as possible to also keep the maintenance of the access control easy. It can help to draw a diagram to get an overview and leave out unnecessary details.

i Note

If you have previously configured virtual workspaces, you may encounter SSO attribute deviations.

These steps outline how to configure the new SSO setup for the SSO self-service, which includes new attributes. The legacy SSO configuration process is being phased out. If you use a legacy SSO setup, we recommend switching to the current setup. For more information, see:

- [Single Sign-On \(SSO\)](#) for details on the new SSO configuration process
- [Migration to the Self-Service SSO Configuration](#) for support with migrating to the new SSO self-service

1. List your access control entities that will be represented by virtual workspaces.
2. Group your users and assign the required access control entities.
3. Define which user groups need read and/or write access to which fact sheets.

Step 2: Create Access Control Entities in SAP LeanIX

Configure access control entities in SAP LeanIX.

1. Navigate to **Administration > Access Control**.
2. Choose **Add Entity**.
3. Enter an access control entity **ID** in uppercase letters with no space. Underscores are allowed.

Each attribute value must match the **ace** value configured in your identity provider. For example, the access control value ID value **FINANCE_DEPT** matches the **ace** value **FINANCE_DEPT**.

ID	Display Name	Description
COMPANY_ONE	Company One	Employees from Company One
COMPANY_TWO	Company Two	Employees from Company Two
FINANCE_DEPT	Finance	All employees of the Finance department

4. Enter a **Display name**.

For example: Finance

The display name is visible in SAP LeanIX when assigning access control entities manually.

5. In the **Description** field, give more details as necessary to make the maintenance of access control entities easier in the future.

6. Repeat to add more access control entities.

Step 3: Update the SSO Configuration with the Access Control Attribute

In the SSO configuration, you need to add a new attribute. You do this in your identity provider, for example Okta, Entra ID, or similar. The exact steps depend on your identity provider. In general, access control is configured in a similar way that you assign roles. To learn more, see [Single Sign-On](#).

1. In your identity provider, create the Active Directory (AD) groups for the assignment of access control entities. Add users to the AD groups.
2. In the identity provider, navigate to the SSO configuration for your SAP LeanIX application.
3. Add the new attribute `ace`.
4. Assign the `ace` values to the corresponding AD user groups.

Ensure the attribute value is uppercase with no spaces. Underscores are allowed.

Each attribute value must match the access control entity ID configured in SAP LeanIX. For example, the `ace` value `FINANCE_DEPT` matches the access control entity ID `FINANCE_DEPT`.

5. Verify that the identity provider sends the correct `ace` value to SAP LeanIX.

Use one of the following options to test the `ace` values:

- Recommended: [Check the Session Details](#)

As this option is more precise it is recommended especially for troubleshooting.

- [Check the Profile Details](#)

Check the Session Details

A SAML tracer helps you understand most SSO issues. Use it to inspect SAML requests and check which SSO attributes your identity provider sends. You can use browser extensions or desktop applications, such as SAML Tracer, to inspect the SAML request. These tools show which SSO attributes are being sent.

1. Install the SAML tracer of your choice.
2. Select a test user with specific user group assignments, or add your own user to the AD user groups.
3. Access your workspace at `https://{SUBDOMAIN}.leanix.net` and log in.
4. Open the SAML tracer browser extension or desktop application.
5. In the SAML tracer, check attribute mappings.

If the SSO configuration is successful, access control entities are displayed.

Check the Profile Details

1. Choose a (test) user that has specific user group assignments or put your own user into the AD user groups.
2. Open an incognito browser window and log in to SAP LeanIX.
3. Select the profile icon.

If the SSO configuration was successful, the access control entities are visible.



Jessica Williams

MEMBER

Permissions: Company Two, Finance

Last login: 5 days ago



My Settings



Store



Switch Workspace



Switch User Role



Invite User



Logout

Step 4: Configure Global Read/Write Settings for New Fact Sheets

Configuring the global access settings reduces maintenance in the future. New fact sheets are automatically created with the correct access settings.

Example: How do the global access settings work?

Let's look at the global settings with a small example.

A member with the access control entities "Finance" and region "AMER" creates a new fact sheet. This is how the access settings for new fact sheets will be populated:

- **Global unrestricted**

The fact sheet is visible to all users from all workspaces. Editing is possible for all users with the member role.

- **Read & Write restricted**

The fact sheet is automatically restricted to “Finance” and region “AMER” users. All members with these access control entities can edit the fact sheet.

- **Write restricted**

The fact sheet is visible to all users from all workspaces.

Editing is automatically restricted to “Finance” and region “AMER” members.

i Note

The global access setting does not apply for users with the admin role.

Admins can always see all fact sheets and configure all access settings. Therefore, fact sheets created by an admin cannot be populated with access control information.

Admins always have to select the access settings when creating a new fact sheet. Otherwise the fact sheet is accessible to everyone across all virtual workspaces.

1. In SAP LeanIX, navigate to **Administration > Meta Model Configuration**.
2. Choose a fact sheet type.
3. Choose **Edit**.
4. In the configuration, choose your preferred setting from the **Access Control** dropdown.

Meta Model: Application

Fields Quality Seal Permissions Conditional Attributes Audit Log

Fact Sheet Name English ▾

Application Add Subtypes Edit

Information

Name and De... *Help text (optional)*

Name
The name is used to identify this fact sheet ...

Release

Alias
Use this field for an alternative name, whic...

External ID
Store your External ID here which uniquely ...

Description

Edit global configuration

Key
Application

Color
#0f7eb5

Maximal Hierarchy Level
3 ▾

On-the-Fly Creation
Enabled ▾

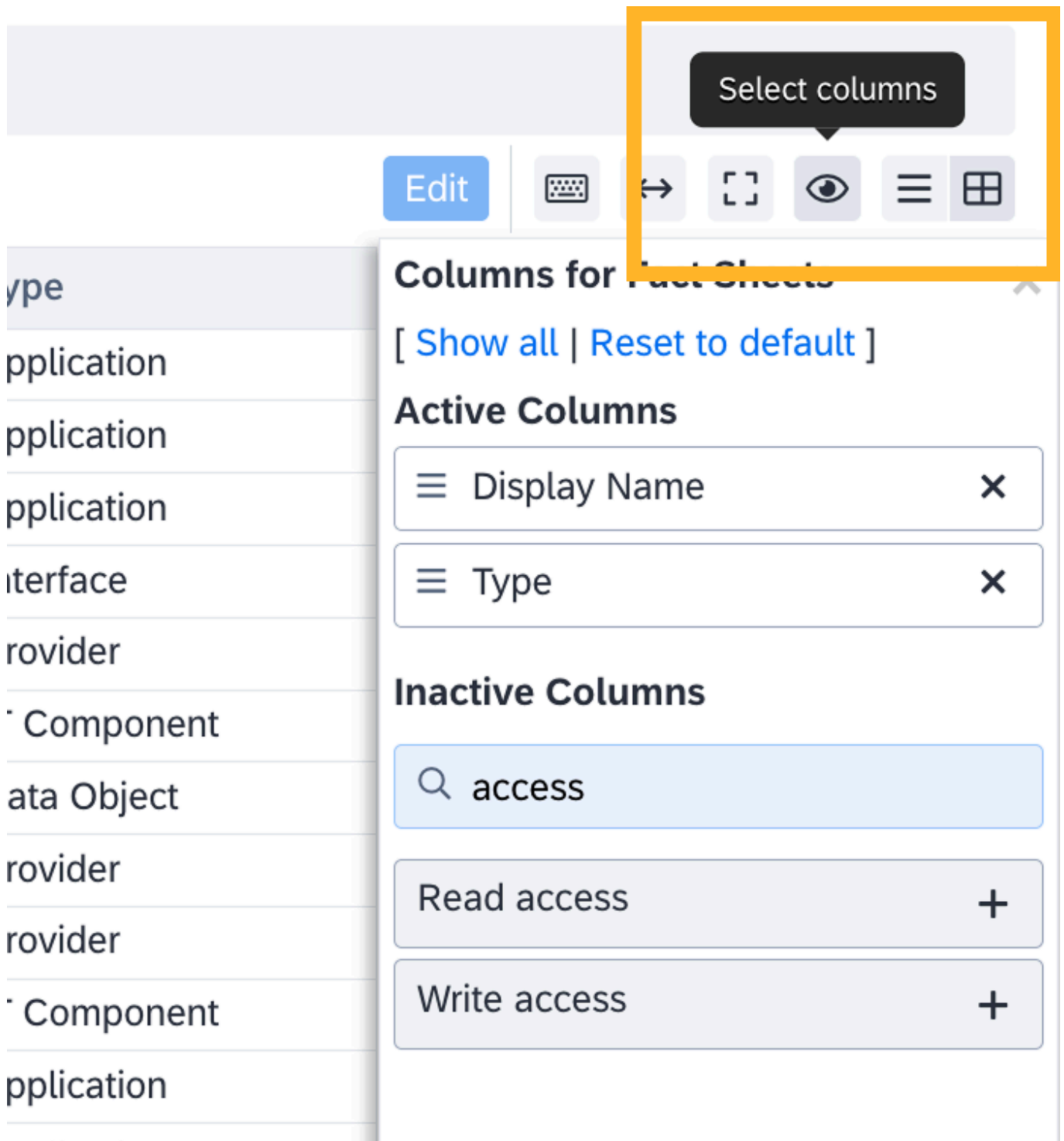
Access Control
Read & Write restricted ▾

Step 5: Assign Access Control to Existing Fact Sheets

Assigning Access Control Entities Through Import

It's best to use the import option to assign the access control entities in bulk for a large number of existing fact sheets. To learn more, see [Importing Fact Sheet Data Through Excel File](#).

1. Navigate to the [Inventory](#).
2. Change to the table view.
3. Select the columns for [Read access](#) and [Write access](#).



4. Export the Excel file.
5. In the Excel file, add the access control entities for each fact sheet.

You can add more than one access control entity per fact sheet.

	A	B	C	D	E	F
1	id	type	name	displayName	permittedReadACL	permittedWriteACL
2	ID	Type	Name	Display Name	Read access	Write access
3	28fe4aa2-6	Application	AC Manage	AC Management	COMPANY_ONE;COMPANY_TWO	
4	2efa37b5-1	Application	AC Manage	AC Management		COMPANY_ONE;ENGINEERING_DEPT
5	109ca295-	Application	AC Manage	AC Management		COMPANY_TWO;ENGINEERING_DEPT
6	688f3195-	Interface	AC Manage	AC Management		HR_DEPT
7	b01f0ca9-f	Provider	accenture	accenture		IT_DEPT
8	9deb9733-	ITComponent	App Mainte	accenture App Ma		
9	85a244d3-	DataObject	Account	Account		
10	beb56daa-	Provider	acctur	acctur	COMPANY_ONE	COMPANY_ONE;IT_DEPT
11	59208e96-	Provider	Acocce	Acocce	COMPANY_TWO	COMPANY_TWO;IT_DEPT
12	65e5f779-b	ITComponent	Application	Acocce Application		
13	2e50479c-	Application	Adobe Crea	Adobe Creative C		
14	7aaf9610-c	Application	Adobe Acro	Adobe Creative C		COMPANY_ONE
15	f9d649e8-8	Application	Adobe Illus	Adobe Creative C		COMPANY_ONE

6. Import the updated Excel file.

Assigning Access Control Entities in the Inventory Table View

The table view of the inventory offers an overview of access permissions. In the table, you have quick access to a dropdown to configure access control entities for each fact sheet and easily navigate between fact sheets.

1. Navigate to the [Inventory](#).
2. Change to the table view.
3. Select the columns for [Read access](#) and [Write access](#).

The screenshot shows the 'Columns for Fact Sheets' dialog in the SAP Fiori interface. The 'Edit' button is highlighted in blue. A yellow box highlights the 'Select columns' button and the column selection icons (keyboard, double arrow, crop, eye, list, grid). The dialog shows 'Active Columns' (Display Name, Type) and 'Inactive Columns' (access, Read access, Write access).

4. Choose [Edit](#).
5. Choose a field with [Read access](#) or [Write access](#) and start typing.

The available access control entities are displayed in a dropdown. You can add more than one access control entity for each fact sheet.

Inventory Add Fact Sheet Save As

AI

40 of 702 results Cancel Save

Display Name ↑	Type	Read access	Write access
AC Management	Application	Company One, Company Two	
AC Management Cloud	Application		Company One, Engineering
AC Management Cloud 2	Application		Company Two, Engineering
AC Management to HR Admin	Interface		HR
accenture	Provider		
accenture App Maintenance & Su...	IT Component		
Account	Data Object		
acctur	Provider	Company One	
Acoce	Provider	Company Two	
Acoce Application Hosting	IT Component		
Adobe Creative Cloud	Application		
Adobe Creative Cloud / Adobe Ac...	Application		Company One
Adobe Creative Cloud / Adobe Ill...	Application		Company One

IT x |
Company One
Company Two
Engineering
Finance
HR

6. Update all required fact sheets manually.

7. Choose **Save**.

Assigning Access Control Entities for a Single Fact Sheet

You can update the access permissions directly in the fact sheet settings.

1. Navigate to the fact sheet.
2. On the Fact Sheet tab, go to the Access Control List (optional) section.
3. Choose Edit.
4. Choose a field with Read access or Write access.

The available access control entities are displayed in a dropdown. You can add more than one access control entity for each fact sheet.

Fact Sheet

Application

Demo

[Edit tags](#)

Confluence is a team collaboration service. Co... [Show more](#)

Fact Sheet

Subscriptions

Comments

To-dos

Resources

Transformations

Metrics

Surveys

Last update (less than a minute ago)

▼ **Information** 12%

Name and Description

Name	Atlassian Confluence
Description	Confluence is a team co... Show more
Product Category	Collaboration & Productivity / Team Collaboration

Access Control List (Optional)



Edit

Read access	Engineering
Write access	

Lifecycle

What is the lifecycle state (e.g. planned, active, or end of life)?

Relations Explorer



Atlassian Confluence

1 IT Components



Atlassian Atlassian Confl

Access to SAP for Me

Learn how to set up SAP for Me access for your organization.

Overview

SAP for Me is the central access point for all SAP activities. It helps you and your team manage products, contacts, support cases, and licensing. S-user administrators set up SAP for Me access for their teams by generating S-user IDs. Whether you're setting up your organization's first administrator or managing user permissions, this page will guide you to the right resources.

What Is an S-User ID?

An S-user ID is a secure login used to access SAP tools and services such as:

- SAP for Me
- SAP Support Portal
- SAP Community

These IDs are essential for submitting support requests, managing workspaces, and accessing product information.

Learn more about S-User IDs in [Get started with your S-user ID](#) .

Who Manages S-User IDs?

An S-user administrator manages your organization's access to SAP systems. This person creates and deletes S-user IDs, assigns permissions (known as authorizations), and helps colleagues access tools such as SAP for Me.

Learn more about how to designate an S-User administrator in [Setting Up Your Organization's S-User Administrator](#).

Finding Your S-User Administrator

Only S-user administrators can create and manage S-user IDs. If you aren't an administrator, work with your S-user administrator to get started.

If you're new to SAP or don't know who your administrator is, try the following:

- Review your welcome email from SAP. It often includes your initial S-user ID and contact information for your administrator.
- Ask your internal IT team or SAP contact. Most organizations have a designated SAP administrator.
- Contact SAP Support through the [SAP Support Portal](#)  to identify your administrator.

Getting Started

To get started, explore the following guides:

- [Setting Up Your Organization's S-User Administrator](#) - Learn how to designate your first administrator using the User Onboarding Token and understand the role of the cloud administrator.
- [Creating and Managing S-Users in SAP For Me](#) - Discover how to create and manage S-user IDs and authorizations.

Setting Up Your Organization's S-User Administrator

This guide describes how set up your organization's first S-user administrator to start managing access to SAP for Me

Overview

S-user administrators control your organization's access to SAP systems. They create and delete S-user IDs, assign permissions, and help colleagues use tools such as SAP for Me.

When your organization becomes an SAP customer, SAP creates the first S-user ID. This ID has the highest authorization level and acts as the super administrator or cloud administrator. This account can then create other administrators roles, for example, the organization's S-user administrator for user management and access.

This guide describes how set up your organization's first S-user administrator to start managing access to SAP for Me.

i Note

After this, SAP does not create or manage more S-user IDs for your organization. For security reasons, your S-user administrators must manage all additional user accounts.

Creating Your First S-User Administrator

Prerequisites

You have access to the SAP Sales Order document and the SAP Onboarding Portal.

Context

Super/cloud administrators can choose a user to handle user management and permissions for their organization. They use a User Onboarding Token for this process. This setup ensures someone in your organization can manage SAP access. Learn more about [User Onboarding Tokens](#) .

i Note

You can only redeem a User Onboarding Token once.

Procedure

1. Open your SAP sales order document.
2. Scroll to the section **AUTHORIZED ADMINISTRATORS** and find the line that says **User Onboarding Token**. Copy the token listed.
3. Go to the [SAP Onboarding Portal](#)  and choose **Start**.
4. Paste the token you copied from the sales order into the field **User Onboarding Token**
5. Enter the new administrator details by filling in the required fields:
 - a. **First and last name**
 - b. **Email address**
 - c. **Company details** (if prompted)
6. Verify your corporate email address and complete your account setup.

Results

SAP updates the administrator in the SAP systems with the name of the person who redeemed the token.

The administrator S-user is created and assigned to the person who redeemed the token. This person will receive all technical product-related notifications, renewal notices, and S-user lifetime notifications.

Creating Additional S-User Administrators

Context

Administrators can duplicate their own S-user authorizations to create new administrator S-user IDs.

→ Tip

SAP recommends creating at least two additional S-user administrators.

To create additional S-user administrators, follow the instructions in [Cloning S-User ID Authorizations](#).

Creating and Managing S-Users in SAP For Me

This guide describes how to create, manage, and troubleshoot S-user IDs to enable seamless access to SAP for Me.

Overview

To access SAP tools and services, you need a secure login called an S-user ID. These IDs are essential for accessing SAP for Me, participating in discussions in the SAP community, provisioning SAP LeanIX workspaces, and submitting support requests.

This guide describes how to create, manage, and troubleshoot S-user IDs to enable seamless access to SAP for Me.

i Note

Only S-user administrators can create and manage S-user IDs.

Prerequisites

Before you begin:

- Ensure you're the designated S-user administrator. Learn more about the administrator concept in [About the User, Authorization and Administrator Concept](#) .
- You must have access to SAP for Me. Learn how to get access in [Access and Authorizations](#) .

Creating S-User IDs

Context

In the user management area of SAP for Me, administrators can request S-user IDs for their colleagues and assign them the required authorizations. For example, you might request S-user IDs for all workspace admins to make sure they can submit

Learn more about authorizations in [About the User, Authorization and Administrator Concept](#) .

i Note

By default, new S-user IDs are valid for 24 months from the date of creation. Administrators can adjust the expiration date if needed. To learn more, refer to the [S-user lifetime overview](#) .

Procedure

1. Go to SAP for Me.
2. In the **User Management** area, go to **Home > Users & Contacts > User Management**.
3. **Choose Request User.**
4. Enter the user's details in the **Request User** form.
5. Optional: If you've set up authorization packages for user groups within your organization, you can also assign them at this time. Choose "+" to select an authorization package.

Learn more about authorization packages in [Creating Authorization Packages](#).

6. Choose **Submit**.
7. On the **User was requested** confirmation screen, choose **OK**.

Results

The new user receives an email from SAP Notification Services with the new S-user ID and an activation link. The activation link is valid for 10 days.

i Note

The user request is not instant and may take some time to complete. You can check the status of the new user on the **Requested Users** tab. Once the user is complete, it moves to the **Users** tab.

Deleting S-User IDs

Procedure

1. Go to SAP for Me.
2. In the **User Management** area, go to **Home > Users & Contacts > User Management**.
3. Select the user that you want to delete.
4. Choose **Delete**.

Editing S-User ID Authorizations

Context

You may want to edit an S-user's authorizations if the permissions they currently have are insufficient for performing their role. You can grant or restrict access to specific functions and fine-tune their access by setting appropriate authorization levels.

Procedure

1. Go to SAP for Me.
2. In the **User Management** area, go to **Home > Users & Contacts > User Management**.
3. Choose a user's name to open their details.
4. Open the **Authorizations** tab in the user details window. Choose the edit (pencil) icon to begin editing.
5. Select the desired authorizations.
6. Optional: Adjust the authorization level for granular control, such as restricting case creation to a specific installation number.

By default, authorizations are granted at the global level for all customer numbers and installations.
7. Choose **Save**.

Creating Authorization Packages

Context

Authorization packages simplify the process of assigning multiple permissions to users by grouping them into reusable bundles. You can apply these packages directly to S-user IDs in your company, making it easier to manage access for users with similar roles or responsibilities.

Example

Suppose your support team needs access to case management features. You can create a package called "Cases" that includes the following authorizations:

- Report Technical Problem
- Send Cases to SAP
- Close Cases

When a new team member joins, assign the "Cases" package to their S-user ID. This automatically grants them all necessary permissions, so you don't need to configure each one manually.

Caution

If you make changes to the package, such as adding, removing, or modifying authorizations, they automatically apply to all users assigned to it.

Procedure

1. Go to SAP for Me.
2. In the **User Management** area, go to **Home > Users & Contacts > User Management**.
3. Choose **Auth. Packages** from the menu.
4. Choose **Create**.
5. On the **Create Authorization Package** screen, enter a name.
6. Optional: Super administrators can turn on **Write Protection** so others can't modify the package.
7. Select all authorizations to be assigned to the package (for example, all case authorizations).
8. Choose **Save**.

Cloning S-User ID Authorizations

Context

Administrators can clone an existing authorization profile by copying it from a source S-user to one or more target S-users. This ensures that all selected users are granted an identical set of authorizations.

Procedure

1. Go to SAP for Me.
2. In the [User Management](#) area, go to the [Users](#), [Requested Users](#), or [Important Contacts](#) tab.
3. Search for the source S-user ID to copy authorizations from. To open the user details screen, choose anywhere in the row for that user.

If you're an administrator and copying your own authorizations to other S-user IDs, select your own S-user ID.

4. Choose [Copy Authorizations](#).

If you don't see this option, you aren't logged in with a super, cloud, or user administrator S-user ID

5. Select the users to whom you want to apply the copied authorizations, and choose [OK](#).
6. On the [Copy Options](#) screen, choose how you want to copy the authorizations.
 - o [Add Authorizations](#): The selected users keep their existing authorizations and receive additional authorizations based on those copied from the source S-user.
 - o [Overwrite Authorizations](#): This replaces the selected users' authorizations with the authorizations from the source S-user.
7. Choose [Continue](#).

Troubleshooting

Issue	Solution
No activation email received	Ask the user to check spam/junk folders. Resend from User Management if needed.
Can't assign authorizations	Ensure you have the required admin role and the same authorizations.
User not appearing in list	Check the Requested Users tab. The user may still be processing.